MSc thesis — report in progress

Energy Storage Bids in Day-Ahead Electricity Markets

Reproducing a state-of-charge-based market design with interval bids, and the cost of imperfect price information

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Version	Working version, 1 November 2026

This is a living document. It records all results obtained so far and is updated as the thesis progresses. Chapters marked as *in progress* contain designs or preliminary results that have not yet been approved.

Abstract

Battery energy storage systems (ESS) are becoming price-setting resources in wholesale electricity markets. Their value comes from moving energy in time, and is therefore governed by their state of charge (SOC). In most day-ahead markets, however, the clearing problem does not represent the SOC: a storage owner submits a charging bid and a discharging offer for every interval, and must anticipate its own SOC through those bids. Baldick has recently proposed a state-of-charge-based design in which the market operator optimises the SOC directly from the technical data of each ESS and a single bid for its end-of-horizon SOC. That design maximises social welfare, but requires a market reform.

This thesis asks whether the outcome of the SOC-based design can be reached in the current, interval-based design through an appropriate choice of bids, what information those bids require, and what is lost when that information is imperfect. Both designs are formulated in a common market model with a DC network, ramping limits and elastic demand, and their Karush–Kuhn–Tucker (KKT) conditions are compared term by term. The main analytical results are the following. (i) Interval bids built from the multipliers of the SOC-based design reproduce its dispatch and nodal prices exactly. (ii) The multipliers are not needed: any optimal dual of the storage owner's own price-taking self-schedule, computed from its technical data and the prices at its node, yields equivalent bids, for any number of storage units. (iii) The set of all equivalent bids is a polyhedron that depends only on those two inputs. Prices pin the bids in every interval except those in which an SOC limit is reached, where a selection rule is needed. The gap between the two designs is therefore a gap in price information, not in market design.

The SOC-based design is also computationally heavier, with a solve time that grows faster than that of the current design in both the number of storage units and the number of intervals. An experimental framework quantifies the welfare cost of imperfect price forecasts. First experiments on Baldick's test case confirm that bids built from the true multipliers and quantities lose nothing, and show that the loss is driven both by forecast errors and by quantity information that interval bids cannot carry: an hourly market cannot coordinate a purchase with the sale it is meant for, and with several owners their spare-capacity bids can cross.

Keywords: energy storage, state of charge, day-ahead market, market design, KKT conditions, equivalent bids, price forecasting, market monitoring.

Status of this document

This report collects every result of the thesis to date, in the order in which a final thesis would present them. It is written in parallel with the technical note on the KKT equivalence, which contains the same results in a compact form; Appendix C maps the numbering of the two documents.

Chapter	**Content**	**Status**
1 Introduction	Motivation, research questions, contributions	drafted
2 State of the art	Market designs for storage, storage models, bidding, monitoring, forecasting	drafted; to be extended
3 Market models	Both designs, Lagrangian, KKT conditions, reference example	complete
4 Equilibrium (MCP)	Participant problems, complete MCP, partition of the KKT system, equilibrium theorem	complete
5 Equivalence	Equivalent bids, Lagrangian relaxation, conditions on stepped bids	complete
6 Bids from technical data and prices	Value of stored energy, segments, Theorem, polyhedron, selection rule, several ESS	complete
7 Computational burden	Problem size, coupling, benchmark	complete
8 Numerical verification	Baldick's three cases, several ESS	complete
9 Experimental design	Bid families, tie-breaker, quantity rule, error model, pipeline, metrics, stages	implemented; *decisions pending*
10 First experiments	Stage 0 (one ESS, Baldick's case); Stage 1 case and zero-error check (two owners)	Stage 0 complete; *Stage 1 on hold*
11 Discussion and outlook	Answers so far, limitations, next steps	drafted

Open decisions (to be agreed with the supervisors).

1. Quantity rule with several owners: keep the spare-capacity blocks, cap them by the SOC of the owner's plan, or use realistic power limits (Sections 9.5 and 10.5.2).
2. Real-time evaluation: whether an ESS's charge is curtailed before load is shed; closest SOC-feasible schedule, redispatch premium $\psi \in \{0, 10, 50\}$ \$/MWh, value of lost load (Section 9.8).
3. Stage 1 grid: two owners with price and terminal-value errors, the full one-ESS grid, or both (Section 9.11).
4. Error model for Stages 1–2 (additive Gaussian with persistence, a common component and a price floor) and its calibration from historical forecasts in Stage 3 (Sections 9.6 and 9.7).
5. Main indicator: welfare loss as a percentage of the storage value of the benchmark, and a second test system whose storage value is not dominated by avoided load shedding (Sections 9.9 and 10.1).
6. Test system and storage fleet of Stages 2 and 3 (Section 9.11).

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Nomenclature

Acronyms	
CD	current design (interval-by-interval bids and offers, no SOC in the clearing)
DA, RT	day-ahead, real-time
DEB	default energy bid (CAISO market power mitigation)
ESS	energy storage system
ISO	independent system operator (market operator)
KKT	Karush–Kuhn–Tucker (optimality conditions)
LMP	locational marginal price
LP	linear program
MCP	mixed complementarity problem
MPM	market power mitigation
SOC	state of charge
SW	social welfare
VOLL	value of lost load
Sets and indices	
$t \in \mathcal{T} = \{1, \dots, T\}$	intervals of length ΔT (h)
$n, m \in \mathcal{N}$; r	buses; reference bus
$(n, m) \in \Lambda$	transmission lines, oriented from bus n to bus m
$i \in \mathcal{G}$, $k \in \mathcal{K}_i$	generating units; offer blocks of unit i
$l \in \mathcal{D}$, $j \in \mathcal{J}_l$	elastic loads; bid blocks of load l
$e \in \mathcal{E}$	storage units (ESS)
$\mathcal{K}^g_e$, $\mathcal{J}^p_e$	discharge-offer and charge-bid blocks of ESS e (CD only)
$\mathcal{G}(n), \mathcal{D}(n), \mathcal{E}(n)$	units, loads and ESS at bus n
$n(i), n(l), n(e)$	bus of unit i, load l, ESS e
Primal variables	
$q_{i,k,t}$; $Q_{i,t} := \sum_k q_{i,k,t}$	output of block k of unit i; total output (MW)
$d_{l,j,t}$	served elastic demand of block j of load l (MW)
$f_{n,m,t}$; $\theta_{n,t}$	flow from n to m (MW); voltage angle at bus n (rad)
$p_{e,t}$, $g_{e,t}$	charging and discharging power of ESS e (MW)
$p_{e,j,t}$, $g_{e,k,t}$	charging and discharging power per bid/offer block (CD, MW)
$s_{e,t}$	state of charge at the end of interval t (MWh)
Data: rest of the market	
$c_{i,k}$, $\bar{q}_{i,k,t}$	offer price (\$/MWh) and size (MW) of block k of unit i
$u_{l,j}$, $\bar{d}_{l,j}$	bid price (\$/MWh) and size (MW) of block j of load l
$\bar{R}_i$, $\underline{R}_i$	ramp-up and ramp-down limits of unit i (MW per interval)
$D_{n,t}$	inelastic demand at bus n (MW)
$B_{n,m}$, $\bar{f}_{n,m}$	susceptance (p.u.) and thermal limit (MW) of line (n, m)
Data: storage	
α_e	charging coefficient: stored MWh per MWh drawn from the grid
β_e	inverse discharging coefficient: stored MWh per MWh delivered; $\beta_e > \alpha_e > 0$
γ_e	retention factor per interval (self-discharge), $0 < \gamma_e \leq 1$
ϵ_e, ζ_e	degradation cost of charging and discharging (\$/MWh)
$\underline{s}_e$, $\bar{s}_e$	minimum and maximum SOC (MWh)

Nomenclature (continued)

$\bar{p}_e$, $\bar{g}_e$	maximum charging and discharging power (MW)
$s_{e,0}$; $\bar{\bar{s}}_{e,T} := \gamma_e^T s_{e,0}$	initial SOC; SOC at T if the ESS stays idle (MWh)
$b_e(\cdot)$, $w_e(\cdot)$	end-of-horizon benefit (concave) and its derivative, the willingness to pay for stored energy (\$/MWh); w_e is a constant for a linear benefit
θ_e	technical data of ESS e: $(\alpha_e, \beta_e, \gamma_e, \epsilon_e, \zeta_e, \underline{s}_e, \bar{s}_e, s_{e,0}, \bar{p}_e, \bar{g}_e, b_e)$
Data: storage bids in the current design	
$u^p_{e,j,t}$, $\hat{p}_{e,j,t}$	charging bid price (\$/MWh) and block size (MW)
$c^g_{e,k,t}$, $\hat{g}_{e,k,t}$	discharging offer price (\$/MWh) and block size (MW)
$u_{e,t}$, $c_{e,t}$	single-block bid and offer prices (block index dropped)
Dual variables (all in \$/MWh)	
$\lambda_{n,t}$	nodal power balance: locational marginal price
$\kappa_{n,m,t}$; ξ_t	DC power flow of line (n,m); reference angle
$\bar{\mu}_{n,m,t}$, $\underline{\mu}_{n,m,t}$	thermal limits of line (n,m)
$\bar{\delta}_{i,k,t}$, $\underline{\delta}_{i,k,t}$	bounds of offer blocks
$\bar{\tau}_{i,t}$, $\underline{\tau}_{i,t}$; $\varrho_{i,t}$	ramp-up and ramp-down limits; ramp wedge
$\bar{\psi}_{l,j,t}$, $\underline{\psi}_{l,j,t}$	bounds of demand blocks
$\pi_{e,t}$	SOC dynamics: marginal value of energy stored at the end of t
$\bar{\nu}_{e,t}$, $\underline{\nu}_{e,t}$	upper and lower SOC limits: SOC scarcity rents
$r_{e,t} := \underline{\nu}_{e,t} - \bar{\nu}_{e,t}$	net SOC rent
$\bar{\rho}_{e,t}, \underline{\rho}_{e,t}$; $\bar{\sigma}_{e,t}, \underline{\sigma}_{e,t}$	charging and discharging power bounds (per block in the CD)
$(\cdot)^\star$	value at a primal–dual optimum of the SOC-based design
Equivalent bids	
$\mathcal{P}^{\mathrm{CD}}$, $\mathcal{P}^{\mathrm{SOC}}$	clearing problems of the current and SOC-based designs
$\mathcal{P}^{\mathrm{SS}}_e(\lambda)$	price-taking self-schedule of ESS e at prices λ
$\mathcal{D}_e(\lambda)$	set of optimal multipliers of $\mathcal{P}^{\mathrm{SS}}_e(\lambda)$
$E(m)$	bids built from multipliers m by the map (E)
$\Phi_e(\theta_e, \lambda)$	set of equivalent bids of ESS e at prices λ
$(p^\circ, g^\circ, s^\circ)$	an optimal schedule of $\mathcal{P}^{\mathrm{SS}}_e(\lambda)$ (the owner's plan)
$\mathcal{U}$, $\mathcal{L}$	intervals in which the upper (lower) SOC row is tight in $(p^\circ, g^\circ, s^\circ)$
$\mathcal{H}_e$; $h_{e,k}$; N_e	intervals with a binding SOC limit; the k-th one; their number
$\mathcal{S}_{e,k}$; $\mathcal{S}^\circ_{e,k}$	segment k of the SOC path; its intervals without a binding limit
$\tilde{t}_{e,k}$; $\tau(t)$	marginal interval of segment k; marginal interval of the segment of t
$v_{e,k}(\lambda)$	value of stored energy of segment k, read from prices
$\pi^-_{e,t}, \pi^+_{e,t}$; $\pi^R_{e,t}$	admissible range of $\pi_{e,t}$; pre-limit selection
$[x]^+$	$\max\{x, 0\}$
Experiments	
$\hat{\lambda}_{e,t}$	price forecast used by ESS e
$\bar{\lambda}$	mean of $\lambda^\star$ over the horizon
σ, ρ, b, ω	error size, persistence (AR(1) coefficient), bias and common share
χ_t, $\eta_{e,t}$; υ_t	common and ESS-specific error processes (unit variance); their shocks
$\underline{\lambda}$	price floor of the forecast (\$/MWh)
$\hat{w}_e$, σ_w	owner's estimate of its terminal value; size of its error (\$/MWh)
κ^{tb}	throughput tie-breaker of the self-schedule (\$/MWh)
δ	price step of the quantity rule (\$/MWh)
ψ	real-time redispatch premium (\$/MWh)
$\mathcal{B}^\star, \mathcal{B}^{\star p}, \mathcal{B}^{\mathrm{fcst}}, \mathcal{B}^{\mathrm{priv}}$	bid families (Section 9.3)
$SW^\star$, SW^{real}, ΔSW	benchmark and realised social welfare; welfare loss
p_k, L_k	probability and loss of regime k of the experiment outcomes

V	value of storage in the benchmark, $SW^{\star} - SW^{\text{noESS}}$

Conventions. Multipliers of constraints written in power (MW) enter the Lagrangian with the factor ΔT; multipliers of constraints written in energy (MWh), namely π, $\bar{\nu}$ and $\underline{\nu}$, enter without it. All multipliers are then in \$/MWh. The complementarity condition $0 \leq a \perp b \geq 0$ means $a, b \geq 0$ and $a^{\top} b = 0$; for a box, $l \leq z \leq u \perp F$ means $z = l \Rightarrow F \geq 0$, $l < z < u \Rightarrow F = 0$, $z = u \Rightarrow F \leq 0$.

Notes on symbols. The technical note uses m both for loads and for the second bus of a line; here loads are indexed by l, so that m always denotes a bus. The experiment slides call the common share κ and the common error ξ_t; here they are ω and χ_t, because κ and ξ are already multipliers of the network constraints. The persistence ρ (no bar) is not to be confused with the multipliers $\bar{\rho}, \underline{\rho}$.

Chapter 1

Introduction

1.1 Context

Battery energy storage systems (ESS) are now a significant and fast-growing part of the generation fleet in several power systems. In the United States, utility-scale battery capacity grew by 66% in 2024 and exceeded 26 GW, with close to 20 GW of additions planned for 2025 [1]. In ERCOT, more than 10 GW of storage had been installed by the end of 2024 [2], and in California the installed storage capacity exceeded 17 GW by mid-2026 [3]. In Europe, the market coupling operators are preparing a dedicated storage order for the single day-ahead coupling, planned for the end of 2027 [4]. At these scales, storage is no longer a price-taker at the margin: it often sets the market price, and the rules under which it bids shape both prices and welfare.

The economic value of storage comes from moving energy in time: buying when energy is cheap and selling when it is expensive [5, 6]. What a battery can sell in one interval depends on what it bought before, net of losses. Its feasible operation is therefore governed by its *state of charge* (SOC), which couples all intervals of the market horizon. This intertemporal coupling is what distinguishes storage from a conventional generator or load, and it is what makes its market participation difficult to design.

Regulation has opened wholesale markets to storage. FERC Order No. 841 required every US market operator to define a participation model through which storage can buy and sell energy and ancillary services [7]. The order did not prescribe how the SOC should be handled, and the resulting rules differ on one key point: *who manages the state of charge in the day-ahead market.*

1.2 Two ways of handling the state of charge

Interval bids (current design). In several markets, including the ERCOT day-ahead market and the European day-ahead coupling, an ESS participates through interval-specific bids and offers, historically as a paired generator and load. For each interval it submits a discharging offer and a charging bid, and the day-ahead clearing problem does not contain its SOC [8, 9]. The intertemporal coupling is left to the owner, who must embed in each interval's bid an estimate of the value of stored energy. To do so, the owner needs a forecast of future prices. In this report this is called the *current design* (CD).

SOC-based design. Baldick [8], building on earlier work on battery representation in unit commitment [10], proposes the opposite allocation. The ESS owner submits only its technical parameters (efficiencies, retention, degradation costs, SOC and power limits) and a single bid for the deviation of its end-of-horizon SOC from the SOC it would have if it stayed idle. The market operator optimises the SOC trajectory together with the rest of the system. The clearing problem then internalises the intertemporal constraints of storage, and its outcome maximises declared social welfare. Intermediate designs exist: CAISO and NYISO enforce the SOC in the clearing but still require interval bids and have no end-of-horizon bid [8].

The difficulty. The SOC-based design is the natural welfare benchmark, but adopting it requires a reform of market rules, clearing software and settlement, and it transfers responsibility for the SOC from the owner to the market operator. Section 1.3 discusses why markets may not adopt it, or not soon. In the meantime, markets continue to run on interval bids. This motivates the central question of the thesis.

Central question. Can the outcome of the SOC-based design (dispatch and prices) be reproduced in the current design through an appropriate choice of storage bids? If so, what information do those bids require, and how much welfare is lost when that information is imperfect?

1.3 Why not adopt the SOC-based design directly?

The SOC-based design is the welfare benchmark, so it is fair to ask why this thesis does not simply recommend it. The reasons below explain why markets may not adopt it, or not soon. They also explain why the comparison between the two designs is useful whichever design is eventually chosen.

- *Reform cost and time.* A new storage model changes market rules, clearing and settlement software, and requires a regulatory and stakeholder process. Baldick notes that implementing his model "would, however, require considerable changes to the structure of existing electricity markets", and that issues such as the treatment of ancillary services remain open [8, §6].
- *Responsibilities move from the owner to the operator.* In the SOC-based design the operator decides when the battery charges and discharges. It thereby takes on tasks that today belong to the owner: managing the SOC, respecting degradation and cycling limits, and answering for schedules that turn out to be infeasible. Owners lose the freedom to manage their own equipment, which they may also use for ancillary services, bilateral contracts or real-time arbitrage. Degradation in particular is rarely linear [11]; Baldick uses a proportional model and sketches SOC-dependent extensions [8, §4.3, App. A.1].
- *Model accuracy.* The operator must model each battery correctly. Efficiency depends on power, SOC and temperature, and hybrid, co-located and aggregated resources are difficult to describe with a few parameters. A mismatch between model and device can produce infeasible schedules.
- *Strategic behaviour moves, it does not disappear.* An ESS can still act strategically through the technical data and the end-of-horizon value it declares. Baldick anticipates the need to mitigate both [8, §4.3, §4.6].
- *SOC deviations and settlement.* If the operator sets the day-ahead SOC, someone must answer for real-time deviations and for the link between consecutive days [8, App. A.5]. CAISO, which already includes the SOC in its day-ahead market while keeping bids, has had to address bid cost recovery and mitigation issues specific to batteries [12].
- *Computational burden.* SOC constraints couple all intervals and make the clearing problem larger and less separable [10]. Chapter 7 quantifies this: the solve time of the SOC-based design grows faster than that of the current design in both the number of storage units and the number of intervals.

Why the comparison is useful anyway. The thesis does not argue for the current design against the SOC-based one. It answers three questions that matter whichever design is chosen.

1. *What is the reform worth?* The welfare gap between the SOC-based design and the current design with well-constructed bids is the value of the reform. If it is small for realistic price forecasts, the reform may not pay for its costs; if it is large, this is a quantitative argument for it. Without an equivalence result, this comparison would not be well posed, since the current design would be compared with arbitrary bids.
2. *Is there an intermediate step?* The equivalent bids derived in Chapter 6 can be computed by the market operator without changing its clearing engine. They can serve as a reference for monitoring storage bids, or as a voluntary default bid for owners who prefer to declare their technical data.
3. *Where does the reform matter?* The analysis shows that the difference between the designs is concentrated in the intervals in which SOC limits bind. This identifies the systems in which the SOC-based design adds most value.

1.4 Research questions and objectives

The thesis addresses four research questions. They refine those of the project plan in light of the results obtained so far.

RQ1 *Equivalence.* Do there exist charging bids and discharging offers in the current design such that its clearing outcome (dispatch and nodal prices) coincides with that of the SOC-based design? Under which conditions?

RQ2 *Information.* What information must those bids embed? Which part is private to the storage owner, and which part is endogenous to the market? Can the owner construct the bids without access to the solution of the SOC-based design?

RQ3 *Cost of imperfect information.* How large is the welfare loss of the current design when storage bids are built from imperfect price forecasts or from technical data only, and which properties of the forecast error (size, persistence, bias, correlation between owners) drive it?

RQ4 *Use by the market operator (extension).* Can the operator use the same construction with its own price forecast to obtain a competitive reference bid for each ESS, and how well does such a reference separate competitive from strategic bids?

Hypotheses. The experimental part tests the following hypotheses; their status after the first experiments is given in Table 9.4.

H1 With the true prices and the benchmark quantities, the welfare loss is zero (follows from Theorem 6.3; a verification gate).

H2 The welfare loss grows with the forecast error and is concentrated in the intervals where SOC limits bind.

H3 Persistent forecast errors cost more than hour-to-hour noise of the same size.

H4 The pre-limit selection rule (Section 6.5) loses less welfare than an arbitrary selection of multipliers.

H5 Errors common to all ESS owners cost more than independent errors of the same size.

H6 Bids built from a price forecast outperform bids built from technical data only when the forecast error is below a threshold that grows with the cost of real-time adjustment.

1.5 Contributions

The contributions obtained so far are the following.

1. A complete formulation of both designs in a common market model with a DC network written in node pairs, generator ramps, elastic demand and several ESS, with the full KKT systems of both clearing problems (Chapter 3). A proof that the SOC-based design is a competitive equilibrium: written as a mixed complementarity problem, its KKT system is partitioned exactly by the optimality conditions of the individual participants, and the storage block is the owner's own self-schedule (Chapter 4).
2. An exact equivalence: interval bids built from the multipliers of the SOC-based design reproduce its dispatch and nodal prices. The current design is a Lagrangian relaxation of the SOC-based design with respect to the SOC constraints (Chapter 5).
3. An information result: the equivalent bids can be computed by each ESS from its own technical data and the nodal prices at its own node, through its price-taking self-schedule, without the multipliers of the SOC-based design and independently of other participants. The result holds for any number of ESS (Chapter 6).
4. A characterisation of all equivalent bids as a polyhedron, showing that prices pin the bids except at the intervals where an SOC limit is reached, and a selection rule for those intervals designed for robustness to price errors (Chapter 6).
5. Closed-form expressions for the equivalent bids in terms of the value of stored energy and of prices at the marginal intervals of the SOC path (Chapter 6).
6. A quantification of the additional computational burden of the SOC-based design (Chapter 7).
7. A verified implementation and an experimental framework to measure the welfare cost of imperfect price information, with first results on one ESS and on two storage owners (Chapters 8 to 10).

1.6 Overview of the approach

Figure 1.1 summarises the logic of the thesis. Both designs are linear programs, so their optimal points are characterised by their KKT conditions. These can also be read as a market equilibrium between price-taking participants (Chapter 4), in which the storage owner's block is its own self-schedule. Comparing the two KKT systems yields equivalent bids. Decomposing the KKT system of the SOC-based design shows that the storage part of it is the KKT system of each ESS's own self-schedule, which removes the need for the operator's multipliers. Finally, replacing the true prices by a forecast turns the equivalence into a bidding strategy, whose welfare cost is measured experimentally.

1.7 Structure of the report

Chapter 2 reviews market designs for storage, storage models in market clearing, storage bidding under uncertainty, market monitoring and price forecasting. Chapter 3 formulates both designs and derives their KKT conditions. Chapter 4 writes the SOC-based design as a mixed complementarity problem and proves that it is a competitive equilibrium. Chapter 5 establishes the equivalence through bids built from multipliers. Chapter 6 expresses the equivalent bids through technical data and prices and characterises the set of all equivalent bids. Chapter 7

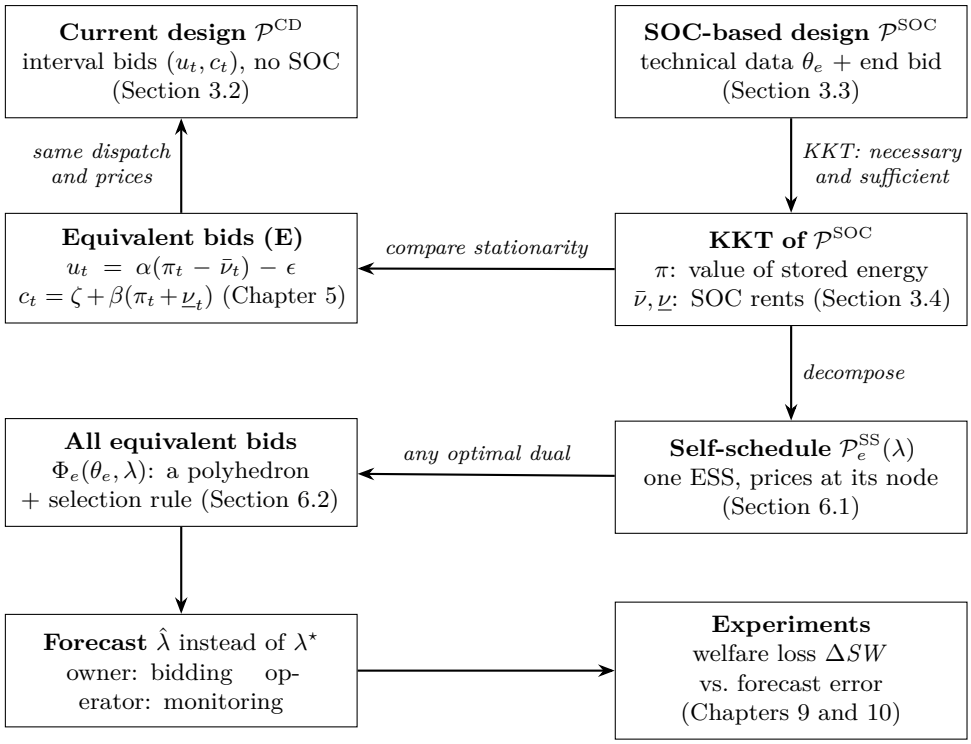


Figure 1.1: Logic of the thesis: from the two clearing problems to the equivalent bids, their computation from technical data and prices, and their use with forecasts.

quantifies the computational burden of the SOC-based design. Chapter 8 verifies all results numerically. Chapter 9 designs the experiments on imperfect price information, and Chapter 10 reports the first experiments. Chapter 11 discusses the results so far and the next steps. The appendices contain the data of the test cases, a description of the code, the correspondence with the technical note and a log of the work.

Chapter 2

State of the art

This chapter places the thesis in the literature and in current market practice. Section 2.1 recalls the principles of day-ahead clearing and marginal pricing on which both designs rest. Section 2.2 describes how markets currently treat storage. Section 2.3 reviews storage models in market clearing, including Baldick's SOC-based design. Section 2.4 reviews storage valuation and bidding under price uncertainty, Section 2.5 market power and monitoring of storage, and Section 2.6 electricity price forecasting. Section 2.7 identifies the gap addressed by the thesis.

Status. This chapter covers the literature used so far. It will be extended during the thesis, in particular on European market rules, on SOC-dependent bids and on the empirical properties of price forecast errors.

2.1 Day-ahead clearing, marginal pricing and equilibrium

Organised day-ahead markets clear by solving an optimisation problem: the market operator maximises declared social welfare, i.e. the value of served demand minus the cost of supply, as declared in bids and offers, subject to power balance, network and unit constraints. The nodal prices are the dual variables of the power balance constraints, the locational marginal prices (LMPs) of spot pricing theory [13, 14]. When the clearing problem is convex, and in particular when it is a linear program (LP), its optimality conditions are the Karush–Kuhn–Tucker (KKT) conditions, which are necessary and sufficient [15, 16].

The KKT conditions admit an economic reading that is central to this thesis. They are equivalent to a market equilibrium, written as a mixed complementarity problem (MCP), in which each participant maximises its own profit at the market prices and the prices clear the market [17, 18]. The welfare-maximising dispatch is therefore a competitive equilibrium, and the nodal prices support it: at those prices no price-taking participant wants to deviate. Two consequences are used repeatedly below. First, two clearing problems with the same KKT system have the same optimal dispatch and prices, whatever their objective functions look like. Second, the part of the KKT system that involves only one participant is the KKT system of that participant's own profit maximisation at the market prices. The second point is the basis of the decomposition in Chapter 6, and relates to the classical decomposition techniques of mathematical programming [19].

LP duals need not be unique. When the dual optimal set is not a point, several price vectors and several multiplier vectors support the same dispatch. This *dual degeneracy* is common in markets with stepped offers and is a recurring theme of the thesis: it determines when prices pin the equivalent bids and when a selection rule is needed (Section 6.2).

2.2 Storage participation in current markets

Table 2.1 summarises how the markets most relevant to this thesis treat the SOC.

Market	**SOC in day-ahead clearing**	**Storage submits**	**Remarks**
ERCOT	no	interval bids and offers	Since 5 December 2025 (RTC+B), storage is a single resource whose real-time dispatch models its SOC; multi-interval SOC optimisation is listed as a future initiative [20]. The market monitor has recommended a multi-interval real-time market that values the SOC over future intervals [2].
CAISO	yes	interval bids and offers	No end-of-horizon bid [8]. Storage bids may be mitigated to a default energy bid that includes an opportunity cost computed from the operator's prices [3, 12].
NYISO	yes	interval bids and offers	No end-of-horizon bid [8].
PJM (pumped hydro)	partly	technical data	SOC and round-trip efficiency modelled without interval bids, not fully integrated in the day-ahead model [8].
Europe (SDAC)	no	simple and block orders	Storage bids through standard orders [9]. A dedicated storage order, meant to let storage bid "with less detailed market forecast", was consulted in 2026 and is planned for the end of 2027 [4].
SOC-based design [8]	yes	technical data and one end-of-horizon bid	Proposal; the operator optimises the SOC.

Table 2.1: Treatment of the state of charge in selected markets (as of October 2026).

United States. FERC Order No. 841 [7] required market operators to accommodate storage, but left the representation of the SOC to each market. Baldick observes that many of the resulting participation models do not represent the SOC or round-trip efficiency and therefore rely on the owner's price forecasts [8, §3]. ERCOT's day-ahead market follows this approach. CAISO and NYISO enforce SOC dynamics and limits in their clearing problems, but still receive interval-by-interval bids and offers and no end-of-horizon valuation [8, §3]. In this intermediate design the operator guarantees feasibility of the SOC path, but the owner still has to price stored energy interval by interval.

Europe. In the single day-ahead coupling, storage participates through the standard order types of the coupling algorithm, which contain no SOC [9]. The European market coupling operators consulted market participants in 2026 on a new order type for assets that buy and sell within the same day, with go-live planned for the fourth quarter of 2027 [4]. Its stated aim, to allow storage to bid with a less detailed price forecast, is precisely the information problem studied in this thesis.

2.3 Storage models in market clearing

Storage in scheduling and unit commitment. Storage models in scheduling and unit commitment represent the SOC through an energy balance with charging and discharging efficiencies and SOC limits [21, 22]. Preventing simultaneous charging and discharging in general requires binary variables. In practice, round-trip losses make simultaneous operation unprofitable whenever stored energy has a non-negative value, so the binaries can often be relaxed and the condition checked after the solve; Section 7.3 gives a precise condition for the model of this thesis. Chen and Baldick [10] study battery formulations in day-ahead security-constrained unit commitment, strengthen the SOC constraints with valid inequalities and assess their effect on solve times on MISO cases. Sioshansi et al. [22] survey storage modelling across planning and operation and identify the representation of storage in market models as an open research area.

Baldick's SOC-based design. Baldick [8] proposes a complete market design in which the operator optimises the SOC. Its main features, all retained in this thesis, are:

- SOC dynamics with separate charging and discharging coefficients and a retention factor that models self-discharge;
- SOC limits written in a *tightened* form that accounts for the energy charged or discharged within an interval, so that the SOC never exceeds its limits during the interval;
- degradation costs proportional to charged and discharged energy;
- a single concave benefit for the deviation of the end-of-horizon SOC from the SOC the ESS would have if it stayed idle, which replaces all interval bids and values the energy carried into the next market horizon;
- relaxation of the charge–discharge exclusivity, justified by round-trip losses.

Baldick illustrates the design on a six-interval example with one ESS (base case and two variants), which is used throughout this report, and discusses large-scale implementation, real-time markets and prudential issues [8, §4.7–4.8, App. A]. In each example it compares the design with interval bids, and argues that interval bids require accurate price forecasts [8, §3, §4.7]. This argument is the starting point of RQ1–RQ3: the thesis characterises exactly which forecast is required and measures what an imperfect one costs.

SOC-dependent bids. An alternative line of work keeps owner bids but lets them depend on the SOC. Zheng et al. [23] propose that storage submit charge and discharge bids for each SOC segment, which the operator applies as the SOC evolves, and show that such bids reflect the opportunity cost of storage better than power-based bids. Chen and Tong [24] show that SOC-dependent bids make the multi-interval dispatch non-convex in general, and identify a condition under which the dispatch remains a linear program. These designs sit between the current design and the SOC-based design: the owner still prices stored energy, but the price may depend on the SOC. The equivalent bids of this thesis are interval-dependent rather than SOC-dependent; Section 6.5 shows that a single interval bid cannot express "this energy is worth one value if I am not yet empty and another if I am", which is exactly what an SOC-dependent bid can express.

2.4 Storage valuation and bidding under uncertainty

Value and welfare effects of storage. Early studies quantified the arbitrage value of storage and its welfare effects. Sioshansi et al. [5] estimate the arbitrage value of storage in PJM, and its effects on consumer and producer surplus. Sioshansi [25] shows that the welfare effects of storage depend on its ownership structure, because a profit-maximising owner with market power operates it differently from a welfare maximiser. Kazempour et al. [6] compare storage technologies in a market-based economy and illustrate how market prices determine their economic viability.

Price-taking and price-making bidding. When storage is a price-taker, its optimal schedule against a known price path is a small LP or a dynamic program over the SOC. The marginal value of stored energy at each interval, i.e. the dual of the SOC dynamics, is its opportunity cost, and bids follow from it. Xu et al. [11] derive marginal costs of battery cycle ageing and show how to include them in market bids. Zheng et al. [26] compute the value function of stored energy by analytical stochastic dynamic programming and derive bids from it under price uncertainty. Krishnamurthy et al. [27] schedule storage under day-ahead and real-time price uncertainty with stochastic programming. Large storage units affect prices; Mohsenian-Rad [28] models coordinated price-maker operation in nodal markets, and Anunrojwong et al. [29] analyse strategic battery behaviour and show that it leads to withholding and market distortions.

Relation to this thesis. These works take the market design as given and ask how an owner should bid. This thesis asks a different question: which bids make the current design reproduce the SOC-based design, and what they require. The price-taking self-schedule plays a central role in both, and Chapter 6 shows that its dual is exactly what an ESS needs to reproduce the SOC-based outcome.

2.5 Market power and monitoring of storage

As storage becomes price-setting, its bids become subject to market power mitigation. CAISO's market power mitigation can replace a storage bid by a default energy bid (DEB) computed by the operator. For storage, the DEB includes an estimate of the intra-day opportunity cost. In real time, for a four-hour battery, this is the fourth-highest day-ahead LMP; the DEB also includes the charging cost, a variable operating cost and a 10% adder [12]. For its extended day-ahead market, CAISO derives the price-based opportunity cost by scaling the outputs of the market power mitigation pass [3]. The DEB thus summarises the value of stored energy with a single price level computed from the operator's own price information, without following the SOC dynamics or the end-of-horizon value of the battery. CAISO's day-ahead market also contains a residual unit commitment that uses the operator's own load forecast rather than bid-in demand [30]. Market operators therefore already compute and use their own forecasts.

Academic work on storage market power includes bid bounds that depend on the SOC and on net-load uncertainty [31], and empirical evidence from CAISO bid data that storage bids are inflated during price spikes and are consistent with economic withholding [32]. RQ4 connects to this literature: the equivalent-bid map of Chapter 6, evaluated by the operator with its own price forecast, gives an SOC-consistent competitive reference bid for every interval.

2.6 Electricity price forecasting

Day-ahead price forecasting is a mature field [33]. Lago et al. [34] review the state of the art and provide an open benchmark with two reference models, a regularised linear model (LEAR) and a deep neural network (DNN), on five markets. Over two-year test periods, the best ensembles achieve a mean absolute error of about 1.7 EUR/MWh in Nord Pool (sMAPE about 5%), 2.8 $/MWh in PJM and 3.4–5.8 EUR/MWh in the EPEX markets (sMAPE about 11–14%). For Gaussian errors, the standard deviation is about 1.25 times the mean absolute error, so these values correspond to errors of roughly 5–15% of the mean price for state-of-the-art forecasts in calm market periods. Errors are larger in volatile periods. Price forecast errors are also heavy-tailed and heteroskedastic [33], and errors made for the same day share common drivers, such as errors in wind and load forecasts, which makes them correlated across hours. These properties inform the error model of Section 9.6 and its calibration in Section 9.7.

2.7 Research gap

Table 2.2 positions the thesis. The literature offers designs that bring the SOC into the clearing (SOC-based or SOC-dependent), methods for an owner to bid under price uncertainty, and heuristic references for monitoring storage bids. What is missing is an exact link between the two main designs: a characterisation of the interval bids that reproduce the SOC-based outcome, of the information they require, and of the welfare cost of not having that information. Without that link, the value of moving from interval bids to SOC-based clearing cannot be quantified on equal terms, and monitoring references for storage remain heuristic.

Work	**SOC in clearing**	**Equivalence of designs**	**Information in bids**	**Cost of forecast errors**
Baldick [8]	✓	–	qualitative	qualitative
Chen and Baldick [10]	✓	–	–	–
Zheng et al. [23], Chen and Tong [24]	SOC-dependent bids	–	–	partly
Xu et al. [11], Zheng et al. [26], Krishnamurthy et al. [27]	–	–	owner's bidding	owner's profit
CAISO DEB [3, 12], Qi and Xu [31]	–	–	reference bids	–
This thesis	both designs	✓	exact	social welfare

Table 2.2: Positioning of the thesis.

Chapter 3

Market models

This chapter formulates the two designs within a common market model and derives their optimality conditions. Section 3.1 describes the common model and the assumptions. Sections 3.2 and 3.3 formulate the current design and the SOC-based design. Section 3.4 derives the full KKT systems; Chapter 4 then shows that the SOC-based design is a market equilibrium. Section 3.5 introduces the reference example of Baldick and interprets the multipliers of the SOC-based design on it.

Constraint labels follow the technical note: (A1)–(A7) are common to both designs, (C1)–(C2) belong to the current design, (A8)–(A12) to the SOC-based design; (S·) are stationarity conditions and (CS·) complementary slackness conditions.

3.1 Common market model and assumptions

Both designs clear a multi-interval day-ahead energy market over T intervals of length ΔT. They share the representation of the rest of the market, so that any difference in outcome can be attributed to the treatment of storage:

- a nodal power balance at every bus and interval, whose multipliers are the nodal prices;
- a DC power flow written in node pairs: each line $(n, m) \in \Lambda$ is oriented from bus n to bus m, with flow proportional to the angle difference, thermal limits and a reference-bus angle;
- generators offering price–quantity blocks, with time-varying block sizes (which represent renewable availability) and inter-temporal ramp limits;
- inelastic demand and elastic demand bidding price–quantity blocks.

The objective is the minimisation of declared cost, i.e. the maximisation of declared social welfare.

Assumption 3.1 (Modelling assumptions)**.**

(i) The clearing problems are convex: there are no unit-commitment decisions and no binary variables. Both are linear programs when the end-of-horizon benefit b_e is linear or concave piecewise linear.

(ii) Charge–discharge exclusivity of the ESS is relaxed, as in [8, §4.7]; Section 7.3 gives a condition under which the relaxation is exact.

(iii) Ramp limits apply for $t \geq 2$; the initial output of each unit is free.

(iv) The problems are feasible and bounded, so that strong duality holds.

(v) All participants are price-takers. Strategic behaviour is outside the scope of the analytical part.

3.2 Current design

Each ESS participates as a paired generator and load with interval-by-interval stepped offers and bids [8, §3]. The SOC is not represented in the clearing. For each ESS, $g_{e,t} := \sum_k g_{e,k,t}$ and

$p_{e,t} := \sum_j p_{e,j,t}$. The clearing problem is

$$(\mathcal{P}^{\mathrm{CD}}):\ \min\ f^{\mathrm{CD}} := \sum_{t\in\mathcal{T}} \Delta T\Big[\sum_{i,k} c_{i,k}q_{i,k,t} - \sum_{l,j} u_{l,j}d_{l,j,t} + \sum_{e\in\mathcal{E}}\Big(\sum_{k\in\mathcal{K}^g_e} c^g_{e,k,t}\,g_{e,k,t} - \sum_{j\in\mathcal{J}^p_e} u^p_{e,j,t}\,p_{e,j,t}\Big)\Big] \tag{3.1}$$

subject to, for all buses n, lines $(n,m)\in\Lambda$, units and blocks, loads and blocks, ESS e and intervals t (multipliers in brackets):

$$\begin{aligned}
&\textstyle\sum_{l\in\mathcal{D}(n),j} d_{l,j,t} + D_{n,t} + \sum_{m:(n,m)\in\Lambda} f_{n,m,t} - \sum_{m:(m,n)\in\Lambda} f_{m,n,t} \\
&\qquad - \textstyle\sum_{i\in\mathcal{G}(n),k} q_{i,k,t} - \sum_{e\in\mathcal{E}(n)}(g_{e,t}-p_{e,t}) = 0 && (\Delta T\lambda_{n,t}) && \text{(A1)}\\
&f_{n,m,t} - B_{n,m}(\theta_{n,t}-\theta_{m,t}) = 0 && (\Delta T\kappa_{n,m,t}) && \text{(A2)}\\
&-\bar{f}_{n,m} \le f_{n,m,t} \le \bar{f}_{n,m} && (\Delta T\underline{\mu}_{n,m,t}, \Delta T\bar{\mu}_{n,m,t}) && \text{(A3)}\\
&\theta_{r,t} = 0 && (\Delta T\xi_t) && \text{(A4)}\\
&0 \le q_{i,k,t} \le \bar{q}_{i,k,t} && (\Delta T\underline{\delta}_{i,k,t}, \Delta T\bar{\delta}_{i,k,t}) && \text{(A5)}\\
&Q_{i,t} - Q_{i,t-1} \le \bar{R}_i, \quad Q_{i,t-1} - Q_{i,t} \le \underline{R}_i, \quad t\ge 2 && (\Delta T\bar{\tau}_{i,t}, \Delta T\underline{\tau}_{i,t}) && \text{(A6)}\\
&0 \le d_{l,j,t} \le \bar{d}_{l,j} && (\Delta T\underline{\psi}_{l,j,t}, \Delta T\bar{\psi}_{l,j,t}) && \text{(A7)}\\
&0 \le g_{e,k,t} \le \hat{g}_{e,k,t}, \qquad \textstyle\sum_k \hat{g}_{e,k,t} \le \bar{g}_e && (\Delta T\underline{\sigma}_{e,k,t}, \Delta T\bar{\sigma}_{e,k,t}) && \text{(C1)}\\
&0 \le p_{e,j,t} \le \hat{p}_{e,j,t}, \qquad \textstyle\sum_j \hat{p}_{e,j,t} \le \bar{p}_e && (\Delta T\underline{\rho}_{e,j,t}, \Delta T\bar{\rho}_{e,j,t}) && \text{(C2)}
\end{aligned}$$

The balance (A1) is written as *consumption minus production*, so that $\lambda_{n,t}$ is the price paid by consumers at bus n. The conditions $\sum_k \hat{g}_{e,k,t} \le \bar{g}_e$ and $\sum_j \hat{p}_{e,j,t} \le \bar{p}_e$ are restrictions on the bids, not constraints of the clearing problem.

What the CD does not contain. There is no SOC variable, no SOC dynamics and no link between intervals for the ESS. The clearing problem is separable by interval except for the ramps (A6). An ESS whose bids are not consistent with its SOC may therefore be scheduled to sell energy it does not have, or to buy energy it cannot store.

3.3 SOC-based design

In Baldick's design [8] the operator optimises the SOC. Each ESS submits its technical data and a terminal benefit b_e, and no interval bids:

$$(\mathcal{P}^{\mathrm{SOC}}):\ \min\ f^{\mathrm{SOC}} := \sum_{t\in\mathcal{T}} \Delta T\Big[\sum_{i,k} c_{i,k}q_{i,k,t} - \sum_{l,j} u_{l,j}d_{l,j,t} + \sum_{e\in\mathcal{E}}\big(\epsilon_e p_{e,t} + \zeta_e g_{e,t}\big)\Big] - \sum_{e\in\mathcal{E}} b_e\big(s_{e,T} - \bar{\bar{s}}_{e,T}\big) \tag{3.2}$$

subject to (A1)–(A7) and, for all e and t (equation numbers of [8] in square brackets):

$$\begin{aligned}
&s_{e,t} - \gamma_e s_{e,t-1} - \Delta T\alpha_e p_{e,t} + \Delta T\beta_e g_{e,t} = 0 && (\pi_{e,t}) \quad [(4)] && \text{(A8)}\\
&\gamma_e s_{e,t-1} + \Delta T\alpha_e p_{e,t} - \bar{s}_e \le 0 && (\bar{\nu}_{e,t}) \quad [(6)] && \text{(A9)}\\
&\underline{s}_e - \gamma_e s_{e,t-1} + \Delta T\beta_e g_{e,t} \le 0 && (\underline{\nu}_{e,t}) \quad [(7)] && \text{(A10)}\\
&0 \le p_{e,t} \le \bar{p}_e && (\Delta T\underline{\rho}_{e,t}, \Delta T\bar{\rho}_{e,t}) \quad [(1)] && \text{(A11)}\\
&0 \le g_{e,t} \le \bar{g}_e && (\Delta T\underline{\sigma}_{e,t}, \Delta T\bar{\sigma}_{e,t}) \quad [(2)] && \text{(A12)}
\end{aligned}$$

with $s_{e,0}$ given and $s_{e,t}$ free.

Reading the constraints. (A8) is the SOC dynamics: a fraction $1-\gamma_e$ of the stored energy is lost in each interval, each MWh drawn from the grid adds α_e MWh to the store, and each MWh delivered to the grid removes β_e MWh from it. Round-trip efficiency is $\alpha_e/\beta_e < 1$. (A9)–(A10) are the tightened SOC limits of [8]: the upper limit is checked as if the whole interval's charging happened before any discharging, and the lower limit as if the whole interval's discharging happened first. The SOC therefore stays within $[\underline{s}_e, \bar{s}_e]$ throughout the interval, not only at its end. Bounds on $s_{e,t}$ itself are implied. The term $\bar{\bar{s}}_{e,T} = \gamma_e^T s_{e,0}$ is the SOC at T if the ESS stayed idle, so $b_e(s_{e,T} - \bar{\bar{s}}_{e,T})$ values only the energy that the ESS adds to (or removes from) its store through market participation. With a linear benefit $b_e(\Delta) = w_e\Delta$, w_e is the price the owner is willing to pay for a MWh stored at the end of the horizon.

3.4 Lagrangian and KKT conditions

3.4.1 Lagrangian of the SOC-based design

Writing $[\cdot]$ for the left-hand side of a constraint in the form $(\cdot) = 0$ or $(\cdot) \le 0$,

$$\begin{aligned}\mathcal{L}^{\mathrm{SOC}} = \; & f^{\mathrm{SOC}} + \sum_{n,t} \Delta T \lambda_{n,t}\,[\mathrm{A1}]_{n,t} + \sum_{(n,m)\in\Lambda,t} \Delta T \kappa_{n,m,t}\,[\mathrm{A2}]_{n,m,t} + \sum_{t} \Delta T \xi_t\,\theta_{r,t} \\ & + \sum_{(n,m)\in\Lambda,t} \Delta T\big[\bar{\mu}_{n,m,t}(f_{n,m,t} - \bar{f}_{n,m}) - \underline{\mu}_{n,m,t}(f_{n,m,t} + \bar{f}_{n,m})\big] \\ & + \sum_{i,k,t} \Delta T\big[\bar{\delta}_{i,k,t}(q_{i,k,t} - \bar{q}_{i,k,t}) - \underline{\delta}_{i,k,t} q_{i,k,t}\big] \\ & + \sum_{i,t\ge 2} \Delta T\big[\bar{\tau}_{i,t}(Q_{i,t} - Q_{i,t-1} - \bar{R}_i) + \underline{\tau}_{i,t}(Q_{i,t-1} - Q_{i,t} - \underline{R}_i)\big] \\ & + \sum_{l,j,t} \Delta T\big[\bar{\psi}_{l,j,t}(d_{l,j,t} - \bar{d}_{l,j}) - \underline{\psi}_{l,j,t} d_{l,j,t}\big] \\ & + \sum_{e,t} \big\{\pi_{e,t}[\mathrm{A8}]_{e,t} + \bar{\nu}_{e,t}[\mathrm{A9}]_{e,t} + \underline{\nu}_{e,t}[\mathrm{A10}]_{e,t}\big\} \\ & + \sum_{e,t} \Delta T\big[\bar{\rho}_{e,t}(p_{e,t} - \bar{p}_e) - \underline{\rho}_{e,t} p_{e,t} + \bar{\sigma}_{e,t}(g_{e,t} - \bar{g}_e) - \underline{\sigma}_{e,t} g_{e,t}\big]. \end{aligned} \tag{3.3}$$

The Lagrangian of $\mathcal{P}^{\mathrm{CD}}$ is the same without the rows of $\pi, \bar{\nu}, \underline{\nu}$, with the ESS cost terms replaced by the bid terms, and with the ESS power bounds written per block.

3.4.2 KKT conditions common to both designs

Stationarity, rest of the market:

$$\partial q_{i,k,t}: \quad c_{i,k} - \lambda_{n(i),t} + \bar{\delta}_{i,k,t} - \underline{\delta}_{i,k,t} + \varrho_{i,t} = 0 \tag{S1}$$

$$\partial d_{l,j,t}: \quad -u_{l,j} + \lambda_{n(l),t} + \bar{\psi}_{l,j,t} - \underline{\psi}_{l,j,t} = 0 \tag{S2}$$

$$\partial f_{n,m,t}: \quad \lambda_{n,t} - \lambda_{m,t} + \kappa_{n,m,t} + \bar{\mu}_{n,m,t} - \underline{\mu}_{n,m,t} = 0 \tag{S3}$$

$$\partial \theta_{n,t}: \quad -\textstyle\sum_{m:(n,m)\in\Lambda} B_{n,m}\kappa_{n,m,t} + \sum_{m:(m,n)\in\Lambda} B_{m,n}\kappa_{m,n,t} + \xi_t\,\mathbf{1}_{\{n=r\}} = 0 \tag{S4}$$

with the *ramp wedge*

$$\varrho_{i,t} := \big(\bar{\tau}_{i,t} - \underline{\tau}_{i,t}\big) - \big(\bar{\tau}_{i,t+1} - \underline{\tau}_{i,t+1}\big), \qquad \bar{\tau}_{i,1} = \underline{\tau}_{i,1} = \bar{\tau}_{i,T+1} = \underline{\tau}_{i,T+1} := 0. \tag{3.4}$$

(S3) states that the flow on line (n,m) is valued at the price difference between its ends, corrected by the multipliers of the flow definition and of the thermal limits. (S4) is the stationarity of the

angle at bus n: it collects the multipliers κ of all lines leaving n (with sign $-B_{n,m}$) and entering n (with sign $+B_{m,n}$).

Complementary slackness, rest of the market:

$$\begin{aligned}
&0 \le \underline{\delta}_{i,k,t} \ \perp \ q_{i,k,t} \ge 0, \quad 0 \le \bar{\delta}_{i,k,t} \ \perp \ \bar{q}_{i,k,t} - q_{i,k,t} \ge 0, \\
&0 \le \underline{\psi}_{l,j,t} \ \perp \ d_{l,j,t} \ge 0, \quad 0 \le \bar{\psi}_{l,j,t} \ \perp \ \bar{d}_{l,j} - d_{l,j,t} \ge 0 && \text{(CS1)} \\
&0 \le \underline{\mu}_{n,m,t} \ \perp \ \bar{f}_{n,m} + f_{n,m,t} \ge 0, \quad 0 \le \bar{\mu}_{n,m,t} \ \perp \ \bar{f}_{n,m} - f_{n,m,t} \ge 0 && \text{(CS2)} \\
&0 \le \bar{\tau}_{i,t} \ \perp \ \bar{R}_i - Q_{i,t} + Q_{i,t-1} \ge 0, \quad 0 \le \underline{\tau}_{i,t} \ \perp \ \underline{R}_i - Q_{i,t-1} + Q_{i,t} \ge 0, \quad t \ge 2 && \text{(CS3)}
\end{aligned}$$

Primal feasibility: (A1)–(A7); λ, κ, ξ free.

3.4.3 KKT conditions of the current design

Stationarity, ESS:

$$\begin{aligned}
\partial g_{e,k,t}: \quad & c^g_{e,k,t} - \lambda_{n(e),t} + \bar{\sigma}_{e,k,t} - \underline{\sigma}_{e,k,t} = 0 && \text{(S5}^{\text{CD}}\text{)} \\
\partial p_{e,j,t}: \quad & -u^p_{e,j,t} + \lambda_{n(e),t} + \bar{\rho}_{e,j,t} - \underline{\rho}_{e,j,t} = 0 && \text{(S6}^{\text{CD}}\text{)}
\end{aligned}$$

Complementary slackness, ESS:

$$\begin{aligned}
&0 \le \underline{\sigma}_{e,k,t} \ \perp \ g_{e,k,t} \ge 0, \quad 0 \le \bar{\sigma}_{e,k,t} \ \perp \ \hat{g}_{e,k,t} - g_{e,k,t} \ge 0, \\
&0 \le \underline{\rho}_{e,j,t} \ \perp \ p_{e,j,t} \ge 0, \quad 0 \le \bar{\rho}_{e,j,t} \ \perp \ \hat{p}_{e,j,t} - p_{e,j,t} \ge 0
\end{aligned} \qquad \text{(CS4}^{\text{CD}}\text{)}$$

Primal feasibility: (A1)–(A7), (C1)–(C2). In *box form*, the ESS rows read

$$0 \le g_{e,k,t} \le \hat{g}_{e,k,t} \ \perp \ c^g_{e,k,t} - \lambda_{n(e),t}, \qquad 0 \le p_{e,j,t} \le \hat{p}_{e,j,t} \ \perp \ \lambda_{n(e),t} - u^p_{e,j,t}, \tag{3.5}$$

i.e. an offer block is fully dispatched if its price is below the nodal price, empty if above, and free if equal; symmetrically for bid blocks.

3.4.4 KKT conditions of the SOC-based design

Stationarity, rest of the market: (S1)–(S4) with (3.4). *Stationarity*, ESS:

$$\begin{aligned}
\partial p_{e,t}: \quad & \epsilon_e + \lambda_{n(e),t} - \alpha_e(\pi_{e,t} - \bar{\nu}_{e,t}) + \bar{\rho}_{e,t} - \underline{\rho}_{e,t} = 0 && \text{(S5)} \\
\partial g_{e,t}: \quad & \zeta_e - \lambda_{n(e),t} + \beta_e(\pi_{e,t} + \underline{\nu}_{e,t}) + \bar{\sigma}_{e,t} - \underline{\sigma}_{e,t} = 0 && \text{(S6)} \\
\partial s_{e,t},\ t < T: \quad & \pi_{e,t} - \gamma_e(\pi_{e,t+1} - \bar{\nu}_{e,t+1} + \underline{\nu}_{e,t+1}) = 0 && \text{(S7)} \\
\partial s_{e,T}: \quad & \pi_{e,T} \in \partial b_e(s_{e,T} - \bar{\bar{s}}_{e,T}) \quad (= w_e(s_{e,T} - \bar{\bar{s}}_{e,T}) \text{ if differentiable}) && \text{(S8)}
\end{aligned}$$

Complementary slackness: (CS1)–(CS3) and

$$\begin{aligned}
&0 \le \underline{\rho}_{e,t} \ \perp \ p_{e,t} \ge 0, \quad 0 \le \bar{\rho}_{e,t} \ \perp \ \bar{p}_e - p_{e,t} \ge 0, \\
&0 \le \underline{\sigma}_{e,t} \ \perp \ g_{e,t} \ge 0, \quad 0 \le \bar{\sigma}_{e,t} \ \perp \ \bar{g}_e - g_{e,t} \ge 0 && \text{(CS4)} \\
&0 \le \bar{\nu}_{e,t} \ \perp \ \bar{s}_e - \gamma_e s_{e,t-1} - \Delta T \alpha_e p_{e,t} \ge 0, \\
&0 \le \underline{\nu}_{e,t} \ \perp \ \gamma_e s_{e,t-1} - \Delta T \beta_e g_{e,t} - \underline{s}_e \ge 0 && \text{(CS5)}
\end{aligned}$$

Primal feasibility: (A1)–(A12); $\lambda, \kappa, \xi, \pi$ free. In box form,

$$0 \le p_{e,t} \le \bar{p}_e \ \perp \ \lambda_{n(e),t} + \epsilon_e - \alpha_e(\pi_{e,t} - \bar{\nu}_{e,t}), \qquad 0 \le g_{e,t} \le \bar{g}_e \ \perp \ \zeta_e + \beta_e(\pi_{e,t} + \underline{\nu}_{e,t}) - \lambda_{n(e),t}. \tag{3.6}$$

Unrolled costate. Iterating (S7) backwards from (S8),

$$\pi_{e,t} = \gamma_e^{T-t}\pi_{e,T} + \sum_{\tau=t+1}^{T} \gamma_e^{\tau-t}(\underline{\nu}_{e,\tau} - \bar{\nu}_{e,\tau}). \tag{3.7}$$

The value of stored energy at the end of t is the terminal value, discounted by retention, plus the SOC rents of all later intervals.

3.5 Reference example and meaning of the multipliers

Baldick's base case [8, §4.7.1] is used throughout the report. It has one bus, $T = 6$ intervals of $\Delta T = 4\,\text{h}$, a renewable unit offering at $0\,\$/\text{MWh}$, a thermal unit of $100\,\text{MW}$ offering at $50\,\$/\text{MWh}$, inelastic demand, and one ESS with $\alpha = 0.9$, $\beta = 1$, $\gamma = 1$, $\epsilon = \zeta = 1\,\$/\text{MWh}$, $\underline{s} = 0$, $\bar{s} = 800\,\text{MWh}$, $s_0 = 0$, $\bar{p} = \bar{g} = 1000\,\text{MW}$ and a linear terminal benefit with $w = 40\,\$/\text{MWh}$. The data are listed in Table 5.2 (Chapter 5) and Appendix A. Two variants change one parameter each: $w = 55\,\$/\text{MWh}$ [8, §4.7.2] and $\bar{s} = 1200\,\text{MWh}$ [8, §4.7.3].

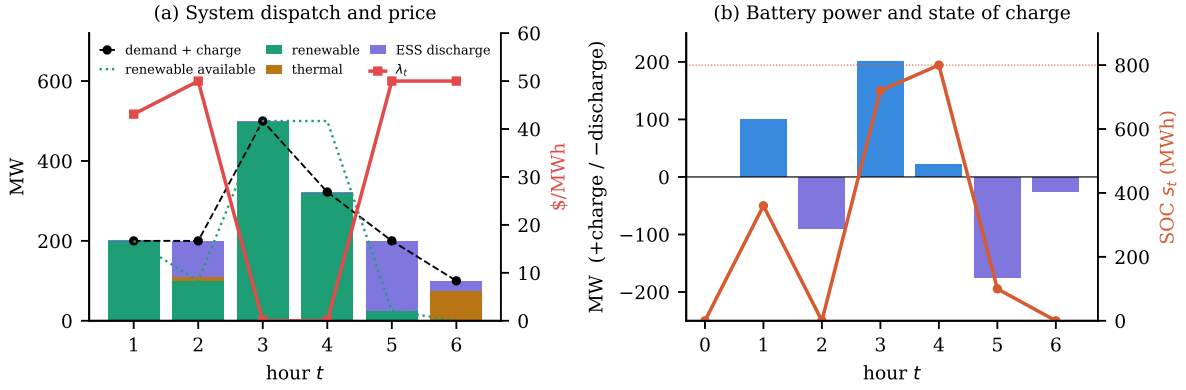


Figure 3.1: Optimal solution of the SOC-based design in Baldick's base case. (a) Dispatch and nodal prices. (b) ESS charging and discharging and SOC. At $t = 4$, renewable energy is curtailed because the ESS is full.

Figure 3.1 shows the optimal solution. The ESS charges when renewable energy is in surplus ($t = 1, 3, 4$) and discharges when the alternative is thermal generation ($t = 2, 5, 6$). The nodal prices are $\lambda^\star = (43.1, 50, 0, 0, 50, 50)\,\$/\text{MWh}$ and the SOC path is $s^\star = (360, 0, 720, 800, 100, 0)\,\text{MWh}$: the ESS is empty at $t = 2$, full at $t = 4$, and ends empty because its terminal value $w = 40$ is below the price at which it can still sell.

Meaning of $\pi_{e,t}$. By the envelope theorem, $\pi_{e,t}$ is the marginal value, in \$/MWh, of one extra MWh of stored energy at the end of interval t, i.e. the decrease of the optimal cost (increase of welfare) when (A8) at t becomes $s_{e,t} = \gamma_e s_{e,t-1} + \Delta T(\alpha_e p_{e,t} - \beta_e g_{e,t}) + \Delta$. It is the *value of stored energy*. In the base case it equals $(43.1 + 1)/0.9 = 49$ at $t = 1$: one more stored MWh at the end of interval 1 saves buying $1/0.9\,\text{MWh}$ at $43.1\,\$/\text{MWh}$ plus degradation.

Meaning of $\bar{\nu}_{e,t}, \underline{\nu}_{e,t}$. They are SOC scarcity rents: the value of relaxing the upper (lower) SOC limit by one MWh. By (S7), the value of stored energy changes between two consecutive intervals only through retention and through these rents. They are positive only when an SOC limit binds.

Non-uniqueness. The value of stored energy is not always unique. Re-solving the base case with $\Delta = \pm 0.5\,\text{MWh}$ at each interval gives the right and left derivatives of the optimal value, which bound the admissible π_t (Table 3.1).

t	1	2	3	4	5	6
right derivative ($\Delta > 0$)	49	1.11	1.11	1.11	49	40
left derivative ($\Delta < 0$)	49	49	1.11	49	49	40
admissible π_t	49	[1.11, 49]	1.11	[1.11, 49]	49	40

Table 3.1: Admissible values of the value of stored energy in the base case ($/MWh).

At $t = 2$ the ESS is empty. One extra MWh only saves later charging at a price of 0, so it is worth the charging degradation cost, $\pi_2^+ = \epsilon/\alpha = 1.11$. One missing MWh cannot be replaced at $t = 3$, because the renewable surplus is fully used there, so the ESS sells one MWh less at $t = 5$: $\pi_2^- = (\lambda_5 - \zeta)/\beta = 49$. At $t = 4$ the ESS is full and the argument is symmetric. Any value between the two derivatives is a valid multiplier; the solver returns one of them. Section 6.2 shows that this non-uniqueness occurs exactly at the intervals where an SOC limit binds and cannot be undone, and that it determines where a selection rule is needed.

Chapter 4

The SOC-based design as a market equilibrium

The SOC-based design is stated by Baldick as a single optimisation problem solved by the market operator: maximise declared welfare subject to all constraints [8, (9)]. Optimality of the operator's dispatch does not by itself say that the participants are content with it. This chapter shows that they are: the dispatch and prices of the SOC-based design form a *competitive equilibrium*, in which every participant, taking the nodal prices as given, maximises its own profit, and the prices clear the market. The argument writes the market as a mixed complementarity problem (MCP) [17].

The chapter has a second purpose, which is the reason it comes before the equivalence results. The storage block of the equilibrium is the optimisation problem of the ESS owner itself, its price-taking self-schedule. This is what later allows the multipliers of the operator's problem to be replaced by the multipliers of the owner's own problem (Lemma 6.2 and Theorem 6.3).

The route is

$$\mathcal{P}^{\mathrm{SOC}} \;\longrightarrow\; \mathrm{KKT}(\mathcal{P}^{\mathrm{SOC}}) \;\xrightarrow[\text{by owner}]{\text{split}}\; \{\text{participant problems (B1)–(B4)}\}$$
$$\longrightarrow\; \textstyle\bigcup_a \mathrm{KKT}(a) + \text{clearing} = \mathrm{MCP} \;\overset{?}{=}\; \mathrm{KKT}(\mathcal{P}^{\mathrm{SOC}}), \quad (4.1)$$

where $\mathcal{P}^{\mathrm{SOC}}$ is stated in Section 3.3, its KKT system in Section 3.4, the participant problems in Section 4.1 and the MCP in Section 4.2. The identity marked "?" is Theorem 4.4; Theorem 4.7 turns it into an equilibrium statement.

Recall: what this chapter uses.
Operator's problem $\mathcal{P}^{\mathrm{SOC}}$ (Section 3.3): minimise declared cost subject to the balance (A1) [$\Delta T\lambda$], network (A2)–(A4), offer and demand blocks (A5), (A7), ramps (A6), and the ESS rows (A8)–(A12).
Its KKT system (Section 3.4): stationarity (S1)–(S8), complementary slackness (CS1)–(CS5), primal feasibility (A1)–(A12).
Notation. $\lambda_{n,t}$: nodal price; $\pi_{e,t}$: value of stored energy; $\bar{\nu}, \underline{\nu}$: SOC rents; b_e: end-of-horizon benefit, $w_e = b'_e$; $0 \le a \perp b \ge 0$: complementarity.

4.1 Participants and their problems

Fix the nodal prices $\lambda = (\lambda_{n,t})$. Each participant solves its own problem, subject only to its own constraints, taking λ as given. No participant problem mentions social welfare.

Generator i. Sells its output at the price of its bus:

$$\max_{q_i} \quad \Delta T \sum_{t\in\mathcal{T}} \sum_{k\in\mathcal{K}_i} (\lambda_{n(i),t} - c_{i,k}) q_{i,k,t} \qquad \text{s.t. (A5), (A6) for } i. \tag{B1}$$

Load l. Buys elastic demand at the price of its bus:

$$\max_{d_l} \quad \Delta T \sum_{t\in\mathcal{T}} \sum_{j\in\mathcal{J}_l} (u_{l,j} - \lambda_{n(l),t}) d_{l,j,t} \qquad \text{s.t. (A7) for } l. \tag{B2}$$

Storage owner e**.** Sells discharged energy and buys charging energy at the price of its bus, pays degradation, and values the energy left at the end of the horizon with its own benefit function:

$$\max_{p_e,g_e,s_e} \ \Delta T \sum_{t\in\mathcal{T}} \Big[\lambda_{n(e),t}(g_{e,t}-p_{e,t})-\epsilon_e p_{e,t}-\zeta_e g_{e,t}\Big]+b_e\big(s_{e,T}-\bar{\bar{s}}_{e,T}\big) \qquad \text{s.t. (A8)–(A12) for } e. \tag{B3}$$

Network. A fictitious transmission agent buys energy at the sending end of each line and sells it at the receiving end, and collects the price difference:

$$\max_{f,\theta} \ \Delta T \sum_{t\in\mathcal{T}} \sum_{(n,m)\in\Lambda} (\lambda_{m,t}-\lambda_{n,t}) f_{n,m,t} \qquad \text{s.t. (A2)–(A4).} \tag{B4}$$

(B4) is a modelling device: no merchant transmission firm exists. It represents the operator enforcing Kirchhoff's laws, written as an agent so that the network part of the market also decentralises.

Market clearing. The prices must make the participants' independent choices consistent: the nodal balance (A1) must hold at every bus and interval. This is the only condition that belongs to no participant.

Definition 4.1 (Price-taking self-schedule)**.** Problem (B3) written as a minimisation, for an arbitrary price path $\lambda=(\lambda_1,\dots,\lambda_T)$, is the *price-taking self-schedule* of ESS e:

$$(\mathcal{P}_e^{\mathrm{SS}}(\lambda)):\ \min_{p,g,s} \sum_{t\in\mathcal{T}} \Delta T\big[(\epsilon_e+\lambda_t)p_{e,t}+(\zeta_e-\lambda_t)g_{e,t}\big]-b_e\big(s_{e,T}-\bar{\bar{s}}_{e,T}\big) \quad \text{s.t. (A8)–(A12) for } e. \tag{4.2}$$

With the multipliers and scaling of $\mathcal{P}^{\mathrm{SOC}}$, its KKT system is (S5)–(S8), (CS4)–(CS5) and (A8)–(A12) for e, with $\lambda_{n(e),t}$ replaced by λ_t. We denote by $\mathcal{D}_e(\lambda)$ its set of optimal multipliers $m=(\pi,\bar{\nu},\underline{\nu},\bar{\rho},\underline{\rho},\bar{\sigma},\underline{\sigma})$.

The participants' optimality conditions. Each of (B1)–(B4) has its own Lagrangian and KKT system. Dualising its constraints with the same multipliers and scaling as in $\mathcal{P}^{\mathrm{SOC}}$:

- (B1) gives (S1) with the ramp wedge (3.4), the first line of (CS1), (CS3), (A5) and (A6);
- (B2) gives (S2), the second line of (CS1), and (A7);
- (B3) gives (S5)–(S8), (CS4)–(CS5) and (A8)–(A12);
- (B4) gives (S3), (S4), (CS2) and (A2)–(A4).

For example, the Lagrangian of (B4) written as a minimisation is

$$\Delta T \sum_{t,(n,m)} (\lambda_{n,t}-\lambda_{m,t}) f_{n,m,t} + \sum_{t,(n,m)} \Delta T \kappa_{n,m,t}\big(f_{n,m,t}-B_{n,m}(\theta_{n,t}-\theta_{m,t})\big) + \sum_t \Delta T \xi_t \theta_{r,t} + \sum_{t,(n,m)} \Delta T\big[\bar{\mu}_{n,m,t}(f_{n,m,t}-\bar{f}_{n,m})-\underline{\mu}_{n,m,t}(f_{n,m,t}+\bar{f}_{n,m})\big],$$

whose derivative in $f_{n,m,t}$ is (S3) and whose derivative in $\theta_{n,t}$ is (S4). The other three are analogous. The conditions are identical to the corresponding conditions of $\mathcal{P}^{\mathrm{SOC}}$, *verbatim*, not merely up to a transformation: this is what the scaling convention of Section 3.4 was chosen for.

4.2 The complete complementarity problem

Each participant's KKT system can be written compactly with one complementarity row per variable, using the following elementary fact.

Lemma 4.2 (Box form)**.** *For a scalar variable z with bounds $l \le u$ (possibly infinite) and a function F, the following are equivalent:*

(i) there exist $\underline{\mu}, \bar{\mu} \ge 0$ with $F(z) - \underline{\mu} + \bar{\mu} = 0$, $\underline{\mu}\,(z - l) = 0$, $\bar{\mu}\,(u - z) = 0$ and $l \le z \le u$;

(ii) $l \le z \le u \perp F(z)$, i.e. $z = l \Rightarrow F \ge 0$, $l < z < u \Rightarrow F = 0$, $z = u \Rightarrow F \le 0$.

A free variable ($l = -\infty$, $u = +\infty$) gives $F(z) = 0$.

Proof. (i)⇒(ii): if $l < z < u$, both multipliers vanish and $F = 0$; if $z = l < u$, then $\bar{\mu} = 0$ and $F = \underline{\mu} \ge 0$; symmetrically at u. (ii)⇒(i): set $\underline{\mu} = [F]^+$ and $\bar{\mu} = [-F]^+$. □

Applying Lemma 4.2 to each bounded variable removes the multipliers of its bounds, and the participants' conditions, together with market clearing, form the following system.

The MCP of the SOC-based design.
Generator i, from (B1):

$$0 \le q_{i,k,t} \le \bar{q}_{i,k,t} \;\perp\; c_{i,k} - \lambda_{n(i),t} + \varrho_{i,t} \tag{M1}$$

$$0 \le \bar{\tau}_{i,t} \perp \bar{R}_i - Q_{i,t} + Q_{i,t-1} \ge 0, \qquad 0 \le \underline{\tau}_{i,t} \perp \underline{R}_i - Q_{i,t-1} + Q_{i,t} \ge 0, \quad t \ge 2 \tag{M2}$$

Load l, from (B2):

$$0 \le d_{l,j,t} \le \bar{d}_{l,j} \;\perp\; \lambda_{n(l),t} - u_{l,j} \tag{M3}$$

Storage owner e, from (B3):

$$0 \le p_{e,t} \le \bar{p}_e \;\perp\; \epsilon_e + \lambda_{n(e),t} - \alpha_e\big(\pi_{e,t} - \bar{\nu}_{e,t}\big) \tag{M4}$$

$$0 \le g_{e,t} \le \bar{g}_e \;\perp\; \zeta_e - \lambda_{n(e),t} + \beta_e\big(\pi_{e,t} + \underline{\nu}_{e,t}\big) \tag{M5}$$

$$s_{e,t} \text{ free} \;\perp\; \pi_{e,t} - \gamma_e\big(\pi_{e,t+1} - \bar{\nu}_{e,t+1} + \underline{\nu}_{e,t+1}\big) = 0, \quad t < T; \qquad \pi_{e,T} \in \partial b_e\big(s_{e,T} - \bar{\bar{s}}_{e,T}\big) \tag{M6}$$

$$0 \le \bar{\nu}_{e,t} \perp \bar{s}_e - \gamma_e s_{e,t-1} - \Delta T \alpha_e p_{e,t} \ge 0, \qquad 0 \le \underline{\nu}_{e,t} \perp \gamma_e s_{e,t-1} - \Delta T \beta_e g_{e,t} - \underline{s}_e \ge 0 \tag{M7}$$

$$\pi_{e,t} \text{ free} \;\perp\; s_{e,t} - \gamma_e s_{e,t-1} - \Delta T \alpha_e p_{e,t} + \Delta T \beta_e g_{e,t} = 0 \tag{M8}$$

Network, from (B4):

$$-\bar{f}_{n,m} \le f_{n,m,t} \le \bar{f}_{n,m} \;\perp\; \lambda_{n,t} - \lambda_{m,t} + \kappa_{n,m,t} \tag{M9}$$

$$\theta_{n,t} \text{ free} \perp \text{(S4)}; \qquad \kappa_{n,m,t} \text{ free} \perp f_{n,m,t} - B_{n,m}(\theta_{n,t} - \theta_{m,t}) = 0; \qquad \xi_t \text{ free} \perp \theta_{r,t} = 0 \tag{M10}$$

Market clearing (no participant):

$$\begin{aligned} \lambda_{n,t} \text{ free} \;\perp\; & \textstyle\sum_{l \in \mathcal{D}(n),j} d_{l,j,t} + D_{n,t} + \sum_{m:(n,m)\in\Lambda} f_{n,m,t} - \sum_{m:(m,n)\in\Lambda} f_{m,n,t} \\ & - \textstyle\sum_{i \in \mathcal{G}(n),k} q_{i,k,t} - \sum_{e \in \mathcal{E}(n)} (g_{e,t} - p_{e,t}) = 0 \end{aligned} \tag{M11}$$

Reading the system. Three properties are visible directly.

- *No objective.* Nothing in (M1)–(M11) is maximised. Social welfare is not a primitive of the MCP; it can be computed from a solution, but no row refers to it.
- *The price is an unknown.* λ appears in the rows of all four participants as a given, and in (M11) as the variable that this row determines. This is what makes price-taking expressible.

- *Coupling is through λ alone.* Delete the λ column and the system falls apart into $|\mathcal{G}| + |\mathcal{D}| + |\mathcal{E}| + 1$ independent blocks, one per participant. Separability by owner is visible as a sparsity pattern.

The system is square: each row is paired with exactly one unknown (the variable written on its left), so there are as many rows as unknowns.

4.3 Correspondence with the operator's optimality conditions

Table 4.1 sorts the KKT conditions of $\mathcal{P}^{\mathrm{SOC}}$ by the variable in which each is stationary, and assigns each to the MCP row and the participant it ends up in.

KKT of $\mathcal{P}^{\mathrm{SOC}}$	**stationary in**	**MCP row**	**owner (problem)**
(S1) + (CS1), first line + (A5)	$q_{i,k,t}$	(M1)	generator i (B1)
(CS3) + (A6)	–	(M2)	generator i (B1)
(S2) + (CS1), second line + (A7)	$d_{l,j,t}$	(M3)	load l (B2)
(S5) + (CS4) (ρ) + (A11)	$p_{e,t}$	(M4)	ESS e (B3)
(S6) + (CS4) (σ) + (A12)	$g_{e,t}$	(M5)	ESS e (B3)
(S7), (S8)	$s_{e,t}$	(M6)	ESS e (B3)
(CS5) + (A9), (A10)	–	(M7)	ESS e (B3)
(A8)	–	(M8)	ESS e (B3)
(S3) + (CS2) + (A3)	$f_{n,m,t}$	(M9)	network (B4)
(S4), (A2), (A4)	$\theta_{n,t}$	(M10)	network (B4)
(A1)	–	(M11)	**nobody**: market clearing

Table 4.1: Each KKT condition of the operator's problem belongs to exactly one participant, except the market-clearing condition. Dual feasibility (signs of the multipliers) travels with the complementarity condition of its constraint.

Assumption 4.3 (Separability by owner)**.** *Every primal variable of $\mathcal{P}^{\mathrm{SOC}}$ and every constraint other than* (A1) *belongs to exactly one participant: generator, load, ESS or network.*

Assumption 4.3 holds for the model of Chapter 3: ramps belong to the generator, SOC rows to the ESS, and flows and angles to the network. It is not stated by Baldick, whose compact form collects everything that is not storage into one polyhedron [8, (9)]; that form cannot be split by owner, which is why Chapter 3 writes every constraint out.

Theorem 4.4 (Partition of the KKT system)**.** *Under Assumption 4.3, a point z (primal variables, prices and multipliers) satisfies the KKT system of $\mathcal{P}^{\mathrm{SOC}}$ if and only if it solves the MCP* (M1)–(M11)*, with the bound multipliers recovered as in Lemma 4.2. Moreover, the correspondence is a* partition*: each KKT condition is used exactly once and belongs to exactly one participant, and the only condition belonging to no participant is* (A1)*.*

Proof. ($\Rightarrow$) Sort the KKT conditions of $\mathcal{P}^{\mathrm{SOC}}$ by the variable each is stationary in, as in Table 4.1. Every stationarity condition involves exactly one primal variable, and every primal variable belongs to exactly one owner by Assumption 4.3, so the sort is well defined and the classes are disjoint. Within each class, Lemma 4.2 converts the stationarity condition and the complementarity conditions of the bounds into one box row; the remaining complementary-slackness and feasibility conditions of that owner are carried over unchanged. This produces

(M1)–(M10). The only condition left over is (A1), which is stationary in no variable and private to no owner: it is (M11).

($\Leftarrow$) Each step above is an identity between the same expressions, so it reverses term by term; the eliminated bound multipliers are recovered by Lemma 4.2(ii)$\Rightarrow$(i).

Exhaustiveness and disjointness follow from the structure of the Lagrangian of $\mathcal{P}^{\mathrm{SOC}}$: once (A1) is dualised, the rest is a sum of one block per owner plus constants, so its gradient has one block per owner and no cross terms. □

4.4 From correspondence to equilibrium

Theorem 4.4 is a statement about equations. To speak of an equilibrium, each participant must actually be *maximising* at the prices, not merely stationary. That requires assumptions.

Definition 4.5 (Competitive equilibrium)**.** A dispatch $x^* = (q^*, d^*, p^*, g^*, s^*, f^*, \theta^*)$ and nodal prices λ^* form a *competitive equilibrium* if (i) at λ^*, each participant's part of x^* is optimal for its own problem (B1)–(B4), and (ii) the market clears, i.e. (A1) holds.

Assumption 4.6 (Equilibrium assumptions)**.**

(E1) Price-taking. *No participant anticipates its effect on λ.*

(E2) Convexity of the market. *No binary decisions (no unit commitment); all constraints are affine.*

(E3) Separability *(Assumption 4.3).*

(E4) Bid validity. *Each end-of-horizon willingness to pay w_e is non-increasing, so that $b_e(\Delta) = \int_0^\Delta w_e$ is concave.*

Theorem 4.7 (The SOC-based design is a competitive equilibrium)**.** *Under Assumption 4.6, let x^* be optimal for $\mathcal{P}^{\mathrm{SOC}}$ with multipliers including λ^*. Then (x^*, λ^*) is a competitive equilibrium. Conversely, every competitive equilibrium in which the participants' multipliers are those of their own problems yields an optimal solution of $\mathcal{P}^{\mathrm{SOC}}$ with prices λ^*.*

Proof. Step 1 (each participant's problem is convex). (B1), (B2) and (B4) have linear objectives over polyhedra. (B3) has a linear part plus $b_e(s_{e,T} - \bar{\bar{s}}_{e,T})$, which is concave by (E4), over the polyhedron (A8)–(A12). Hence the KKT conditions of each participant are *sufficient* for its global optimality. *Step 2 (the operator's KKT conditions hold).* By (E2), $\mathcal{P}^{\mathrm{SOC}}$ is convex with affine constraints, so a constraint qualification holds automatically and an optimal x^* has multipliers satisfying the KKT system. *Step 3 (correspondence).* By (E3) and Theorem 4.4, these multipliers solve the MCP. *Step 4 (equilibrium).* Each participant's block of the MCP is its own KKT system at λ^*; by Step 1 its part of x^* is optimal for its own problem. Row (M11) is market clearing. This is Definition 4.5. *Converse.* If each participant is optimal at λ^*, then (by Step 1 and the affine constraints) its KKT conditions hold with some multipliers; together with (A1) they solve the MCP, hence by Theorem 4.4 the KKT system of $\mathcal{P}^{\mathrm{SOC}}$, which is sufficient for optimality since $\mathcal{P}^{\mathrm{SOC}}$ is convex. □

Why (E4) matters. Steps 3 and 4 rest on different assumptions: separability gives the partition, convexity gives the economics, and neither implies the other. Without (E4), the conditions of the storage block would only say that a derivative vanishes: the owner would be *stationary*, not necessarily *maximising*, and an allocation from which a participant wants to deviate is not an equilibrium, however many first-order conditions it satisfies. (E4) is not always

natural. If the owner earns a discrete benefit from ending above an SOC threshold, for example a capacity or reserve obligation with a minimum energy, the marginal value of the MWh that crosses the threshold carries the whole benefit, so w_e increases there and b_e is not concave. Such an owner cannot declare its true valuation in the format of the SOC-based design at all; this is the same non-convexity as a start-up cost. For the linear benefit used throughout this report, $b_e(\Delta) = w_e\Delta$, (E4) holds trivially.

What is not claimed. The MCP is an equilibrium *of the model*. It is an equilibrium of the actual market only if the participant problems are the participants' real problems: participants are price-takers (E1); the declared w_e is the owner's true valuation, otherwise this is an equilibrium of the reported game, and truthfulness is a separate question; and the whole horizon is financially binding, since if only part of it settles, (B3) is not the owner's actual problem.

Corollary 4.8 (Individual rationality and revenue adequacy)**.** *Under Assumption 4.6, at the equilibrium of Theorem 4.7:*

(i) every generator and load earns a nonnegative surplus;

(ii) every ESS whose idle trajectory $s_{e,t} = \gamma_e^t s_{e,0}$ respects its SOC limits earns at least $b_e(0) = 0$, i.e. participating is never worse than staying idle;

(iii) the operator's merchandising surplus, $\Delta T \sum_t \sum_n \lambda^*_{n,t}(\text{consumption} - \text{production})_{n,t}$, equals the network agent's profit $\Delta T \sum_{t,(n,m)}(\lambda^*_{m,t} - \lambda^*_{n,t}) f^*_{n,m,t}$, and is nonnegative.

Proof. In each case a zero action is feasible for the participant's problem: $q = 0$ for (B1) (ramps allow a constant output), $d = 0$ for (B2), the idle trajectory for (B3), and $f = 0$, $\theta = 0$ for (B4). Its value is zero, so the optimum is at least zero. For (iii), multiplying (A1) by $\lambda^*_{n,t}$ and summing over buses gives $\sum_n \lambda^*_{n,t}(\text{consumption} - \text{production})_{n,t} = \sum_{(n,m)}(\lambda^*_{m,t} - \lambda^*_{n,t}) f^*_{n,m,t}$, since each flow is counted as consumption at its sending bus and production at its receiving bus. □

Existence. An equivalence between two empty sets would say nothing. For a linear or concave piecewise-linear b_e, $\mathcal{P}^{\text{SOC}}$ is an LP; if it is feasible and bounded, an optimal solution with multipliers exists, and by Theorem 4.4 so does a solution of the MCP. Existence can also be argued on the MCP directly, without reference to $\mathcal{P}^{\text{SOC}}$: its mapping is the sum of a skew-symmetric linear part and the (monotone) gradient of the convex terms, and is monotone exactly when every b_e is concave [17].

4.5 Operator's problem and MCP: same equations, different objects

By Theorem 4.4 the two systems coincide as sets of equations. As mathematical objects they still differ in five respects.

1. *Objective.* $\mathcal{P}^{\text{SOC}}$ has an objective, declared welfare; the MCP has none.
2. *Status of the price.* In $\mathcal{P}^{\text{SOC}}$, λ is an output (a multiplier); in the MCP it is an unknown, determined by (M11).
3. *Ownership of multipliers.* In $\mathcal{P}^{\text{SOC}}$ all multipliers belong to the operator; in the MCP each belongs to the participant whose constraint it prices. The value of stored energy π_e and the rents $\bar{\nu}_e, \underline{\nu}_e$ belong to the ESS.

4. *Coupling.* In $\mathcal{P}^{\mathrm{SOC}}$ the participants are coupled through a shared feasible set; in the MCP only through λ.
5. *Where convexity is needed.* $\mathcal{P}^{\mathrm{SOC}}$ needs it once, for its own KKT conditions to be sufficient; the MCP needs it once per participant, for each participant to be maximising.

4.6 Consequence: the owner's problem gives the multipliers

Point 3 above is the property used in the rest of the report.

Corollary 4.9 (The storage block is the owner's own problem)**.** *Let* (x^*, λ^*) *and the multipliers of* $\mathcal{P}^{\mathrm{SOC}}$ *solve the MCP. Then:*

(i) the ESS rows (M4)–(M8) *of each* e *are exactly the KKT system of* $\mathcal{P}_e^{\mathrm{SS}}(\lambda^*_{n(e)})$*, so the operator's storage multipliers* $(\pi^*_e, \bar{\nu}^*_e, \underline{\nu}^*_e)$ *are optimal multipliers of the owner's own problem,* $\in \mathcal{D}_e(\lambda^*_{n(e)})$*;*

(ii) no row outside the block of e *contains* π_e*,* $\bar{\nu}_e$ *or* $\underline{\nu}_e$*. Replacing them by* any other *element of* $\mathcal{D}_e(\lambda^*_{n(e)})$ *therefore leaves a solution of the MCP, and hence (Theorem 4.4) of the KKT system of* $\mathcal{P}^{\mathrm{SOC}}$*.*

Proof. (i) is the storage row of Table 4.1 together with Definition 4.1. (ii): by inspection of (M1)–(M11), the storage multipliers appear only in (M4)–(M8). Another element of $\mathcal{D}_e(\lambda^*_{n(e)})$ satisfies these rows together with the owner's optimal schedule (p^*_e, g^*_e, s^*_e) by the standard property of LP duality that any optimal primal and any optimal dual satisfy the KKT conditions together (fact (F1) of Section 6.1); all other rows are unchanged. □

This is the bridge to the main result of the thesis. The equivalent bids (E) of Chapter 5 are written with the operator's multipliers. Corollary 4.9 says that these multipliers are the owner's own, and that the owner may use *any* optimal multipliers of its own problem instead. Lemma 6.2 and Theorem 6.3 in Chapter 6 make this precise for the bids, and show that the owner needs only its technical data and the prices at its bus.

The current design as an MCP. The same construction applies to $\mathcal{P}^{\mathrm{CD}}$. The generator, load, network and clearing rows are identical. The ESS block is different: by (3.5), each bid and offer block of each interval contributes one row of the form $0 \le g_{e,k,t} \le \hat{g}_{e,k,t} \perp c^g_{e,k,t} - \lambda_{n(e),t}$ or $0 \le p_{e,j,t} \le \hat{p}_{e,j,t} \perp \lambda_{n(e),t} - u^p_{e,j,t}$, and there are no rows in s, π or ν. In the equilibrium of the current design the ESS is therefore a collection of independent per-interval sellers and buyers that react only to the price of their own interval; all intertemporal information must be carried by the bid prices and quantities. Comparing (M4)–(M5) with these rows is exactly how the equivalent bids (E) arise: setting $c^g_{e,t} = \zeta_e + \beta_e(\pi_{e,t} + \underline{\nu}_{e,t})$ and $u^p_{e,t} = \alpha_e(\pi_{e,t} - \bar{\nu}_{e,t}) - \epsilon_e$ turns the storage rows of one MCP into those of the other (Chapter 5).

Chapter 5

Equivalence between the two designs

This chapter answers RQ1. It compares the KKT systems of the two designs, constructs interval bids from the multipliers of the SOC-based design, and shows that they reproduce its dispatch and nodal prices (Section 5.2). It then interprets the current design as a Lagrangian relaxation of the SOC-based design (Section 5.3), gives necessary and sufficient conditions on general stepped bids (Section 5.4), and identifies the information embedded in the equivalent bids (Section 5.5).

Recall: what this chapter uses.

Stationarity of the ESS (Section 3.4), with $\lambda := \lambda_{n(e),t}$:

- CD: $\lambda = u^p_{e,j,t} - \bar{\rho}_{e,j,t} + \underline{\rho}_{e,j,t}$ (S6^{CD}); $\lambda = c^g_{e,k,t} + \bar{\sigma}_{e,k,t} - \underline{\sigma}_{e,k,t}$ (S5^{CD})
- SOC: $\lambda = \alpha_e(\pi_{e,t} - \bar{\nu}_{e,t}) - \epsilon_e - \bar{\rho}_{e,t} + \underline{\rho}_{e,t}$ (S5); $\lambda = \zeta_e + \beta_e(\pi_{e,t} + \underline{\nu}_{e,t}) + \bar{\sigma}_{e,t} - \underline{\sigma}_{e,t}$ (S6)
- Costate: $\pi_{e,t} = \gamma_e(\pi_{e,t+1} - \bar{\nu}_{e,t+1} + \underline{\nu}_{e,t+1})$ (S7); $\pi_{e,T} = w_e$ for linear b_e (S8)

Notation. π: value of stored energy; $\bar{\nu}, \underline{\nu}$: SOC rents (upper, lower); $\bar{\rho}, \underline{\rho}$ and $\bar{\sigma}, \underline{\sigma}$: multipliers of the charging and discharging power bounds; u^p, c^g: bid and offer prices; $\hat{p}, \hat{g}$: block sizes; $(\cdot)^\star$: optimum of $\mathcal{P}^{\mathrm{SOC}}$.

5.1 The two KKT systems side by side

Table 5.1 lists the KKT conditions of both designs. The rows of the rest of the market are identical. The designs differ only in the stationarity of the ESS power variables, in what bounds the ESS power (bid quantities or physical ratings), and in the SOC rows, which exist only in the SOC-based design.

Condition	**Current design**	**SOC-based design**
(S1)–(S4), (CS1)–(CS3), (A1)–(A7)	identical	identical
ESS charging stationarity	$\lambda_{n(e),t} = u^p_{e,j,t} - \bar{\rho}_{e,j,t} + \underline{\rho}_{e,j,t}$	$\lambda_{n(e),t} = \big[\alpha_e(\pi_{e,t} - \bar{\nu}_{e,t}) - \epsilon_e\big] - \bar{\rho}_{e,t} + \underline{\rho}_{e,t}$
ESS discharging stationarity	$\lambda_{n(e),t} = c^g_{e,k,t} + \bar{\sigma}_{e,k,t} - \underline{\sigma}_{e,k,t}$	$\lambda_{n(e),t} = \big[\zeta_e + \beta_e(\pi_{e,t} + \underline{\nu}_{e,t})\big] + \bar{\sigma}_{e,t} - \underline{\sigma}_{e,t}$
ESS power bounds	$\hat{p}_{e,j,t}$, $\hat{g}_{e,k,t}$ (bid)	$\bar{p}_e$, $\bar{g}_e$ (physical)
SOC rows (A8)–(A10), (S7)–(S8), (CS5)	absent	present

Table 5.1: KKT conditions of the two designs.

The bracketed terms in the right column play exactly the role of the bid and offer prices in the left column. This suggests the construction of the next section.

5.2 Equivalent bids

Let each ESS submit one block per interval ($|\mathcal{K}_e^g| = |\mathcal{J}_e^p| = 1$, block index dropped) with

$$\boxed{u_{e,t}^{p\star} := \alpha_e(\pi_{e,t}^\star - \bar{\nu}_{e,t}^\star) - \epsilon_e, \quad \hat{p}_{e,t} = \bar{p}_e; \qquad c_{e,t}^{g\star} := \zeta_e + \beta_e(\pi_{e,t}^\star + \underline{\nu}_{e,t}^\star), \quad \hat{g}_{e,t} = \bar{g}_e} \tag{E}$$

Economic reading. Charging one MWh from the grid stores α_e MWh, worth $\alpha_e\pi_{e,t}^\star$, minus the rent $\alpha_e\bar{\nu}_{e,t}^\star$ that measures the scarcity of space in the store, minus the degradation cost. The bid is the highest price at which the ESS still wants to charge. Discharging one MWh to the grid uses β_e stored MWh, worth $\beta_e\pi_{e,t}^\star$, plus the rent $\beta_e\underline{\nu}_{e,t}^\star$ that measures the scarcity of energy in the store, plus degradation. The offer is the lowest price at which the ESS still wants to discharge.

Proposition 5.1 (KKT equivalence)**.** *Let $z^\star$ be primal–dual optimal for $\mathcal{P}^{\mathrm{SOC}}$. Under the bids* (E)*, the point $(q^\star, d^\star, f^\star, \theta^\star, p^\star, g^\star)$ with multipliers $(\lambda^\star, \kappa^\star, \xi^\star, \mu^\star, \delta^\star, \tau^\star, \psi^\star, \rho^\star, \sigma^\star)$ satisfies* (S1)–(S4)*,* (S5^{CD})–(S6^{CD})*,* (CS1)–(CS3)*,* (CS4^{CD})*,* (A1)–(A7) *and* (C1)–(C2)*. Hence it is optimal for $\mathcal{P}^{\mathrm{CD}}$, with the same nodal prices $\lambda^\star$.*

Proof. Substituting (E) into (S5)–(S6) gives (S6^{CD})–(S5^{CD}) with the same $\rho^\star, \sigma^\star$. Since $\hat{p} = \bar{p}$ and $\hat{g} = \bar{g}$, (CS4) coincides with (CS4^{CD}), and (C1)–(C2) hold because $p^\star, g^\star$ satisfy (A11)–(A12). All other rows are common to both systems. $\mathcal{P}^{\mathrm{CD}}$ is an LP, so the KKT conditions are sufficient for optimality. □

The proof uses nothing but the comparison of Table 5.1. It holds for any network, any number of ESS, any ramps and any elastic demand, because these enter only through rows that are common to both designs.

5.3 The current design as a Lagrangian relaxation

The equivalence has a second, global reading. If the SOC constraints of $\mathcal{P}^{\mathrm{SOC}}$ are moved to the objective with their optimal multipliers, the problem separates by interval and becomes the current design.

Proposition 5.2 (Lagrangian relaxation)**.** *Dualise* (A8)–(A10) *in $\mathcal{P}^{\mathrm{SOC}}$ at $(\pi^\star, \bar{\nu}^\star, \underline{\nu}^\star)$. Under* (E)*,*

$$\min_s \Big\{ f^{\mathrm{SOC}} + \sum_{e,t} \big(\pi_{e,t}^\star [A8]_{e,t} + \bar{\nu}_{e,t}^\star [A9]_{e,t} + \underline{\nu}_{e,t}^\star [A10]_{e,t}\big) \Big\} = f^{\mathrm{CD}} + K, \qquad K := \sum_{e\in\mathcal{E}} K_e, \tag{5.1}$$

where

$$K_e := -\pi_{e,0}^\star s_{e,0} - \bar{s}_e \sum_t \bar{\nu}_{e,t}^\star + \underline{s}_e \sum_t \underline{\nu}_{e,t}^\star + \min_s \big\{ \pi_{e,T}^\star s - b_e(s - \bar{\bar{s}}_{e,T}) \big\}, \qquad \pi_{e,0}^\star := \gamma_e(\pi_{e,1}^\star - \bar{\nu}_{e,1}^\star + \underline{\nu}_{e,1}^\star). \tag{5.2}$$

Consequently, (i) $f^{\mathrm{SOC}\star} = f^{\mathrm{CD}\star} + K$, and (ii) a CD-optimal point is SOC-optimal if and only if some s satisfies (A8)–(A10) *with complementary slackness* (CS5) *with respect to $\bar{\nu}^\star, \underline{\nu}^\star$.*

Proof. In the relaxed objective, the coefficient of $s_{e,t}$ for $t < T$ is the left-hand side of (S7), which vanishes at $(\pi^\star, \bar{\nu}^\star, \underline{\nu}^\star)$, so these variables drop out. The coefficients of $p_{e,t}$ and $g_{e,t}$ become $\Delta T(\epsilon_e - \alpha_e(\pi_{e,t}^\star - \bar{\nu}_{e,t}^\star)) = -\Delta T u_{e,t}^{p\star}$ and $\Delta T(\zeta_e + \beta_e(\pi_{e,t}^\star + \underline{\nu}_{e,t}^\star)) = \Delta T c_{e,t}^{g\star}$. The remaining terms are constants and the terms in $s_{e,T}$, which together give K. (i) follows from strong duality; (ii) from the optimality conditions of Lagrangian relaxation [19]. □

The constant K explains why the two optimal values differ even though the dispatch and prices coincide: the current design does not see the value of the initial stored energy, the rents collected at the SOC limits, or the terminal benefit. Part (ii) is a warning: the relaxation may have optimal points that the SOC-based design does not, and these may violate the SOC constraints (Remark 5.6).

5.4 Conditions on general stepped bids

Proposition 5.1 uses one block per interval at full capacity. The next result characterises all stepped bids under which the dispatch of the SOC-based design is optimal in the current design at the same prices.

Proposition 5.3 (Necessary and sufficient conditions on stepped bids)**.** *For arbitrary CD bid and offer blocks, $z^\star$ restricted to the CD variables is CD primal–dual optimal if and only if, for every e and t, with $\lambda^\star := \lambda^\star_{n(e),t}$,*

$$\sum_{j:\, u^p_{e,j,t} > \lambda^\star} \hat{p}_{e,j,t} \;\le\; p^\star_{e,t} \;\le\; \sum_{j:\, u^p_{e,j,t} \ge \lambda^\star} \hat{p}_{e,j,t}, \qquad \sum_{k:\, c^g_{e,k,t} < \lambda^\star} \hat{g}_{e,k,t} \;\le\; g^\star_{e,t} \;\le\; \sum_{k:\, c^g_{e,k,t} \le \lambda^\star} \hat{g}_{e,k,t}. \tag{5.3}$$

Proof. Given $\lambda^\star$, the box form (3.5) decouples by block. Bid blocks with $u^p > \lambda^\star$ and offer blocks with $c^g < \lambda^\star$ must be full; bid blocks with $u^p < \lambda^\star$ and offer blocks with $c^g > \lambda^\star$ must be empty; blocks priced exactly at $\lambda^\star$ may take any value within their size. Summing over blocks gives (5.3). The rows of the rest of the market hold at $z^\star$. □

In words, the schedule of the SOC-based design must lie on each ESS's submitted bid and offer curves, evaluated at the price of the SOC-based design. This is the familiar condition that a supply curve passes through the equilibrium point, applied to each ESS and interval.

5.5 Information embedded in the equivalent bids

Substituting the unrolled costate (3.7) into (E) shows what the bids contain.

Corollary 5.4 (Information embedded in the equivalent bids)**.**

$$c^{g\star}_{e,t} = \zeta_e + \beta_e\Big[\gamma_e^{T-t}\pi^\star_{e,T} + \sum_{\tau=t+1}^{T} \gamma_e^{\tau-t}\big(\underline{\nu}^\star_{e,\tau} - \bar{\nu}^\star_{e,\tau}\big) + \underline{\nu}^\star_{e,t}\Big], \tag{5.4}$$

$$u^{p\star}_{e,t} = \alpha_e\Big[\gamma_e^{T-t}\pi^\star_{e,T} + \sum_{\tau=t+1}^{T} \gamma_e^{\tau-t}\big(\underline{\nu}^\star_{e,\tau} - \bar{\nu}^\star_{e,\tau}\big) - \bar{\nu}^\star_{e,t}\Big] - \epsilon_e. \tag{5.5}$$

The terms fall into two groups.

- Private to ESS e: *$\alpha_e, \beta_e, \gamma_e, \epsilon_e, \zeta_e$ and w_e, i.e. exactly the data that e submits in the SOC-based design.*
- Endogenous: *the SOC rents $\bar{\nu}^\star_{e,\tau}, \underline{\nu}^\star_{e,\tau}$ of e for $\tau \ge t$, and the final SOC $s^\star_{e,T}$ through $\pi^\star_{e,T} = w_e(s^\star_{e,T} - \bar{\bar{s}}_{e,T})$ when b_e is nonlinear.*

The endogenous terms depend on all offers and bids, on the network and on the ramps. Chapter 6 shows that they depend on all of these *only through the nodal price path at the bus of e*.

Corollary 5.5 (Equivalence with private information only)**.** *If $b_e(\Delta) = w_e\Delta$ and no SOC limit of ESS e binds in $\{t, \dots, T\}$, then*

$$c^{g\star}_{e,t} = \zeta_e + \beta_e \gamma_e^{T-t} w_e, \qquad u^{p\star}_{e,t} = \alpha_e \gamma_e^{T-t} w_e - \epsilon_e. \tag{5.6}$$

Otherwise the equivalent bids require the rents $\bar{\nu}^\star_e, \underline{\nu}^\star_e$, i.e. information on the intra-horizon prices [8, §3].

Corollary 5.5 gives the *technical-data-only* bids used as the lower bound of the experiments (Section 9.3). It also identifies when information matters: an ESS that never reaches an SOC limit needs no price forecast at all.

5.6 Remarks

Remark 5.6 (Uniqueness and quantities). By Proposition 5.2(ii), the CD-optimal set contains the SOC-optimal set but may be larger, and may contain SOC-infeasible dispatches. This can happen only in intervals in which an ESS block is priced exactly at $\lambda^\star_{n(e),t}$, i.e. in which the ESS is price-setting. Choosing blocks so that the bounds of Proposition 5.3 are tight ($\hat{p}_{e,t} = p^\star_{e,t}$ at $u^p_{e,t} > \lambda^\star$, $\hat{g}_{e,t} = g^\star_{e,t}$ at $c^g_{e,t} < \lambda^\star$) pins the schedule, but may change $\lambda^\star$ where the ESS is price-setting. Exact reproduction of both schedule and prices therefore requires price *and* quantity information. The experiments address this with a quantity rule (Section 9.5).

Remark 5.7 (Charge–discharge exclusivity). Under (E), $c^{g\star}_{e,t} - u^{p\star}_{e,t} = \epsilon_e + \zeta_e + (\beta_e - \alpha_e)\pi^\star_{e,t} + \beta_e \underline{\nu}^\star_{e,t} + \alpha_e \bar{\nu}^\star_{e,t}$. If this is positive, the box form (3.5) gives $p_{e,t} > 0 \Rightarrow \lambda_{n(e),t} \le u^{p\star}_{e,t}$ and $g_{e,t} > 0 \Rightarrow \lambda_{n(e),t} \ge c^{g\star}_{e,t}$, which cannot hold together. Hence $p_{e,t} g_{e,t} = 0$ in the current design.

Remark 5.8 (Current design with SOC enforcement). If the current design also includes (A8)–(A10) but keeps interval bids and no terminal bid, as in CAISO and NYISO [8, §3], then for a linear b_e and single blocks priced at (5.6) the objective equals f^{SOC} identically. Indeed, $s_{e,T} - \bar{\bar{s}}_{e,T} = \Delta T \sum_t \gamma_e^{T-t}(\alpha_e p_{e,t} - \beta_e g_{e,t})$, so $\sum_t \Delta T(c^g_{e,t} g_{e,t} - u^p_{e,t} p_{e,t}) = \sum_t \Delta T(\epsilon_e p_{e,t} + \zeta_e g_{e,t}) - w_e(s_{e,T} - \bar{\bar{s}}_{e,T})$. The two problems then coincide, and the rents are computed by the market rather than by the owner: *with SOC enforcement, technical-data-only bids are exact.* For a nonlinear b_e, replacing w_e by $\pi^\star_{e,T}$ gives KKT equivalence only.

5.7 Worked example: Baldick's base case

Table 5.2 collects, for the base case, the data, the solution of the SOC-based design and its multipliers, the price-based derivation of π (Section 6.4), and the resulting bids. It should be read from top to bottom: rows (a) are inputs, rows (b)–(c) are outputs of $\mathcal{P}^{\mathrm{SOC}}$, rows (d) rebuild π from prices only, rows (e) build the bids, and row (f) checks the clearing of the current design.

Reading the table. Row (f) shows that every $\lambda^\star_t$ lies on an edge of the bid–offer band: the ESS is price-setting in every interval, so the cleared quantity is determined by the rest of the market, which is why quantities may need a tie-breaking rule (Remark 5.6). The band $c^g_t - u^p_t$ is narrow where π does not jump ($t = 1, 3, 5$: 6.9, 2.11 and 6.9 \$/MWh) and wide where an SOC limit binds ($t = 2, 4, 6$: 50, 50 and 15 \$/MWh). Chapter 6 explains this pattern and shows how to compute rows (d) and (e) without row (c).

	$t=1$	$t=2$	$t=3$	$t=4$	$t=5$	$t=6$
(a) System data						
demand D_t (MW)	100	200	300	300	200	100
renewable available (MW)	200	100	500	500	25	0
(b) Solution of the SOC-based design						
renewable dispatched (MW)	200	100	500	322.2	25	0
thermal dispatched (MW)	0	10	0	0	0	75
ESS charge $p_t^\star$ (MW)	100	0	200	22.2	0	0
ESS discharge $g_t^\star$ (MW)	0	90	0	0	175	25
SOC $s_t^\star$ at end of t (MWh)	360	0	720	800	100	0
nodal price $\lambda_t^\star$ (\$/MWh)	43.1	50	0	0	50	50
(c) Multipliers of the SOC-based design (solver) (\$/MWh)						
$\underline{\nu}_t^\star$ (lower limit)	0	47.89	0	0	0	9
$\bar{\nu}_t^\star$ (upper limit)	0	0	0	47.89	0	0
rent $r_t^\star = \underline{\nu}_t^\star - \bar{\nu}_t^\star$	0	47.89	0	−47.89	0	9
$\pi_t^\star$ (value at end of t)	49	1.11	1.11	49	49	40
$\pi_{t-1}^\star/\gamma$ (value at start of t)	49†	49	1.11	1.11	49	49
(d) Price-based derivation of π (Proposition 6.13)						
binding SOC limit in t	–	empty	–	full	–	empty
segment	$\mathcal{S}_1$	$\mathcal{S}_2$	$\mathcal{S}_2$	$\mathcal{S}_3$	$\mathcal{S}_3$	$\mathcal{S}_4$ (last)
marginal interval?	(C)	no	(C)	no	(D)	no
price-based value	$\frac{43.1+1}{0.9}$	–	$\frac{0+1}{0.9}$	–	$\frac{50-1}{1}$	w
value of the segment	49	1.11	1.11	49	49	40
(e) Equivalent bids (Corollaries 6.8 and 6.14)						
$\max\{\pi_t^\star, \pi_{t-1}^\star/\gamma\}$	49	49	1.11	49	49	49
$\min\{\pi_t^\star, \pi_{t-1}^\star/\gamma\}$	49	1.11	1.11	1.11	49	40
offer $c_t^g = 1 + 1 \cdot \max$ (\$/MWh)	**50**	**50**	**2.11**	**50**	**50**	**50**
bid $u_t^p = 0.9 \cdot \min -1$ (\$/MWh)	**43.1**	**0**	**0**	**0**	**43.1**	**35**
block sizes	$\hat{p}_t = \bar{p} = 1000\,\text{MW}$, $\hat{g}_t = \bar{g} = 1000\,\text{MW}$					
(f) Clearing of the current design at $\lambda^\star$						
position of $\lambda_t^\star$ in the band	$= u_t^p$	$= c_t^g$	$= u_t^p$	$= u_t^p$	$= c_t^g$	$= c_t^g$
ESS action	charge	discharge	charge	charge	discharge	discharge

Table 5.2: Baldick's base case step by step. $^\dagger\,\pi_0/\gamma := \pi_1 + \underline{\nu}_1 - \bar{\nu}_1 = 49$ (convention of Proposition 5.2). (C)/(D): charge- or discharge-marginal interval (Definition 6.9). The bids in (e) coincide with those obtained directly from (E) with the multipliers in (c).

Chapter 6

Equivalent bids from technical data and prices

The equivalent bids of Chapter 5 are built from the multipliers of the SOC-based design, which exist only inside that design. This chapter answers RQ2: it shows that every ESS can compute equivalent bids from two inputs only, its own technical data and the nodal prices at its own bus. Section 6.1 proves this through a decomposition of the KKT system. Section 6.2 characterises the set of all equivalent bids. Sections 6.3 and 6.4 give closed-form expressions that show which prices determine the bids. Section 6.5 introduces a selection rule for the intervals where the bids are not unique. Section 6.6 extends the results to several ESS, and Section 6.7 discusses who can compute the bids.

Recall: what this chapter uses.
Equivalent bids (E): $u_{e,t} = \alpha_e(\pi_{e,t} - \bar{\nu}_{e,t}) - \epsilon_e, \quad c_{e,t} = \zeta_e + \beta_e(\pi_{e,t} + \underline{\nu}_{e,t}), \quad$ blocks $\bar{p}_e, \bar{g}_e$.
SOC rows: (A8) $s_t = \gamma s_{t-1} + \Delta T(\alpha p_t - \beta g_t)$ $[\pi_t]$;
(A9) $\gamma s_{t-1} + \Delta T \alpha p_t \le \bar{s}$ $[\bar{\nu}_t]$; (A10) $\gamma s_{t-1} - \Delta T \beta g_t \ge \underline{s}$ $[\underline{\nu}_t]$.
Costate: (S7) $\pi_{t-1} = \gamma(\pi_t - \bar{\nu}_t + \underline{\nu}_t)$; (S8) $\pi_T = w$ (linear b).
Self-schedule (Definition 4.1): $\mathcal{P}_e^{\mathrm{SS}}(\lambda)$ maximises the profit of e at prices λ subject to (A8)–(A12); $\mathcal{D}_e(\lambda)$ is its set of optimal multipliers.
Notation. θ_e: technical data; $\lambda^\star_{n(e)}$: prices of the SOC-based design at the bus of e; $[x]^+ = \max\{x, 0\}$.

6.1 Bids from technical data and prices

Define the bid map and the set of equivalent bids at prices λ:

$$\begin{aligned} &E(m) := (u, c), \qquad u_t = \alpha_e(\pi_t - \bar{\nu}_t) - \epsilon_e, \qquad c_t = \zeta_e + \beta_e(\pi_t + \underline{\nu}_t); \\ &\Phi_e(\theta_e, \lambda) := \{E(m) : m \in \mathcal{D}_e(\lambda)\}, \end{aligned} \tag{6.1}$$

always with one block per interval and block sizes $\hat{p}_{e,t} = \bar{p}_e$, $\hat{g}_{e,t} = \bar{g}_e$.

Assumption 6.1. *b_e is concave and piecewise linear (in particular, linear), $\bar{p}_e, \bar{g}_e > 0$, and $\mathcal{P}_e^{\mathrm{SS}}(\lambda)$ is feasible and bounded. Then $\mathcal{P}_e^{\mathrm{SS}}(\lambda)$ is an LP and strong duality holds.*

Two standard facts of linear programming are used [15, §5.5], [16, Ch. 4].

(F1) Every primal optimal solution and every dual optimal solution satisfy the KKT conditions *together*.

(F2) For any fixed primal optimal solution $\hat{x}$, the dual optimal set is the set of dual-feasible points that satisfy complementary slackness with $\hat{x}$.

(F2) follows from (F1) and from the fact that complementary slackness with a feasible $\hat{x}$ closes the duality gap.

Lemma 6.2 (Decomposition)**.** *Let $z^\star$ be primal–dual optimal for $\mathcal{P}^{\mathrm{SOC}}$, with nodal prices $\lambda^\star$. For every ESS e, the schedule $(p_e^\star, g_e^\star, s_e^\star)$ is optimal for $\mathcal{P}_e^{\mathrm{SS}}(\lambda^\star_{n(e)})$, and $(\pi_e^\star, \bar{\nu}_e^\star, \underline{\nu}_e^\star, \bar{\rho}_e^\star, \underline{\rho}_e^\star, \bar{\sigma}_e^\star, \underline{\sigma}_e^\star) \in \mathcal{D}_e(\lambda^\star_{n(e)})$.*

Proof. The variables of e appear in $\mathcal{P}^{\mathrm{SOC}}$ only in (A1) at bus $n(e)$ and in (A8)–(A12) of e. The KKT system of $\mathcal{P}_e^{\mathrm{SS}}(\lambda^\star_{n(e)})$ is therefore a subset of the KKT system of $\mathcal{P}^{\mathrm{SOC}}$, with the price fixed

at $\lambda^\star_{n(e)}$: it is the storage block of the MCP (Corollary 4.9). Since $z^\star$ satisfies the latter, its ESS components satisfy the former. $\mathcal{P}_e^{\mathrm{SS}}$ is an LP, so the KKT conditions are sufficient for optimality of both its primal and its dual components. □

Theorem 6.3 (Equivalent bids from technical data and prices)**.** *Let $z^\star$ be primal–dual optimal for $\mathcal{P}^{\mathrm{SOC}}$, with dispatch $x^\star = (q^\star, d^\star, f^\star, \theta^\star, p^\star, g^\star)$ and nodal prices $\lambda^\star$, and let Assumption 6.1 hold. For every ESS e, take* any *$m_e \in \mathcal{D}_e(\lambda^\star_{n(e)})$ and submit $(u_e, c_e) = E(m_e)$ with blocks $\bar{p}_e, \bar{g}_e$. Then:*

(i) $x^\star$ is optimal for $\mathcal{P}^{\mathrm{CD}}$, and $\lambda^\star$ is an optimal vector of CD nodal prices;

(ii) the bids of e depend on the instance only through θ_e and $\lambda^\star_{n(e)}$: they lie in $\Phi_e(\theta_e, \lambda^\star_{n(e)})$, which depends neither on the other participants' offers, nor on the network, nor on the ramps, nor on which multipliers the operator's solver returns;

(iii) the bids (E) *of Proposition 5.1 are the particular selection $m_e = (\pi^\star_e, \bar{\nu}^\star_e, \underline{\nu}^\star_e, \bar{\rho}^\star_e, \underline{\rho}^\star_e, \bar{\sigma}^\star_e, \underline{\sigma}^\star_e)$.*

Proof. Step 1. By Lemma 6.2, $(p^\star_e, g^\star_e, s^\star_e)$ is optimal for $\mathcal{P}_e^{\mathrm{SS}}(\lambda^\star_{n(e)})$. *Step 2.* By (F1), this schedule and the chosen m_e satisfy (S5)–(S8), (CS4) and (CS5) together, at the price $\lambda^\star_{n(e)}$. *Step 3.* Substituting $E(m_e)$ into (S5)–(S6) gives exactly ($\mathrm{S6^{CD}}$)–($\mathrm{S5^{CD}}$) at $(p^\star_e, g^\star_e, \lambda^\star_{n(e)})$, with the power-bound multipliers taken from m_e. Because $\hat{p}_{e,t} = \bar{p}_e$ and $\hat{g}_{e,t} = \bar{g}_e$, (CS4) is ($\mathrm{CS4^{CD}}$). *Step 4.* The rows of the rest of the market, (S1)–(S4), (CS1)–(CS3) and (A1)–(A7), contain no ESS multiplier, so they hold at $x^\star$ with the multipliers of $z^\star$. (C1)–(C2) hold because $p^\star, g^\star$ satisfy (A11)–(A12). *Step 5.* Steps 3–4 give the full KKT system of $\mathcal{P}^{\mathrm{CD}}$ at $x^\star$ and $\lambda^\star$. $\mathcal{P}^{\mathrm{CD}}$ is an LP, so $x^\star$ is primal optimal and $\lambda^\star$ is part of a dual optimal solution. This proves (i). Part (ii) follows from the definition (6.1), since $\mathcal{P}_e^{\mathrm{SS}}$ contains only θ_e and the price path. Part (iii) is Lemma 6.2. □

Interpretation. An ESS does not need to know the multipliers of the SOC-based design. It needs its own technical data, which it knows, and the nodal prices at its own bus. With these, it solves its own self-schedule, reads *any* optimal dual, and plugs it into (E). The gap between the two designs is therefore a gap in price information, not in market design.

Remark 6.4 (What Theorem 6.3 does and does not claim)*.*

(a) The ESS needs only the prices, not the multipliers of the SOC-based design: any optimal dual of its own self-schedule works.

(b) The ESS's own optimal schedule may differ from $(p^\star_e, g^\star_e)$, because $\mathcal{P}_e^{\mathrm{SS}}(\lambda^\star)$ is often degenerate. In the base case it may, for instance, charge 222.2 MW at $t = 4$ only and discharge 200 MW at $t = 5$, with the same profit. The bids are unaffected, because $\mathcal{D}_e$ does not depend on which primal optimum is used (F2). The quantities, however, are affected (Remark 5.6); this is the origin of the quantity-information loss measured in Chapter 10.

(c) $\lambda^\star$ is *a* CD clearing price. When the CD dual is not unique, the operator's pricing rule may select another price vector for the same dispatch. In Baldick's single-bus cases this does not happen: the CD prices equal $\lambda^\star$ for every admissible choice of bids (Chapter 8).

(d) $x^\star$ is *a* CD optimum. As in Remark 5.6, the CD may have other optima, so quantities still need a tie-breaking rule.

6.2 The set of all equivalent bids

Theorem 6.3 says that any element of Φ_e works. The next result describes Φ_e explicitly from one optimal schedule.

Proposition 6.5 (All equivalent bids, from prices and one schedule)**.** *Let Assumption 6.1 hold, fix $\lambda \in \mathbb{R}^T$, and let $(p^\circ, g^\circ, s^\circ)$ be any optimal solution of $\mathcal{P}_e^{\mathrm{SS}}(\lambda)$, with $s_0^\circ := s_{e,0}$. Define the sets of intervals in which the SOC rows are tight,*

$$\mathcal{U} := \{t : \gamma_e s_{t-1}^\circ + \Delta T \alpha_e p_t^\circ = \bar{s}_e\}, \qquad \mathcal{L} := \{t : \gamma_e s_{t-1}^\circ - \Delta T \beta_e g_t^\circ = \underline{s}_e\}. \tag{6.2}$$

Then $(u, c) \in \Phi_e(\theta_e, \lambda)$ if and only if $u = \alpha_e(\pi - \bar{\nu}) - \epsilon_e$ and $c = \zeta_e + \beta_e(\pi + \underline{\nu})$ for some $\pi \in \mathbb{R}^T$ and $\bar{\nu}, \underline{\nu} \in \mathbb{R}_+^T$ such that

$$\begin{aligned}
&(a)\ \ \pi_T \in \partial b_e(s_T^\circ - \bar{\bar{s}}_{e,T}) \quad (\pi_T = w_e \text{ for linear } b_e),\\
&(b)\ \ \pi_t = \gamma_e(\pi_{t+1} - \bar{\nu}_{t+1} + \underline{\nu}_{t+1}), \quad t < T,\\
&(c)\ \ \bar{\nu}_t = 0 \text{ for } t \notin \mathcal{U}, \qquad \underline{\nu}_t = 0 \text{ for } t \notin \mathcal{L},\\
&(d)\ \ u_t \begin{cases} \le \lambda_t, & p_t^\circ = 0 \\ = \lambda_t, & 0 < p_t^\circ < \bar{p}_e \\ \ge \lambda_t, & p_t^\circ = \bar{p}_e \end{cases} \qquad c_t \begin{cases} \ge \lambda_t, & g_t^\circ = 0 \\ = \lambda_t, & 0 < g_t^\circ < \bar{g}_e \\ \le \lambda_t, & g_t^\circ = \bar{g}_e. \end{cases}
\end{aligned} \tag{6.3}$$

In particular, $\Phi_e(\theta_e, \lambda)$ is a polyhedron, computed from θ_e and λ alone (one LP for the schedule, then linear conditions), and it is the same for every optimal schedule.

Proof. By (F2), $m \in \mathcal{D}_e(\lambda)$ if and only if m is dual feasible and satisfies complementary slackness with $(p^\circ, g^\circ, s^\circ)$. Row by row: (a) is (S8); (b) is (S7); (c), together with $\bar{\nu}, \underline{\nu} \ge 0$, is (CS5). For the power bounds, (S5) reads $u_t - \lambda_t = \bar{\rho}_t - \underline{\rho}_t$ and (S6) reads $c_t - \lambda_t = \underline{\sigma}_t - \bar{\sigma}_t$. With $\bar{\rho}, \underline{\rho}, \bar{\sigma}, \underline{\sigma} \ge 0$ and (CS4) at (p°, g°): if $p_t^\circ = 0$ then $\bar{\rho}_t = 0$ and $u_t - \lambda_t = -\underline{\rho}_t \le 0$; if $0 < p_t^\circ < \bar{p}_e$ both vanish and $u_t = \lambda_t$; if $p_t^\circ = \bar{p}_e$ then $\underline{\rho}_t = 0$ and $u_t - \lambda_t = \bar{\rho}_t \ge 0$. The discharging side is symmetric. Conversely, given (a)–(d), set $\bar{\rho}_t = [u_t - \lambda_t]^+$, $\underline{\rho}_t = [\lambda_t - u_t]^+$, $\underline{\sigma}_t = [c_t - \lambda_t]^+$ and $\bar{\sigma}_t = [\lambda_t - c_t]^+$. These are nonnegative, satisfy (S5)–(S6) and, by (d), (CS4). Independence of the schedule follows from (F2). □

Corollary 6.6 (When the prices pin π)**.** *Under Proposition 6.5:*

(i) if $t \ge 2$ and $t \notin \mathcal{U} \cup \mathcal{L}$, then $\pi_{t-1} = \gamma_e \pi_t$;

(ii) if $0 < p_t^\circ < \bar{p}_e$ and $t \notin \mathcal{U}$, then $\pi_t = (\lambda_t + \epsilon_e)/\alpha_e$; if $0 < g_t^\circ < \bar{g}_e$ and $t \notin \mathcal{L}$, then $\pi_t = (\lambda_t - \zeta_e)/\beta_e$;

(iii) for linear b_e, $\pi_T = w_e$.

Call a chain *a maximal set of consecutive intervals linked by (i). On every chain that contains an interval of type (ii) or the interval T, π is unique and equal to $\gamma_e^{\tau - t}$ times the pinned value at τ. On any other chain, π is determined only up to an interval, bounded by (d) and by the sign of the rents at the ends of the chain: $r_t \ge 0$ if $t \in \mathcal{L} \setminus \mathcal{U}$ and $r_t \le 0$ if $t \in \mathcal{U} \setminus \mathcal{L}$, where $r_t := \underline{\nu}_t - \bar{\nu}_t$.*

Proof. (i): by (c), $\bar{\nu}_t = \underline{\nu}_t = 0$, and (b) at $t-1$ gives $\pi_{t-1} = \gamma_e \pi_t$. (ii): by (c), $\bar{\nu}_t = 0$ (respectively $\underline{\nu}_t = 0$), and (d) gives $u_t = \alpha_e \pi_t - \epsilon_e = \lambda_t$ (respectively $c_t = \zeta_e + \beta_e \pi_t = \lambda_t$). (iii) is (a). The statements on chains follow by iterating (i) from the pinned interval; the sign conditions follow from (c). □

Reading. Prices pin the value of stored energy wherever the ESS trades a partial quantity without being stopped by an SOC limit, and this value propagates (up to retention) to all intervals not separated from it by a tight SOC row. The value is *not* pinned on chains that are cut off on both sides by SOC limits and contain no partial trade: there an SOC limit has been reached and cannot be undone within the chain. These are the intervals where a selection rule is needed (Section 6.5).

Worked derivation: the base case from θ_e and $\lambda^\star$ only. Take $\lambda^\star = (43.1, 50, 0, 0, 50, 50)$ and the optimal schedule $p^\circ = (100, 0, 200, 22.2, 0, 0)$, $g^\circ = (0, 90, 0, 0, 175, 25)$, $s^\circ = (360, 0, 720, 800, 100, 0)$. Then (6.2) gives $\mathcal{U} = \{4, 5\}$ and $\mathcal{L} = \{1, 2, 3, 6\}$. Every interval is tight, so every chain is a single interval.

t	1	2	3	4	5	6
tight row	$\mathcal{L}$	$\mathcal{L}$	$\mathcal{L}$	$\mathcal{U}$	$\mathcal{U}$	$\mathcal{L}$
type (Corollary 6.6)	(ii) charge	–	(ii) charge	–	(ii) discharge	(iii)
π_t	$\frac{43.1+1}{0.9} = 49$	$[1.11, 49]$	$\frac{0+1}{0.9} = 1.11$	$[1.11, 49]$	$\frac{50-1}{1} = 49$	$w = 40$

At $t = 2$, $g^\circ_2 = 90$ is interior but $2 \in \mathcal{L}$, so the price does not pin π_2. Its bounds come from the rents: $r_2 = \pi_1 - \pi_2 \geq 0$ and $r_3 = \pi_2 - \pi_3 \geq 0$ (both in $\mathcal{L} \setminus \mathcal{U}$), so $1.11 \leq \pi_2 \leq 49$. At $t = 4$, $r_4 = \pi_3 - \pi_4 \leq 0$ and $r_5 = \pi_4 - \pi_5 \leq 0$ (both in $\mathcal{U} \setminus \mathcal{L}$), so $1.11 \leq \pi_4 \leq 49$. Condition (d) adds no tighter bound. At $t = 1$, $\underline{\nu}_1$ enters only c_1, so any $c_1 \geq 50$ is admissible: an empty ESS may offer at any price at or above $\lambda^\star_1$. These ranges coincide with the dual optimal face of $\mathcal{P}^{\mathrm{SS}}_e(\lambda^\star)$ and with the perturbation analysis of Table 3.1.

6.3 Closed form: bids and the value of stored energy

The polyhedron of Proposition 6.5 is what an implementation computes. The closed forms of this section and the next are not needed for computation, but they show *which* prices determine the bids, and therefore how a forecast error moves them.

Assumption 6.7 (One SOC limit per interval)**.** *For every ESS and interval, at most one of* (A9)–(A10) *is active, i.e. no ESS goes from its minimum to its maximum SOC, or vice versa, within a single interval. A sufficient condition is* $\Delta T(\alpha_e \bar{p}_e + \beta_e \bar{g}_e) < \bar{s}_e - \underline{s}_e$ *(see Step 4 of the proof of Corollary 6.8).*

Terminal value. By (S8), the value of stored energy at the end of the horizon is the end-of-horizon willingness to pay: $\pi^\star_{e,T} = w_e(s^\star_{e,T} - \bar{\bar{s}}_{e,T})$, which equals w_e for a linear b_e.

Net SOC rent. Only the difference of the two SOC multipliers enters (S7). Define the net rent

$$r^\star_{e,t} := \underline{\nu}^\star_{e,t} - \bar{\nu}^\star_{e,t} = \frac{\pi^\star_{e,t-1}}{\gamma_e} - \pi^\star_{e,t}, \qquad t \in \mathcal{T}, \tag{6.4}$$

with $\pi^\star_{e,0}$ as in Proposition 5.2. The net rent is the jump of the value of stored energy across interval t.

Corollary 6.8 (Bids in terms of the value of stored energy)**.** *Under Assumption 6.7,*

$$c^{g\star}_{e,t} = \zeta_e + \beta_e \max\Big\{\pi^\star_{e,t}, \frac{\pi^\star_{e,t-1}}{\gamma_e}\Big\}, \qquad u^{p\star}_{e,t} = \alpha_e \min\Big\{\pi^\star_{e,t}, \frac{\pi^\star_{e,t-1}}{\gamma_e}\Big\} - \epsilon_e. \tag{6.5}$$

Proof. Drop the index e and the superscript $\star$. *Step 1.* By (E), $c_t = \zeta + \beta(\pi_t + \underline{\nu}_t)$ and $u_t = \alpha(\pi_t - \bar{\nu}_t) - \epsilon$. *Step 2.* (S7) for the pair $(t-1, t)$ reads $\pi_{t-1} = \gamma(\pi_t - \bar{\nu}_t + \underline{\nu}_t)$, together with (CS5). *Step 3.* Dividing by γ gives $r_t = \underline{\nu}_t - \bar{\nu}_t = \pi_{t-1}/\gamma - \pi_t$, which is (6.4). *Step 4.* If $\underline{\nu}_t > 0$ and $\bar{\nu}_t > 0$, both SOC rows are tight: $\gamma s_{t-1} + \Delta T \alpha p_t = \bar{s}$ and $\gamma s_{t-1} - \Delta T \beta g_t = \underline{s}$. Subtracting, $\Delta T(\alpha p_t + \beta g_t) = \bar{s} - \underline{s}$, which Assumption 6.7 excludes. Hence $\underline{\nu}_t \bar{\nu}_t = 0$. *Step 5.* With $\underline{\nu}_t, \bar{\nu}_t \geq 0$ and Step 4: if $\underline{\nu}_t > 0$ (the ESS empties), then $r_t = \underline{\nu}_t > 0$; if $\bar{\nu}_t > 0$ (the ESS fills), then $r_t = -\bar{\nu}_t < 0$; otherwise $r_t = 0$. In all three cases,

$$\underline{\nu}_t = \max\{0, r_t\} = [r_t]^+, \qquad \bar{\nu}_t = -\min\{0, r_t\} = [-r_t]^+. \tag{6.6}$$

Step 6. Using $a + \max\{x, y\} = \max\{a+x, a+y\}$ with $a = \pi_t$: $\pi_t + \underline{\nu}_t = \pi_t + \max\{0, \pi_{t-1}/\gamma - \pi_t\} = \max\{\pi_t, \pi_{t-1}/\gamma\}$, and similarly $\pi_t - \bar{\nu}_t = \pi_t + \min\{0, \pi_{t-1}/\gamma - \pi_t\} = \min\{\pi_t, \pi_{t-1}/\gamma\}$. *Step 7.* Inserting both expressions into Step 1 gives (6.5). □

The bids of interval t depend only on the value of stored energy at the start of t, $\pi^\star_{e,t-1}/\gamma_e$, and at its end, $\pi^\star_{e,t}$. The ESS offers at the *higher* of the two values and bids at the *lower*. When the two values coincide, the bid–offer band is at its narrowest; it widens only where the value jumps, i.e. where an SOC limit binds. In the base case, at $t = 2$, $\pi_1 = 49$ and $\pi_2 = 1.11$, so $c_2 = 1 + \max\{1.11, 49\} = 50$ and $u_2 = 0.9 \cdot 1.11 - 1 = 0$.

6.4 Segments, marginal intervals and prices

Segments of the SOC path. Let $\mathcal{H}_e = \{h_{e,1} < \cdots < h_{e,N_e}\}$ be the intervals in which (A9) or (A10) is active (with a positive multiplier) at the chosen optimum. They split the horizon into segments

$$\mathcal{S}_{e,k} = \{h_{e,k-1}, \ldots, h_{e,k} - 1\}, \quad k = 1, \ldots, N_e, \quad h_{e,0} := 1; \qquad \mathcal{S}_{e,N_e+1} = \{h_{e,N_e}, \ldots, T\}. \tag{6.7}$$

By (6.4), $r^\star_{e,t} = 0$ for $t \notin \mathcal{H}_e$, so within a segment the value of stored energy changes only through retention: $\pi^\star_{e,t} = \gamma_e \pi^\star_{e,t+1}$ whenever $t, t+1 \in \mathcal{S}_{e,k}$.

Why segments. Between two binding intervals the ESS can shift energy freely from one interval to another, so one MWh is worth the same in all of them, up to retention. A binding limit breaks this link: energy held before the ESS empties (or fills) cannot be carried past that point, so its value before and after the limit can differ. Each segment therefore carries a single number, its value of stored energy, and the vector $\pi^\star_e$ is described by $N_e + 1$ numbers instead of T.

Where a segment starts. A binding interval h opens a new segment. The SOC rows at h involve $s_{e,h-1}$, so the rent $r^\star_{e,h}$ sits between the value at the end of $h-1$ (old segment) and the value at the end of h (new segment).

In the base case, with the solver multipliers, $\mathcal{H} = \{2, 4, 6\}$ and the segments are $\{1\}$, $\{2, 3\}$, $\{4, 5\}$ and $\{6\}$. Since $\gamma = 1$, $\pi^\star$ is constant inside each segment (1.11 in $\{2, 3\}$, 49 in $\{4, 5\}$) and jumps by the rent at $t = 2, 4, 6$ (Table 5.2).

Definition 6.9 (Marginal interval)**.** An interval $\tilde{t} \in \mathcal{S}_{e,k}$ is a *marginal interval* of segment k if

(D) *discharge-marginal:* $0 < g^\star_{e,\tilde{t}} < \bar{g}_e$ and $\underline{\nu}^\star_{e,\tilde{t}} = 0$, or

(C) *charge-marginal:* $0 < p^\star_{e,\tilde{t}} < \bar{p}_e$ and $\bar{\nu}^\star_{e,\tilde{t}} = 0$.

We write $\tilde{t}_{e,k}$ for a chosen marginal interval of segment k.

Both conditions are needed. If the ESS discharges at full power, it would discharge more if it could, so the price only bounds its value. If it discharges partially but empties in that interval, the SOC limit, not the price, stops it, so again the price only bounds the value.

Lemma 6.10 (The price reveals the value at a marginal interval)**.** *If $\tilde{t}$ is a marginal interval of segment k, then $\pi^\star_{e,\tilde{t}} = v_{e,k}(\lambda^\star)$, where*

$$v_{e,k}(\lambda) := \begin{cases} (\lambda_{n(e),\tilde{t}} - \zeta_e)/\beta_e, & \textit{case (D)}, \\ (\lambda_{n(e),\tilde{t}} + \epsilon_e)/\alpha_e, & \textit{case (C)}. \end{cases} \tag{6.8}$$

Proof. Case (D): $0 < g^\star_{e,\tilde{t}} < \bar{g}_e$ and (CS4) give $\underline{\sigma}^\star = \bar{\sigma}^\star = 0$; with $\underline{\nu}^\star_{e,\tilde{t}} = 0$, (S6) reduces to $\zeta_e - \lambda^\star_{n(e),\tilde{t}} + \beta_e \pi^\star_{e,\tilde{t}} = 0$. Case (C) is symmetric with (S5). □

Discharging one more stored MWh delivers $1/\beta_e$ MWh to the grid at price λ and costs ζ_e/β_e in degradation, so $(\lambda - \zeta_e)/\beta_e$ is what one stored MWh earns now. Charging one more stored MWh requires $1/\alpha_e$ MWh from the grid, at cost $(\lambda + \epsilon_e)/\alpha_e$. At a marginal interval the ESS is indifferent between trading that MWh now and keeping it.

Lemma 6.11 (The value is carried through the segment)**.** *If t and t' belong to the same segment, then $\pi^\star_{e,t} = \gamma_e^{t'-t} \pi^\star_{e,t'}$.*

Proof. Let $t < t'$. For every j with $t < j \le t'$, $j \notin \mathcal{H}_e$, so $r^\star_{e,j} = 0$ and $\pi^\star_{e,j-1} = \gamma_e \pi^\star_{e,j}$ by (6.4). Chaining these equalities gives the result. □

Lemma 6.12 (Which price sets the value)**.** *Let $\mathcal{S}^\circ_{e,k} := \mathcal{S}_{e,k} \setminus \mathcal{H}_e$, and for $t \in \mathcal{S}^\circ_{e,k}$ let $u_t := \alpha_e \pi^\star_{e,t} - \epsilon_e$ and $c_t := \zeta_e + \beta_e \pi^\star_{e,t}$. Then, for $t \in \mathcal{S}^\circ_{e,k}$, $p^\star_{e,t} = \bar{p}_e$ if $\lambda^\star_{n(e),t} < u_t$, $p^\star_{e,t} = 0$ if $\lambda^\star_{n(e),t} > u_t$, and $g^\star_{e,t} = \bar{g}_e$ if $\lambda^\star_{n(e),t} > c_t$, $g^\star_{e,t} = 0$ if $\lambda^\star_{n(e),t} < c_t$. Consequently, for any reference interval $\tau \in \mathcal{S}_{e,k}$, with $\mathcal{C}_k := \{t \in \mathcal{S}^\circ_{e,k} : p^\star_{e,t} > 0\}$ and $\mathcal{G}_k := \{t \in \mathcal{S}^\circ_{e,k} : g^\star_{e,t} > 0\}$,*

$$\max_{t \in \mathcal{C}_k} \gamma_e^{t-\tau} \frac{\lambda^\star_{n(e),t} + \epsilon_e}{\alpha_e} \;\le\; \pi^\star_{e,\tau} \;\le\; \min_{t \in \mathcal{G}_k} \gamma_e^{t-\tau} \frac{\lambda^\star_{n(e),t} - \zeta_e}{\beta_e}, \tag{6.9}$$

with equality on the left if some charging interval is partial, and on the right if some discharging interval is partial.

Proof. For $t \in \mathcal{S}^\circ_{e,k}$ both SOC multipliers vanish, so (S5) reads $u_t - \lambda^\star_{n(e),t} = \bar{\rho}^\star_{e,t} - \underline{\rho}^\star_{e,t}$. If $\lambda^\star < u_t$ then $\bar{\rho}^\star > 0$ and $p^\star = \bar{p}_e$; if $\lambda^\star > u_t$ then $\underline{\rho}^\star > 0$ and $p^\star = 0$; if the charge is partial both vanish and $\lambda^\star = u_t$. The discharging side is symmetric. For (6.9), $p^\star_{e,t} > 0$ implies $\lambda^\star_{n(e),t} \le u_t = \alpha_e \gamma_e^{\tau-t} \pi^\star_{e,\tau} - \epsilon_e$ by Lemma 6.11; symmetrically for discharging. □

Inside a segment the ESS behaves like a price-taker facing two fixed thresholds: it charges at full power when the price is below u_t, discharges at full power when it is above c_t, and is idle in between. With $\gamma_e = 1$, *the value of the segment is set by the most expensive interval in which the ESS still charges* (and by the cheapest interval in which it still discharges), wherever that interval lies in the segment. For example, with $\alpha = 0.9$, $\beta = 1$, $\gamma = 1$, $\epsilon = \zeta = 1$, $\bar{p} = \bar{g} = 100\,\text{MW}$, $\bar{s} = 200\,\text{MWh}$, $s_0 = 0$, $w = 0$ and prices $(0, 10, 5, 50, 50)$, the ESS charges 100, 22.2 and 100 MW in the first three intervals and discharges 100 MW in the last two. The interval priced at 10 is the only partial charge, so the value of the first segment is $(10 + 1)/0.9 = 12.22$, although it is not the last charging interval.

Proposition 6.13 (Value of stored energy from prices)**.** *Under Assumption 6.7, if every segment $k \le N_e$ contains a marginal interval $\tilde{t}_{e,k}$, then*

$$\pi^\star_{e,t} = \begin{cases} \gamma_e^{\tilde{t}_{e,k}-t}\, v_{e,k}(\lambda^\star), & t \in \mathcal{S}_{e,k},\ k \le N_e, \\ \gamma_e^{T-t}\, w_e\big(s^\star_{e,T} - \bar{\bar{s}}_{e,T}\big), & t \in \mathcal{S}_{e,N_e+1}. \end{cases} \tag{6.10}$$

Proof. For $k \le N_e$, Lemma 6.10 gives $\pi^\star_{e,\tilde{t}_{e,k}} = v_{e,k}(\lambda^\star)$, and Lemma 6.11 carries it through the segment. The last segment contains T, where (S8) gives the terminal value, carried backwards by Lemma 6.11. □

Every step uses only stationarity and complementary slackness, so the result holds for any optimal multipliers with that solution's $\mathcal{H}_e$. If a segment has several marginal intervals, their prices are tied together by Lemma 6.11. If a segment has no marginal interval, prices only bound the value (Corollary 6.6). The last segment needs no marginal interval: its value comes from the terminal bid. In the base case, the marginal intervals are $t = 1$ (charging at 43.1), $t = 3$ (charging at 0) and $t = 5$ (discharging at 50), which give $\pi^\star = (49, 1.11, 1.11, 49, 49, 40)$, the solver's multipliers (Table 5.2, rows (d)).

Bids as functions of prices. Let $\tau(t)$ be the marginal interval of the segment that contains t, with $\tau(t) := T$ and $v_{e,T} := w_e(s^\star_{e,T} - \bar{\bar{s}}_{e,T})$ for the last segment, and $\tau(0) := \tau(1)$. Then $\pi^\star_{e,t} = \gamma_e^{\tau(t)-t} v_{e,\tau(t)}(\lambda^\star)$ and, combining with Corollary 6.8:

Corollary 6.14 (Equivalent bids as functions of prices)**.** *Under the conditions of Proposition 6.13,*

$$c^{g\star}_{e,t} = \zeta_e + \beta_e \max\Big\{\gamma_e^{\tau(t)-t}\, v_{e,\tau(t)}(\lambda^\star),\ \gamma_e^{\tau(t-1)-t}\, v_{e,\tau(t-1)}(\lambda^\star)\Big\}, \tag{6.11}$$

$$u^{p\star}_{e,t} = \alpha_e \min\Big\{\gamma_e^{\tau(t)-t}\, v_{e,\tau(t)}(\lambda^\star),\ \gamma_e^{\tau(t-1)-t}\, v_{e,\tau(t-1)}(\lambda^\star)\Big\} - \epsilon_e. \tag{6.12}$$

If $t-1$ and t lie in the same segment, the two arguments coincide.

Proof. By Proposition 6.13, $\pi^\star_{e,t} = \gamma_e^{\tau(t)-t} v_{e,\tau(t)}$ and $\pi^\star_{e,t-1}/\gamma_e = \gamma_e^{\tau(t-1)-t} v_{e,\tau(t-1)}$. Substitute into (6.5). □

What the equivalent bids depend on.

- the private parameters $\alpha_e, \beta_e, \gamma_e, \epsilon_e, \zeta_e, w_e$;
- the nodal prices at the *marginal intervals* of the SOC path only;
- the SOC path, through the binding intervals and, for a nonlinear b_e, the final SOC.

An error in the price of a marginal interval shifts the bids of the whole segment. An error in any other interval matters only if it changes the SOC path, i.e. which intervals are marginal or binding.

6.5 Non-uniqueness and the pre-limit selection rule

By Corollary 6.6, the admissible values of $\pi_{e,t}$ form an interval $[\pi^-_{e,t}, \pi^+_{e,t}]$ that is a point except on chains cut off by SOC limits. Its ends are the right and left derivatives of the optimal value with respect to a perturbation of (A8) at t (Table 3.1). All admissible choices give the same nodal prices under perfect information, but different bids, and therefore different behaviour when the price forecast is wrong.

The rule. We select the multipliers sequentially by the *pre-limit rule*:

$$\pi^R_{e,t} = \mathrm{Proj}_{[\pi^-_{e,t},\, \pi^+_{e,t}]}(\pi^R_{e,t-1}/\gamma_e) = \min\Big\{\max\Big\{\frac{\pi^R_{e,t-1}}{\gamma_e},\ \pi^-_{e,t}\Big\},\ \pi^+_{e,t}\Big\}, \qquad t \in \mathcal{T}, \tag{6.13}$$

where the projection moves the carried-in value to the nearest admissible point. The value carried in from the previous interval is kept whenever it is admissible, so the rent is placed in the *following* interval. In an implementation, $[\pi^-_{e,t}, \pi^+_{e,t}]$ is computed by minimising and maximising π_t over the polyhedron of Proposition 6.5, with the values already selected for earlier intervals fixed.

Rationale: robustness to price errors. In the base case the solver returns $\pi = (49, 1.11, 1.11, 49, 49, 40)$, while the pre-limit rule gives $\pi^R = (49, 49, 1.11, 1.11, 49, 40)$, with bids

$$u^p = (43.1,\ 43.1,\ 0,\ 0,\ 0,\ 35), \qquad c^g = (50,\ 50,\ 50,\ 2.11,\ 50,\ 50).$$

Both choices clear the current design at the same prices. They differ in the *margin*: the distance from $\lambda^\star_t$ to the nearest bid edge that would trigger an unwanted action. An unwanted action is harmless if it is feasible, but infeasible if it means selling when empty or buying when full. Table 6.1 compares three choices.

choice (π_2, π_4)	m_2	m_3 (sell when empty)	m_4	m_5 (buy when full)
solver multipliers (1.11, 49)	50.0	**2.11**	50.0	**6.9**
midpoint (25.06, 25.06)	28.5	26.1	26.1	28.5
pre-limit rule (49, 1.11)	6.9	50.0	2.11	50.0

Table 6.1: Price margins ($/MWh) in the base case. A small margin means that a small forecast error flips the action of the ESS. Bold: margins whose violation makes the schedule SOC-infeasible.

With the solver's multipliers, both infeasibility risks have small margins. The pre-limit rule removes them (margin 50) at the cost of small margins at $t = 2, 4$, where the unwanted action is feasible and only costs welfare. No choice makes all margins large, because a single bid per interval cannot say "this energy is worth 49 if I am not yet empty and 1.11 if I am". The SOC-based design does not have this problem because the operator tracks the SOC; SOC-dependent bids [23] address it directly. The slack does not create the bid–offer spread, which exists anyway through ϵ_e, ζ_e and $\alpha_e < \beta_e$; it decides *where* the band is widened (Figure 6.1).

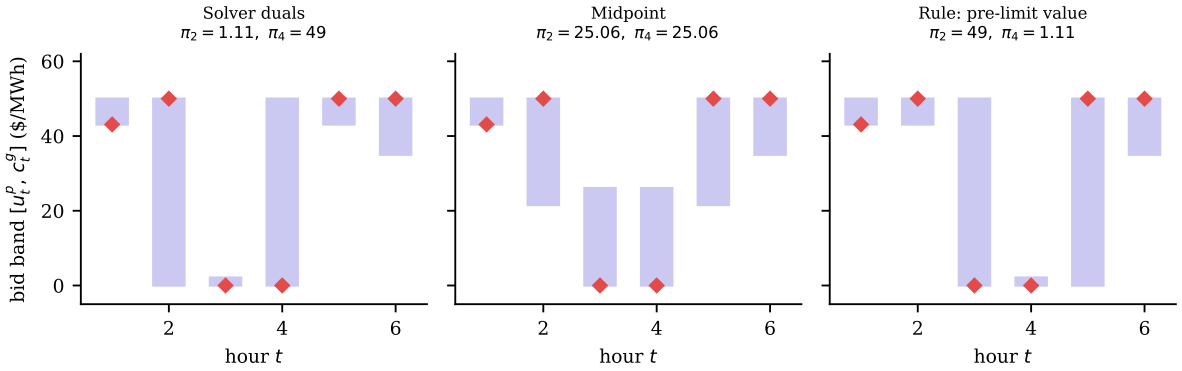


Figure 6.1: Bid–offer bands $[u_t, c_t]$ of the base case for three admissible choices of (π_2, π_4). Diamonds: prices $\lambda^\star_t$. Only the edge that the ESS is not meant to use moves with the choice.

6.6 Several storage units

Theorem 6.3 is stated for every ESS simultaneously, so it covers any number of ESS. It is worth making this explicit, because the bids of each ESS are built independently.

Corollary 6.15 (Several ESS)**.** *Let the SOC-based design with ESS $e \in \mathcal{E}$ have an optimal solution with prices $\lambda^\star$. If each ESS e, independently of the others, solves $\mathcal{P}_e^{\mathrm{SS}}(\lambda^\star_{n(e)})$, picks any $m_e \in \mathcal{D}_e(\lambda^\star_{n(e)})$ and submits $E(m_e)$, then the dispatch of the SOC-based design is optimal in the current design and $\lambda^\star$ is a vector of CD prices.*

Proof. This is Theorem 6.3: Steps 1–3 of its proof are carried out for each e separately, using only θ_e and $\lambda^\star_{n(e)}$; Step 4 does not involve any ESS multiplier. □

The ESS interact only through the prices. Given $\lambda^\star$, there is no coordination problem: no ESS needs to know the data, schedules or bid choices of the others. This is verified numerically with five heterogeneous ESS and 200 random joint choices of multipliers in Section 8.3. The interaction reappears, however, as soon as the prices are forecast rather than known: if several ESS make the same forecast error, their bids move together and move the price (hypothesis H5).

6.7 Who can compute the equivalent bids

Φ_e maps technical data and a price path to the bids of a competitive, price-taking ESS. Two parties can evaluate it.

The ESS owner. With its own forecast $\hat{\lambda}$, the owner can bid $\Phi_e(\theta_e, \hat{\lambda})$. By Theorem 6.3 this reproduces the outcome of the SOC-based design when $\hat{\lambda} = \lambda^\star$. This is the forecast-based bidder of the experiments (Chapter 9).

The market operator. If the ESS registers θ_e, as it would in the SOC-based design, the operator can evaluate $\Phi_e(\theta_e, \hat{\lambda}^{\mathrm{ISO}})$ with its own forecast, without changing its clearing engine. The result is a competitive reference bid for every interval, similar in purpose to the storage default energy bids of CAISO's market power mitigation, which embed an opportunity cost computed from the operator's own price information [3, 12]. Unlike those bids, Φ_e follows the SOC dynamics and the end-of-horizon value of the ESS. Possible uses are:

- *Screening:* compare submitted bids with $\Phi_e(\theta_e, \hat{\lambda}^{\mathrm{ISO}})$. Systematic markups on discharge offers, or bids that move capacity out of the day-ahead market, are the behaviours that theory and data associate with strategic storage [29, 32].
- *Mitigation:* when a market power test fails, replace the bid by an element of Φ_e.
- *A natural tolerance band:* Φ_e is a set, not a point. Where an SOC limit binds, a range of bids is consistent with competitive behaviour, so only bids outside the set are suspicious. The non-uniqueness found above becomes useful for monitoring.

The reference is only as good as the operator's forecast: a forecast error moves the reference, so screening must separate strategic markups from forecast errors. This is RQ4, and it uses the same models as the experiments.

Chapter 7

Computational burden of the SOC-based design

The SOC-based design adds variables and constraints that couple all intervals of each ESS. This chapter quantifies the additional size (Section 7.1), explains why the structure matters more than the size (Section 7.2), shows that no binary variables are needed (Section 7.3), and reports a benchmark of solve times (Section 7.4).

7.1 Problem size

Consider E ESS and T intervals, and let R_0, C_0 and Z_0 be the number of rows, columns and nonzeros ("nnz": nonzero entries of the constraint matrix) of the rest of the market, which are identical in both designs. In the current design each ESS adds the variables $p_{e,t}, g_{e,t}$ with bounds only; each appears once, in the balance of its own interval (one block per interval). The SOC-based design adds the variables $s_{e,t}$ and the rows (A8)–(A10), with 4, 2 and 2 nonzeros respectively (one fewer at $t = 1$, where $s_{e,0}$ is data). Hence

$$\begin{aligned} &\text{rows}^{\text{CD}} = R_0, && \text{cols}^{\text{CD}} = C_0 + 2ET, && \text{nnz}^{\text{CD}} = Z_0 + 2ET, \\ &\text{rows}^{\text{SOC}} = R_0 + 3ET, && \text{cols}^{\text{SOC}} = C_0 + 3ET, && \text{nnz}^{\text{SOC}} = Z_0 + 10ET - 3E, \end{aligned} \tag{7.1}$$

i.e. $\Delta\text{rows} = 3ET$, $\Delta\text{cols} = ET$ and $\Delta\text{nnz} = 8ET - 3E$. The growth is linear in ET.

7.2 Coupling structure

Size is not the main difference; structure is. In the current design, the ESS columns of interval t appear only in the balance row of interval t, so the ESS add nothing that links intervals. In the SOC-based design, solving the recursion (A8),

$$s_{e,t} = \gamma_e^t s_{e,0} + \Delta T \sum_{\tau=1}^{t} \gamma_e^{t-\tau} (\alpha_e p_{e,\tau} - \beta_e g_{e,\tau}), \tag{7.2}$$

shows that the SOC limit at t depends on every decision in $1, \ldots, t$. Written in (p, g) only, (A9)–(A10) form a dense lower-triangular block with $2T(T + 1)$ nonzeros per ESS, i.e. $O(ET^2)$ in total. Keeping the variables s restores sparsity, $O(ET)$, but the problem can no longer be decomposed by interval. With ramps, the rest of the market is already coupled in time, but much more loosely: each ramp row links only two consecutive intervals of one unit.

7.3 No binary variables are needed

Preventing simultaneous charging and discharging in general requires binaries [10, 21]. In the model of this thesis they are not needed. By Remark 5.7, the equivalent bid–offer band of interval t has width

$$c_{e,t} - u_{e,t} = \epsilon_e + \zeta_e + (\beta_e - \alpha_e)\pi_{e,t} + \beta_e \underline{\nu}_{e,t} + \alpha_e \bar{\nu}_{e,t}, \tag{7.3}$$

and the same quantity governs the SOC-based design through (3.6). With round-trip losses ($\beta_e > \alpha_e$) and nonnegative rents, it is positive whenever

$$\pi_{e,t} > -\frac{\epsilon_e + \zeta_e}{\beta_e - \alpha_e} \qquad \text{for all } t, \tag{7.4}$$

which holds whenever stored energy has a nonnegative value. Simultaneous charging and discharging is then never optimal, and the LP relaxation of the exclusivity constraint is exact. It could fail only with strongly negative prices that make it profitable to waste energy through losses; in that case exclusivity is checked after the solve, as in [8].

7.4 Benchmark

Set-up. The benchmark (script `computational_burden.py`) uses a single-bus day of 24 hours at hourly, 15-minute and 5-minute resolution ($T = 24, 96, 288$, $\Delta T = 24/T$), with 20 thermal offer blocks of 80 MW (prices drawn uniformly in $[10, 120]$ \$/MWh) with ramp limits of 50% of their size per interval, a renewable unit with a solar-like profile, and a sinusoidal demand between 600 and 1400 MW. The E ESS are four-hour batteries with random sizes, scaled so that the total energy capacity is about 2000 MWh independently of E, with $\alpha = 0.9$, $\beta = 1.05$, $\gamma = 0.999$, $\epsilon = \zeta = 1$ \$/MWh, $s_0 = \bar{s}/2$ and $w = 45$ \$/MWh. The SOC-based design is solved first; the current design is then cleared with the equivalent bids (E) built from its multipliers, so that both problems give the same dispatch and prices (the prices agree to numerical precision in all instances). Both are solved with HiGHS [35] through SciPy, with the dual simplex and the interior-point method; times are medians of three runs on one core of the computing environment used for this report, and only their ratios are meaningful.

		rows		nonzeros		dual simplex (s)			interior point (s)		
T	E	SOC	CD	SOC	CD	SOC	CD	ratio	SOC	CD	ratio
24	1	1 016	944	2 581	2 392	0.013	0.008	1.7	0.018	0.013	1.3
24	10	1 664	944	4 714	2 824	0.021	0.007	3.0	0.035	0.014	2.6
24	50	4 544	944	14 194	4 744	0.044	0.020	2.2	0.158	0.028	5.7
24	200	15 344	944	49 744	11 944	0.199	0.031	6.5	0.788	0.044	18.0
96	10	6 776	3 896	19 186	11 536	0.055	0.027	2.1	0.163	0.056	2.9
96	50	18 296	3 896	57 466	19 216	0.387	0.067	5.8	1.262	0.086	14.6
96	200	61 496	3 896	201 016	48 016	1.505	0.117	12.8	6.299	0.142	44.4
288	50	54 968	11 768	172 858	57 808	2.163	0.155	14.0	7.378	0.289	25.5
288	200	184 568	11 768	604 408	144 208	44.699	0.398	112.3	47.532	0.607	78.3

Table 7.1: Problem size and solve times of the two designs. The differences in rows and nonzeros match $3ET$ and $8ET - 3E$ exactly.

Results. Table 7.1 and Figure 7.1 show that the size differences follow the formulas of Section 7.1 exactly. The time ratio grows with both E and T: from about 1.7 for one ESS at hourly resolution to 13 for 200 ESS at 15-minute resolution and 112 for 200 ESS at 5-minute resolution, where the SOC-based design takes 44.7 s against 0.4 s. The ratio is not monotone at small sizes, where times are dominated by overheads. The interior-point method shows the same trend.

Discussion. Baldick argues that the additional burden is surmountable, citing work that represents well over 100 ESS in a system of the size of MISO within day-ahead time limits [8, §4.8], [10]. The benchmark does not contradict this: absolute times remain small for this simple

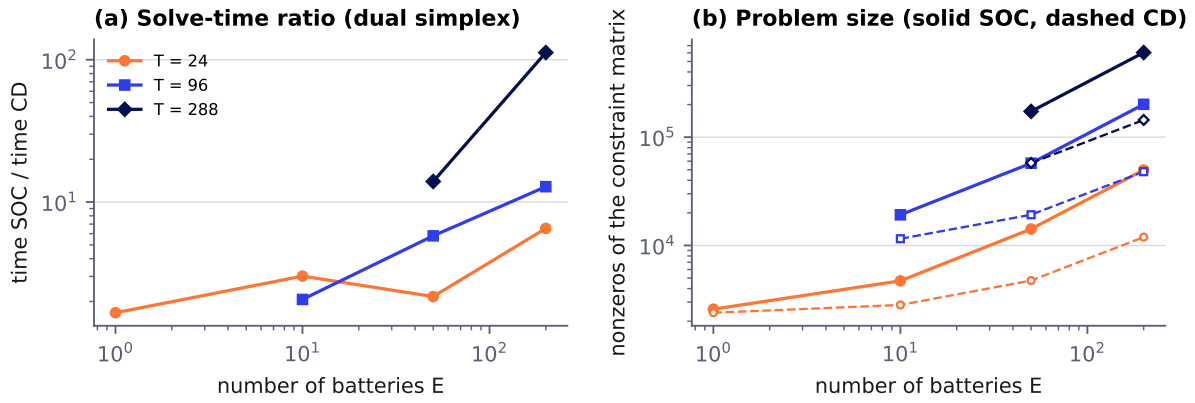


Figure 7.1: (a) Ratio of solve times (SOC-based over current design) with the dual simplex. (b) Nonzeros of the constraint matrix.

system. It shows, however, that the burden grows faster than that of the current design, and that the growth comes from the intertemporal coupling. For this thesis, the main consequence is methodological: the burden matters much less *after* the market has cleared. The SOC-based design is therefore well suited as an *ex post* benchmark against which the outcome of the current design is measured, which is how it is used in the experiments.

Note on the benchmark. The benchmark uses a single bus, linear costs and no unit commitment. With binaries for unit commitment and a large network, absolute times and ratios may be different. These cases are outside the scope of the thesis.

Chapter 8

Numerical verification

This chapter verifies every analytical result of Chapters 5 and 6 numerically (Stage 0 of the project plan). Section 8.1 describes the implementation, Section 8.2 the checks on Baldick's three cases, and Section 8.3 the check with several ESS.

8.1 Implementation

All models are implemented in Python with sparse matrices and solved as LPs with HiGHS [35] through SciPy. The code is organised in four scripts (Appendix B):

- `verify_equivalence.py` builds both clearing problems from a common case description (buses, lines in node pairs, generator blocks with ramps, elastic and inelastic demand, ESS), solves them, and extracts all multipliers with the sign and scaling conventions of Chapter 3. It evaluates the residual of every KKT row.
- `verify_price_based.py` implements the self-schedule $\mathcal{P}_e^{\mathrm{SS}}(\lambda)$, the polyhedron of Proposition 6.5, the admissible ranges $[\pi_{e,t}^{-}, \pi_{e,t}^{+}]$, the closed forms of Sections 6.3 and 6.4 and the pre-limit rule.
- `numerical_check.py` runs all checks on Baldick's three cases and produces Table 8.1 and Figures 8.1 and 8.2.
- `computational_burden.py` runs the benchmark of Chapter 7.

Earlier versions of the scripts were also tested on a three-bus network with ramping limits, elastic demand and two ESS with self-discharge and a positive minimum SOC; the analysis now concentrates on Baldick's cases, and the multi-bus tests will be resumed in Stage 3.

8.2 Baldick's three cases

The checks use the three single-bus cases of [8, §4.7]: the base case ($\bar{s} = 800\,\mathrm{MWh}$, $w = 40\,\$/\mathrm{MWh}$, Table 5.2), a higher terminal value ($w = 55\,\$/\mathrm{MWh}$) and a larger battery ($\bar{s} = 1200\,\mathrm{MWh}$), with all other data as in the base case. Table 8.1 lists the residual of every check. All residuals are at machine precision.

Observations. (i) In all three cases the CD prices equal $\lambda^\star$ for every admissible choice of bids, so the caveat of Remark 6.4(c) does not arise. (ii) The admissible ranges are non-trivial only at the intervals in which the ESS becomes empty or full, as predicted by Corollary 6.6. (iii) In the case $\bar{s} = 1200$, the residual of Lemma 6.2 is $7 \cdot 10^{-12}$ rather than zero; this is the tolerance of the LP solver on a larger objective value.

8.3 Several storage units

Corollary 6.15 is checked on Baldick's system with five heterogeneous ESS replacing the original one (script `multi_ess.py`). Their data are given in Table 8.2; the total energy capacity is

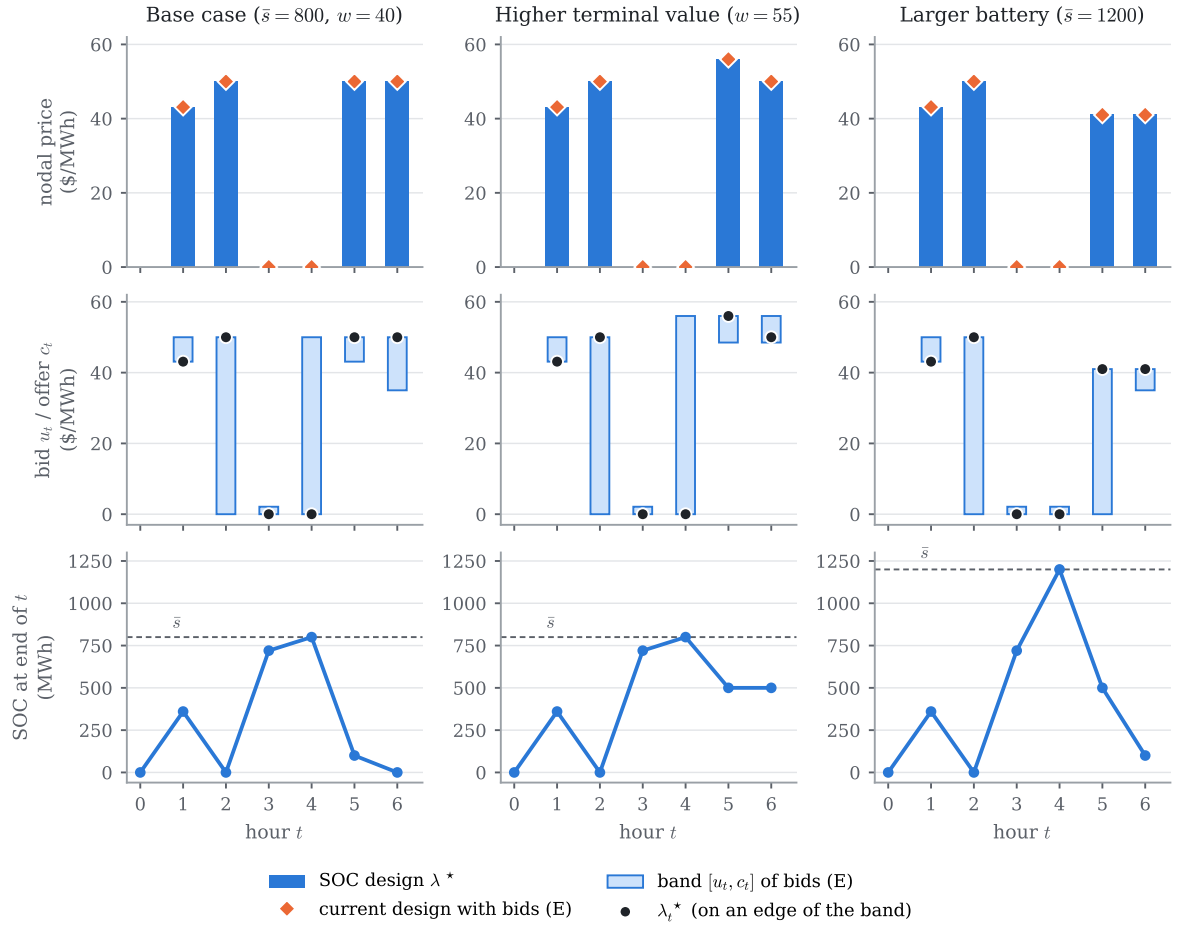


Figure 8.1: Same outcome in both designs, for Baldick's three cases. *Top*: nodal prices of the SOC-based design (bars) and of the current design cleared with the bids (E) (diamonds); they coincide in every interval. *Middle*: band between the bid u_t and the offer c_t. The price always lies on an edge of the band, which is the condition of Proposition 5.3. *Bottom*: SOC of the SOC-based design. The band is narrow while the battery moves freely and wide where an SOC limit binds. The CD solver may itself return another dispatch with the same cost and prices (ties at the band edges, Remark 5.6); the check is that the benchmark dispatch attains the CD optimum.

Check	Result	Base case	$w = 55$	$\bar{s} = 1200$
KKT residual of $\mathcal{P}^{\mathrm{SOC}}$, (S1)–(S8)	solver	$3 \cdot 10^{-15}$	$3 \cdot 10^{-15}$	$3 \cdot 10^{-15}$
$\max_t \lvert\lambda_t^{\mathrm{CD}} - \lambda_t^\star\rvert$ with bids (E)	Proposition 5.1	$3 \cdot 10^{-15}$	$3 \cdot 10^{-15}$	0
$f^{\mathrm{CD}}(x^\star) - f^{\mathrm{CD}\star}$ (relative)	Proposition 5.1	0	0	0
$f^{\mathrm{SOC}\star} - (f^{\mathrm{CD}\star} + K)$ (relative)	Proposition 5.2	0	0	$2 \cdot 10^{-15}$
$\max_t \lvert\pi_t^{\mathrm{prices}} - \pi_t^\star\rvert$	Proposition 6.13	0	0	0
self-schedule optimum minus value of benchmark schedule	Lemma 6.2	0	0	$7 \cdot 10^{-12}$
range of π_t: polyhedron vs. dual optimal face	Proposition 6.5	$7 \cdot 10^{-15}$	$7 \cdot 10^{-15}$	$4 \cdot 10^{-15}$
CD KKT violation at $(x^\star, \lambda^\star)$, all extreme bids	Theorem 6.3	$7 \cdot 10^{-15}$	$7 \cdot 10^{-15}$	$7 \cdot 10^{-15}$

Table 8.1: Residual of each check (absolute values in \$/MWh unless marked relative). Theorem 6.3 is checked for every extreme point of the admissible ranges of Figure 8.2.

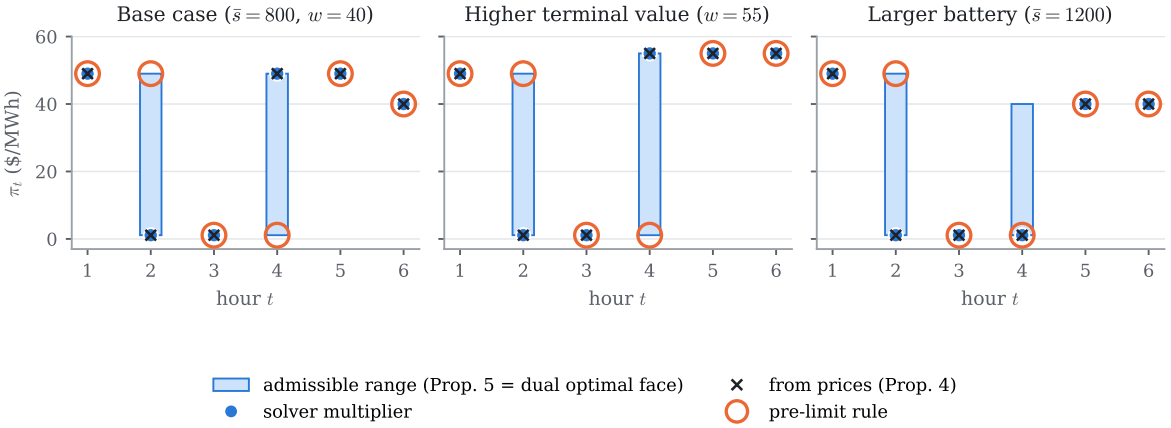


Figure 8.2: Value of stored energy π_t in Baldick's three cases. Bars: admissible range from Proposition 6.5, which coincides with the dual optimal face of the self-schedule. Dots: multipliers returned by the solver. Crosses: values rebuilt from prices with Proposition 6.13. Rings: pre-limit rule (6.13). The value is unique except at SOC limits, where every value in the range gives the same prices.

900 MWh.

ESS	α	β	γ	ϵ	ζ	$\underline{s}$	$\bar{s}$	s_0	$\bar{p}$	$\bar{g}$	w
1	0.90	1.00	1.000	1.0	1.0	0	250	0	300	300	40
2	0.85	1.05	0.995	2.0	1.5	0	150	50	80	80	45
3	0.92	1.02	1.000	0.5	0.5	20	200	100	60	60	35
4	0.88	1.00	0.990	1.5	1.0	0	120	0	100	100	50
5	0.95	1.03	1.000	3.0	2.0	0	180	90	45	45	30

Table 8.2: Five heterogeneous ESS used to check Corollary 6.15 (costs and w in \$/MWh, energies in MWh, powers in MW).

Procedure. (1) The SOC-based design is solved with all five ESS, giving $\lambda^\star = (0, 2.11, 0, 0, 50, 50)$ \$/MWh. (2) Each ESS, alone, solves its own self-schedule at $\lambda^\star$ and computes its polyhedron Φ_e; the admissible range of $\pi_{e,t}$ is non-trivial for every ESS in one to three intervals. (3) In each of 200 trials, every ESS independently picks a random point of its own polyhedron (a random convex combination of two random extreme points), and the current design is cleared with the five resulting bid sets.

Results. Over the 200 trials, the largest violation of the CD KKT conditions at the benchmark dispatch and prices is $1.7 \cdot 10^{-14}$ \$/MWh, the largest relative gap between the CD cost of the benchmark dispatch and the CD optimum is $1.3 \cdot 10^{-16}$, and the largest difference between the CD prices returned by the solver and $\lambda^\star$ is $7.1 \cdot 10^{-15}$ \$/MWh. The gap of Lemma 6.2 is at most $9 \cdot 10^{-13}$ for every ESS. The equivalence therefore holds with several ESS choosing their multipliers independently, as Corollary 6.15 states.

Chapter 9

Experimental design: the cost of imperfect price information

Status: implemented; design decisions pending. This chapter describes the experiments that answer RQ3, as they are implemented in the experiment code (Appendix B). Stage 0 has been run and its gate passes; Stage 1 (two storage owners) is set up, but its first check has raised two design questions that must be agreed with the supervisors before the full grid is run (Chapter 10). The open decisions are listed on the status page at the beginning of the report.

Recall: what this chapter uses.
Equivalent-bid map: $\Phi_e(\theta_e, \lambda)$, the bids of a competitive ESS at prices λ (Theorem 6.3 and Proposition 6.5). **Pre-limit rule**: $\pi_t^R = \min\{\max\{\pi_{t-1}^R/\gamma, \pi_t^-\}, \pi_t^+\}$ (6.13).
Technical-data-only bids (Corollary 5.5): $u_t = \alpha\gamma^{T-t}w - \epsilon$, $c_t = \zeta + \beta\gamma^{T-t}w$.
Quantities: when λ_t equals an ESS bid or offer, the CD is indifferent to the ESS quantity (Remark 5.6).

9.1 Objective

The analytical chapters establish what perfect information achieves: with the true prices $\lambda^\star$, every ESS can build bids that reproduce the SOC-based outcome, also with many ESS (Theorem 6.3 and Corollary 6.15); if no SOC limit binds, technical data alone suffice (Corollary 5.5). They leave open what imperfect information costs. The experiments answer four questions:

1. How much welfare is lost when the bids are built from a forecast $\hat{\lambda}$ instead of $\lambda^\star$?
2. Which properties of the error matter most: its size, its persistence over the day, its bias, or the share of it that is common to all ESS?
3. When is a forecast worth more than no forecast at all, i.e. than bids built from technical data only?
4. How much of the loss is due to price information and how much to quantity information?

The welfare gap between the two designs is the value of the SOC-based reform under a given quality of price information: if it is small, well-constructed interval bids are enough.

9.2 Structure of one experiment

Every experiment compares one benchmark with many realised outcomes. The SOC-based design is solved once per test system and gives the yardstick: prices $\lambda^\star$, schedules, welfare $SW^\star$ and the storage value V. Each *draw* then perturbs the prices that the storage owners believe, lets each owner build its bids from them, clears the current design with the true system, and repairs in real time what the ESS cannot deliver. Table 9.1 lists the optimisation problems involved.

Three principles hold throughout.

	Problem	**Role**
M1	SOC-based design $\mathcal{P}^{\mathrm{SOC}}$	benchmark, solved once: $\lambda^\star$, $(p^\star, g^\star, s^\star)$, multipliers, $SW^\star$
M2	current design $\mathcal{P}^{\mathrm{CD}}$ with the bids (E)	equivalence check (Chapter 8)
M3	self-schedule $\mathcal{P}_e^{\mathrm{SS}}(\lambda)$	each owner's own plan at the prices it uses (Section 9.4)
M4	polyhedron of Proposition 6.5	valid multipliers for that plan; a selection rule picks one
M5	day-ahead clearing of the CD	one merit-order auction per interval, true system, all ESS bids, load shedding at VOLL
M6	real-time repair of each ESS	closest SOC-feasible schedule to the day-ahead schedule
M7	generator redispatch	covers the difference at offer price plus a premium ψ, shedding at VOLL

Table 9.1: Optimisation problems of one experiment draw. All are linear programs solved with HiGHS.

1. *One welfare yardstick.* The benchmark and every realised outcome are evaluated with the same welfare function (9.7), always with the true ESS parameters, never with the bids.
2. *Errors enter only the owners' bids.* The forecast $\hat{\lambda}_e$ (and, with several owners, the estimate $\hat{w}_e$ of the terminal value) is used by owner e to build its bid. The clearing, the real-time stage and the welfare use the true system.
3. *Reproducible draws.* The random numbers of draw d depend only on d (Section 9.6), so every setting faces the same error shapes and any single draw can be recomputed on its own.

With the six-interval test systems of Chapter 10, one draw takes a few hundredths of a second.

9.3 Bid families

Four families of ESS bids are compared, ordered by the information they require (Table 9.2).

Family	**What the ESS knows**	**How the bids are built**	**Role**
$\mathcal{B}^\star$	the benchmark multipliers and quantities	(E) with $(\pi^\star, \bar{\nu}^\star, \underline{\nu}^\star)$; plan = benchmark schedule	gate: $\Delta SW = 0$ (check of H1)
$\mathcal{B}^{\star p}$	true prices $\lambda^\star$	$\Phi_e(\theta_e, \lambda^\star)$ with a selection rule; plan = own self-schedule at $\lambda^\star$	isolates the quantity-information loss
$\mathcal{B}^{\mathrm{fcst}}$	a forecast $\hat{\lambda}_e$ (and $\hat{w}_e$)	$\Phi_e(\theta_e, \hat{\lambda}_e)$ with a selection rule; plan = own self-schedule at $\hat{\lambda}_e$	main object of the experiments
$\mathcal{B}^{\mathrm{priv}}$	technical data only (and $\hat{w}_e$)	(9.1), full capacity at the bid and offer	lower bound: "no forecast"

Table 9.2: Bid families.

$\mathcal{B}^\star$ and $\mathcal{B}^{\mathrm{priv}}$ bracket every result. A forecast is useful only where $\mathcal{B}^{\mathrm{fcst}}$ lies between them, and the comparison of $\mathcal{B}^{\mathrm{fcst}}$ with $\mathcal{B}^{\mathrm{priv}}$ gives the value of a forecast. The technical-data-only bids,

$$u_{e,t} = \alpha_e \gamma_e^{T-t} w_e - \epsilon_e, \qquad c_{e,t} = \zeta_e + \beta_e \gamma_e^{T-t} w_e, \tag{9.1}$$

are exact if no SOC limit binds (Corollary 5.5), and also exact in a current design that enforces the SOC (Remark 5.8). In a current design without SOC enforcement they are the bids of an owner who prices stored energy at its terminal value and has no information on intra-day prices.

Selection rule. Where the polyhedron of Proposition 6.5 leaves a choice of multipliers, the *pre-limit* rule (6.13) is used by default. As an alternative, the *solver* rule takes the vertex of the polyhedron that minimises $\sum_t \pi_t$, which mimics the arbitrary multipliers that an LP solver returns. Comparing the two tests H4.

9.4 The owner's plan and a tie-breaker

The quantities of $\mathcal{B}^{\star p}$ and $\mathcal{B}^{\text{fcst}}$ come from the owner's own plan, the optimal schedule of $\mathcal{P}_e^{\text{SS}}$ at the prices it uses. At the true prices this problem is often degenerate: in Baldick's base case, for instance, a round trip between the first two intervals is exactly break-even and energy is free in two intervals, so many plans give the same profit (Remark 6.4). Which of them the solver returns is arbitrary, yet they are not equally good for the system.

The plan is therefore made unique by a throughput tie-breaker: the self-schedule is solved with the additional objective term

$$-\kappa^{\text{tb}}\,\Delta T \sum_t (p_{e,t} + g_{e,t}), \qquad \kappa^{\text{tb}} = 10^{-3}\ \$/\text{MWh}, \tag{9.2}$$

which selects, among the plans that are optimal for the true profit, one with the largest throughput. The term is a tie-breaker only: κ^{tb} is halved until the plan is still optimal for the true objective, so it never trades profit for activity, and it does not enter any welfare calculation. Since the plan is optimal for $\mathcal{P}_e^{\text{SS}}$, Proposition 6.5 applies to it unchanged. Its effect is discussed in Section 10.4.

9.5 Quantity rule

The problem. With equivalent bids, in every interval in which the ESS trades a partial quantity, the clearing price lands exactly on its bid ($\lambda_t = u_{e,t}$) or offer ($\lambda_t = c_{e,t}$). At that price the operator is indifferent to the ESS quantity: any quantity between zero and the block size gives the same cost (Remark 5.6). The solver then picks a quantity arbitrarily, which may be one that the ESS cannot deliver.

The rule. Each ESS splits its bid and offer of every interval into two blocks:

$$(u_{e,t} + \delta,\ p^{\circ}_{e,t}),\ \ (u_{e,t} - \delta,\ \bar{p}_e - p^{\circ}_{e,t}); \qquad (c_{e,t} - \delta,\ g^{\circ}_{e,t}),\ \ (c_{e,t} + \delta,\ \bar{g}_e - g^{\circ}_{e,t}), \tag{9.3}$$

where $(p^{\circ}_e, g^{\circ}_e)$ is the ESS's own plan and $\delta = 0.01\,\$/\text{MWh}$. The planned charging quantity is bid slightly *above* $u_{e,t}$, so the ESS is slightly keener to buy it; the rest of its capacity is bid slightly below. Symmetrically, the planned discharging quantity is offered slightly *below* $c_{e,t}$ and is dispatched first. If the price ends at $u_{e,t}$, the first block clears fully and the second does not, so the ESS receives exactly its plan, while the price moves by at most δ. By Proposition 5.3, these blocks keep the plan on the bid and offer curves. $\mathcal{B}^{\text{priv}}$ uses a single full-capacity block per side.

Example. In Baldick's base case the plan selected by (9.2) charges 222.2 MW in the first interval, where $u_1 = \lambda_1^{\star} = 43.1\,\$/\text{MWh}$. The ESS bids (43.11 \$/MWh, 222.2 MW) and (43.09 \$/MWh, 777.8 MW). Only 100 MW clear: that is the renewable surplus of the interval,

and beyond it the next supply is the thermal unit at 50 \$/MWh, above the bid. The benchmark also charges 100 MW there; the plan and the market disagree, and the market prevails.

Why the plan may differ from the benchmark. The self-schedule is often degenerate where prices are flat: many schedules give the ESS the same profit (Remark 6.4(b)). They are not equally good for the system, because the system needs the ESS in specific amounts in the intervals where it sets the price. An interval bid cannot express this, and no price information can recover it. This is why $\mathcal{B}^{\star p}$ is part of the design.

Several owners. The second block offers all the remaining power, whatever energy the ESS holds. With one ESS this is harmless. With several owners it is not: two owners' blocks can cross, and the market then clears a trade between the two ESS that neither can deliver. This is the first finding of Stage 1 (Section 10.5.2); how to change the rule is an open decision.

9.6 Error model

The forecast used by ESS e is

$$\hat{\lambda}_{e,t} = \max(\underline{\lambda},\ \lambda^{\star}_t + b\,\bar{\lambda} + \sigma\, z_{e,t}), \qquad z_{e,t} = \sqrt{\omega}\,\chi_t + \sqrt{1-\omega}\,\eta_{e,t}, \tag{9.4}$$

where $\bar{\lambda}$ is the mean of $\lambda^{\star}$ over the horizon and $\underline{\lambda}$ is a price floor. The size σ is given either as a fraction of $\bar{\lambda}$ or directly in \$/MWh (the latter is used in Stage 0); the bias b is a fraction of $\bar{\lambda}$. The common component χ_t and the ESS-specific components $\eta_{e,t}$ are independent AR(1) processes with unit variance,

$$x_t = \rho\, x_{t-1} + \sqrt{1-\rho^2}\,\upsilon_t, \qquad \upsilon_t \sim \mathcal{N}(0,1) \text{ i.i.d.}, \qquad x_1 \sim \mathcal{N}(0,1), \tag{9.5}$$

where x stands for χ and for each η_e, each with its own shocks υ_t. Hence $z_{e,t}$ has unit variance for every ρ and ω, and $\mathrm{corr}(z_{e,t}, z_{e',t}) = \omega$ for $e \neq e'$. The errors of all owners are drawn together, once per draw, so that χ is truly shared.

Parameter	Controls	Values
σ	size of the error	0, 2.5, 5, 10, 20, 40% of $\bar{\lambda}$; Stage 0: 0, 1, 2, 5, 10, 20 \$/MWh
ρ	persistence: interval-to-interval noise ($\rho = 0$) vs. level errors ($\rho = 0.9$)	0, 0.5, 0.9
b	systematic bias	-10, 0, $+10$% of $\bar{\lambda}$
ω	share of the error common to all ESS	0, 0.5, 1
$\underline{\lambda}$	price floor	0 \$/MWh
σ_w	error of the owner's terminal value (Section 9.6.1)	0, 2, 5, 10 \$/MWh

Table 9.3: Parameters of the error model.

Why these parameters. Each parameter isolates one property of a forecast (Figure 9.1). The size σ answers how accurate a forecast must be before bids built on it beat technical-data-only bids. The persistence ρ separates errors in the shape of the price curve, which change when the ESS trades, from errors in its level, which change how much stored energy is worth. The bias b represents a systematically optimistic or pessimistic forecaster. The common share ω represents ESS owners that use the same forecast or the same input data: they make the same mistake

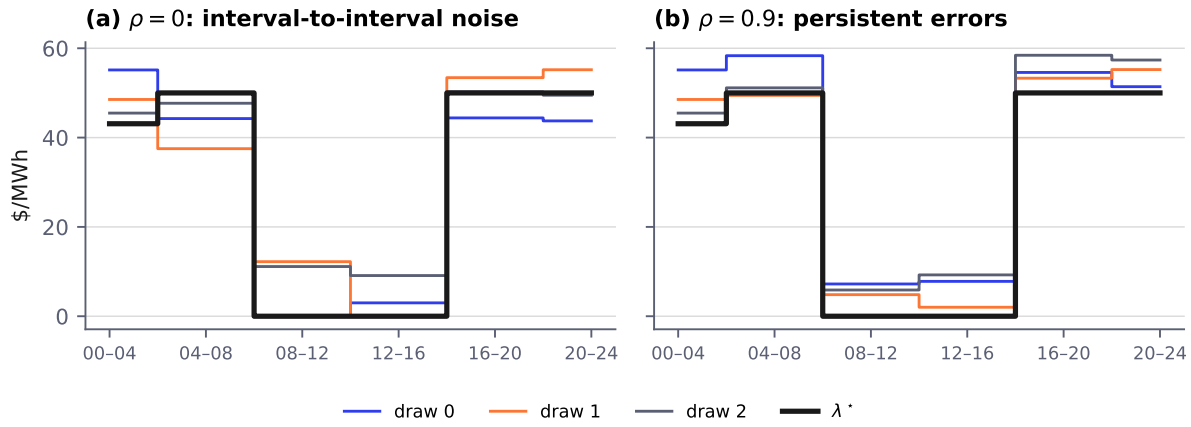


Figure 9.1: Forecasts generated by (9.4) with $\sigma = 10\,\$/\text{MWh}$ around the benchmark price of Baldick's base case (three draws, price floor 0). Interval-to-interval noise changes *when* the ESS should trade; persistent errors shift the whole day and change *how much* stored energy is worth. In the two free intervals the floor keeps every forecast at or above zero.

together and move the price together. The model is additive and Gaussian, which is simple, controllable and reproducible. Its limitations (no heavy tails, no dependence of the error on the price level) are addressed by the calibration below.

Price floor. A forecaster does not predict prices that the market cannot produce. In the test systems the cheapest supply is curtailable renewable energy offered at $0\,\$/\text{MWh}$, so $\lambda^\star \geq 0$ and the forecast is floored at $\underline{\lambda} = 0$. The floor matters in the intervals where $\lambda^\star_t = 0$ exactly: there an unbounded Gaussian error is negative with probability one half whatever σ is. A negative forecast makes the charge bid negative, so the ESS buys nothing in the free intervals and may reach a scarce interval empty. In early runs without the floor, about one draw in five ended in load shedding at every σ, and this unphysical event hid the effect of the error size. For a market with negative prices, $\underline{\lambda}$ is set to its minimum clearing price.

Seeds and common random numbers. The random numbers of draw d are generated from a seed that depends only on a base seed and on d, not on the setting. Draw d therefore has the same error shape z in every setting: changing σ, b or ω scales, shifts or recombines the same random numbers, and changing ρ reuses them through (9.5). Settings can then be compared draw by draw (paired differences), which removes most of the sampling noise from comparisons between settings (Section 10.3).

9.6.1 Error in the owner's terminal value

With several owners (Stage 1 onwards), each owner also estimates what its stored energy is worth at the end of the day:

$$\hat{w}_e = \max\bigl(0,\ w_e + \sigma_w\, z_{w,e}\bigr), \qquad z_{w,e} \sim \mathcal{N}(0,1) \text{ i.i.d.}, \tag{9.6}$$

one value per ESS and draw, constant over the day and independent of the price error, with its own stream of random numbers. The terminal value is the owner's view of the following day; the SOC-based design asks the owner to submit it, while in the current design it never reaches the operator but fixes $\pi_{e,T}$ in the owner's self-schedule. $\hat{w}_e$ enters only the owner-built bids ($\mathcal{B}^{\text{fcst}}$ and $\mathcal{B}^{\text{priv}}$); the benchmark and the welfare keep the true w_e.

9.7 Calibration from historical data

The parameters of Table 9.3 can be estimated from historical data, as long as one first fixes *whose* forecast is meant. Realised day-ahead prices are public, for instance for the Danish bidding zones DK1 and DK2 from the Energi Data Service of Energinet or the ENTSO-E Transparency Platform, but price forecasts generally are not: market operators publish load and renewable forecasts, not price forecasts, and commercial forecasts are proprietary. The standard approach is therefore to build the forecaster and backtest it.

Procedure.

1. For each day of a historical period, forecast the 24 day-ahead prices using only information available before gate closure (the day before, around noon), on a rolling window.
2. Use two forecasters that represent two types of ESS owner: a *naive* one (the price of the same hour one week earlier, or of the previous day), and a *state-of-the-art* one (LEAR or DNN from the open benchmark of [34]).
3. Store the residuals $e_t = \lambda_t - \hat{\lambda}_t$ and estimate:
   - the bias b as the mean of e_t, relative to the mean price;
   - the size σ as the standard deviation of e_t, relative to the mean price;
   - the persistence ρ as the lag-one autocorrelation of e_t within the day;
   - the common share ω as the correlation between the residuals of two *different* forecasters on the same day, since a single forecaster cannot reveal how similarly different owners err.

Expected values. The benchmark of [34] suggests σ of roughly 5–15% of the mean price for state-of-the-art forecasts in calm periods (Section 2.6), which falls inside the grid of Table 9.3. Larger errors are expected in volatile periods, so the choice of the historical period matters.

Empirical errors. Besides the parametric grid, Stage 3 adds an *empirical* error setting: whole historical days of residuals are resampled (a day-block bootstrap) and added to $\lambda^\star$. This keeps the real autocorrelation, heavy tails and dependence on the hour without assuming a distribution. The parametric grid then explains how the loss depends on each property of the error, and the bootstrap gives the cost of realistic forecast quality.

Caveats. (i) The experiments perturb $\lambda^\star$, the prices of the SOC-based design, while historical residuals are errors on prices of the current market. Using one for the other assumes that the error process does not depend on the design. (ii) Historical prices already reflect the current behaviour of storage. With one ESS this effect is small; with a large fleet it should be discussed.

9.8 Pipeline

One Monte Carlo draw proceeds as follows.

1. *Benchmark* (once per instance). Solve the SOC-based design: prices $\lambda^\star$, schedule $x^\star$, welfare $SW^\star$, and the welfare without storage SW^{noESS}.
2. *Forecast.* Draw the errors and build $\hat{\lambda}_e$ (and $\hat{w}_e$) for every ESS by (9.4) and (9.6).

3. *Bids.* Each ESS solves $\mathcal{P}_e^{\mathrm{SS}}(\hat{\lambda}_e)$ with the tie-breaker (9.2), obtaining its plan (p_e°, g_e°); computes the polyhedron of Proposition 6.5; selects the multipliers with the pre-limit rule (or the solver rule); forms the bids (E); and splits them with the quantity rule (9.3).
4. *Day-ahead clearing.* The operator clears the current design with the true system data and these bids, obtaining a day-ahead schedule and prices λ^{CD}. Without ramps or a network the intervals are independent, and each is cleared as a merit-order auction. Load shedding at VOLL $= 1000\,\$/\mathrm{MWh}$ keeps the clearing feasible.
5. *Real time.* The day-ahead schedule of an ESS may violate its SOC limits. Each ESS delivers the SOC-feasible schedule closest to its day-ahead schedule, in the sense of the smallest total absolute deviation (an LP over (A8)–(A12)); the deviation is the *undeliverable energy.* The rest of the system covers the difference: generators move from their day-ahead schedule at their offer price plus a premium ψ per MWh moved up or down, and any shortage is shed at VOLL.
6. *Evaluation.* Compute the realised welfare, the welfare loss, the undeliverable and shed energy, the price distortion and the ESS profit.

Realised welfare. With the true ESS parameters (not the bids),

$$\begin{aligned} SW^{\mathrm{real}} = -\Big(&\text{generation cost} + \psi \cdot \text{redispatched energy} + \text{VOLL} \cdot \text{shed energy} \\ &+ \text{ESS degradation cost} - \textstyle\sum_e w_e\, s_{e,T}\Big). \end{aligned} \tag{9.7}$$

The benchmark welfare $SW^\star$ is evaluated with the same formula; it has no redispatch and no shedding.

Role of ψ. The premium ψ represents the cost of adjusting the system after the day-ahead market. With $\psi = 0$, real-time adjustment is free, and an ESS that commits to an undeliverable schedule causes no cost beyond the generation it shifts; technical-data-only bids then look artificially good (Section 10.3). The premium is therefore a parameter of the experiments: $\psi \in \{0, 10, 50\}\,\$/\mathrm{MWh}$, i.e. frictionless adjustment, a value close to real imbalance penalties, and a stress test.

9.9 Metrics

Welfare loss. The main metric is

$$\Delta SW = SW^\star - SW^{\mathrm{real}} \geq 0, \qquad \text{reported as } \frac{\Delta SW}{V} \times 100\%, \quad V := SW^\star - SW^{\mathrm{noESS}}, \tag{9.8}$$

i.e. as a percentage of the value that storage creates in the benchmark. The loss is never negative: the benchmark maximises welfare over all feasible schedules, and the realised outcome is a feasible schedule evaluated with additional costs (the code checks this for every draw). A loss above 100% means that the system would have been better off without the ESS. Normalising by V makes results comparable across systems, storage sizes and days, but V itself depends on the system: where storage mainly avoids load shedding, V is large and the percentages small (Section 10.1).

Decomposition. Since $\mathcal{B}^{\star p}$ uses the true prices, its loss is due to quantities alone. The difference between the losses of $\mathcal{B}^{\mathrm{fcst}}$ and $\mathcal{B}^{\star p}$ is therefore the additional loss caused by the price

error:

$$\Delta SW(\mathcal{B}^{\text{fcst}}) = \underbrace{\Delta SW(\mathcal{B}^{\star p})}_{\text{quantity information}} + \underbrace{\Delta SW(\mathcal{B}^{\text{fcst}}) - \Delta SW(\mathcal{B}^{\star p})}_{\text{price forecast}}. \tag{9.9}$$

The identity is an accounting split, not a separate result: the two effects can interact, and the split depends on the order in which they are added. This should be stated whenever it is reported.

Secondary metrics.

- undeliverable energy (MWh): total absolute deviation between the day-ahead and the delivered ESS schedule;
- shed energy (MWh);
- price distortion: mean $|\lambda_t^{\text{CD}} - \lambda_t^{\star}|$;
- ESS profit at the day-ahead prices compared with the benchmark: who bears the loss;
- value of a forecast: the error size σ^* at which $\mathcal{B}^{\text{fcst}}$ and $\mathcal{B}^{\text{priv}}$ lose the same welfare;
- loss by interval, split into intervals with and without a binding SOC limit in the benchmark (test of H2).

For every setting the mean with a 95% confidence interval, the median and the 95th percentile are reported. Two settings are compared through the paired difference of their losses, draw by draw.

9.10 Hypotheses and how they are tested

Table 9.4 maps each hypothesis of Section 1.4 to its test and to its status after Stage 0.

	Hypothesis and test	**Status after Stage 0 (Chapter 10)**
H1	True prices and quantities give no loss: $\Delta SW(\mathcal{B}^{\star}) = 0$.	confirmed, with one and with two ESS
H2	The loss grows with σ and comes from intervals where SOC limits bind: loss by interval.	grows with σ (paired steps resolved); split by interval to do
H3	Persistent errors cost more than noise: $\rho = 0.9$ vs. $\rho = 0$.	not yet tested ($\rho = 0$ only)
H4	The pre-limit rule loses less than the solver rule.	at $\lambda^{\star}$ the rule makes no difference; under errors not yet tested
H5	Common errors cost more than independent ones: $\omega = 1$ vs. $\omega = 0$.	Stage 1 (two owners)
H6	A forecast beats technical data only below a threshold σ^* that grows with ψ.	not in the base case at $\psi = 10$: technical-data-only bids lose less at every σ; the ψ grid is needed

Table 9.4: Hypotheses, tests and status.

9.11 Stages and parameter grid

The experiments proceed in stages of increasing realism, so that the source of any change in the conclusions can be identified.

Stage 0

Gate: one ESS, Baldick's base case. $\mathcal{B}^{\star}$ must give zero loss; $\mathcal{B}^{\star p}$, $\mathcal{B}^{\text{priv}}$ at $\psi \in \{0, 10, 50\}$, and $\mathcal{B}^{\text{fcst}}$ with $\sigma \in \{0, 1, 2, 5, 10, 20\}$ \$/MWh, $\rho = 0$, $\psi = 10$, pre-limit rule, 150 draws. *Done* (Chapter 10).

Stage 1

Two storage owners on the same system, with independent bids and both errors of Section 9.6. Proposed grid: $\sigma \in \{0, 1, 2, 5, 10, 20\}$ \$/MWh $\times$ $\sigma_w \in \{0, 2, 5, 10\}$ \$/MWh $\times$ $\omega \in \{0, 0.5, 1\}$, i.e. 72 settings of 150 draws, with $\rho = 0$, no bias, $\psi = 10$ and the pre-limit rule, plus the deterministic families. *Set up; on hold* until the quantity rule and the real-time priority are agreed (Section 10.5.2). An alternative or complement is the full one-ESS grid σ (6) $\times$ ρ (3) $\times$ b (3) $\times$ selection rule (2) $\times$ ψ (3) $= 324$ settings of 500 draws, which tests H2–H4 and H6 without the interaction between owners.

Stage 2

A fleet of ESS (5 to 20) at 10 to 40% of peak load: common share ω, fleet size and interaction through the price. Needs data for a heterogeneous fleet.

Stage 3

Realistic system and errors. A single-bus reduction of an area of the RTS-GMLC system [36] with many days, higher renewable shares and the calibrated and empirical errors of Section 9.7; a simplified multi-node version for congestion effects. A larger system also removes the exact ties that dominate the six-interval cases (Section 10.4).

The numerical verification of the analytical results (Chapter 8) precedes all stages.

Chapter 10

First experiments: one ESS and two storage owners

Status: Stage 0 complete; Stage 1 on hold. This chapter reports the experiments run so far with the experiment code (Appendix B). Stage 0 runs the whole pipeline of Chapter 9 on Baldick's base case with one ESS and passes its gate. Stage 1 splits the ESS between two owners; its first check, at zero forecast error, shows that the quantity rule must be revised before the error grid is run. All numbers are reproduced by the scripts and the Stage 0 notebook of the code.

10.1 Test system and benchmark

Both stages use Baldick's single-bus base case (Appendix A): six intervals of four hours, a renewable unit at 0 \$/MWh, a thermal unit of 100 MW at 50 \$/MWh, inelastic demand and, in Stage 0, one ESS of 800 MWh with $w = 40$ \$/MWh. The system is small enough that every number of an experiment draw can be checked by hand, and its benchmark solution is already verified (Chapter 8). Its optimal solution is shown in Figure 3.1 and Table 5.2; the quantities used below are

Interval	00–04h	04–08h	08–12h	12–16h	16–20h	20–24h
$\lambda_t^\star$ (\$/MWh)	43.1	50	0	0	50	50
$s_t^\star$ (MWh)	360	0	720	800	100	0
$\pi_t^\star$ (\$/MWh)	49	1.11	1.11	49	49	40
$\bar{\nu}_t^\star$, $\underline{\nu}_t^\star$ (\$/MWh)	0, 0	0, 47.9	0, 0	47.9, 0	0, 0	0, 9

with $SW^\star = -19{,}449$ \$, $SW^{\text{noESS}} = -360{,}000$ \$ and a storage value $V = 340{,}551$ \$ per day.

What the storage value measures here. Without the ESS, the interval 16–20h has 25 MW of renewable and 100 MW of thermal capacity for 200 MW of demand: 75 MW are shed for four hours, at a cost of 300,000 \$. Most of V is therefore avoided load shedding, not price arbitrage, and losses expressed in % of V are small in this system. A second test system without a reliability shortfall is needed before the percentages are compared across systems.

Ties. The ESS is large compared with the load (up to 1000 MW and 800 MWh against at most 300 MW of demand) and sets the price in most intervals. The true prices contain exact ties: the round trip between the first two intervals is exactly break-even ($(43.1 + \epsilon)/\alpha = 49 = 50 - \zeta$), and energy is free in two intervals. These ties shape every result below.

10.2 Stage 0: settings

- Deterministic settings: $\mathcal{B}^\star$ ($\psi = 0$) and $\mathcal{B}^{\text{priv}}$ at $\psi \in \{0, 10, 50\}$ \$/MWh.
- Stochastic settings: $\mathcal{B}^{\star p}$ and $\mathcal{B}^{\text{fcst}}$ with $\sigma \in \{0, 1, 2, 5, 10, 20\}$ \$/MWh, i.e. 0, 3.1, 6.2, 15.5, 31.1 and 62.2% of $\bar{\lambda} = 32.2$ \$/MWh; $\rho = 0$; no bias; one ESS, so ω does not apply; price floor 0.

- Pre-limit rule, tie-breaker $\kappa^{\text{tb}} = 10^{-3}$ \$/MWh, quantity rule with $\delta = 0.01$ \$/MWh, real-time premium $\psi = 10$ \$/MWh, VOLL $= 1000$ \$/MWh.
- 150 draws per stochastic setting, base seed 42, common random numbers: 16 settings and 1,506 draws, about a minute on a laptop.

10.3 Stage 0: results

Table 10.1 reports every setting and Figure 10.1 the forecast-based bids.

Setting	welfare loss (% of V)				undeliv.	$\lvert\Delta\lambda\rvert$	ESS profit
	mean	95% CI	median	p95	(MWh)	(\$/MWh)	(\$)
$\mathcal{B}^{\star}$	0.00		0.00		0	0.00	38,308
$\mathcal{B}^{\star p}$	1.30		1.30		40	1.51	31,843
$\mathcal{B}^{\text{fcst}}$, $\sigma = 0$	1.30		1.30		40	1.51	31,843
$\mathcal{B}^{\text{fcst}}$, $\sigma = 1$	5.80	5.30–6.29	4.47	9.31	976	1.21	30,244
$\mathcal{B}^{\text{fcst}}$, $\sigma = 2$	5.80	5.30–6.29	4.47	9.31	976	1.77	28,623
$\mathcal{B}^{\text{fcst}}$, $\sigma = 5$	6.22	5.63–6.80	4.47	9.31	1,002	3.44	22,224
$\mathcal{B}^{\text{fcst}}$, $\sigma = 10$	7.11	6.37–7.86	7.49	16.46	1,050	6.14	11,720
$\mathcal{B}^{\text{fcst}}$, $\sigma = 20$	7.85	7.07–8.64	8.22	16.46	1,091	11.18	−4,871
$\mathcal{B}^{\text{priv}}$, $\psi = 0 / 10 / 50$	0.00 / 3.09 / 15.43				1,051	17.52	−12,000

Table 10.1: Stage 0 on Baldick's base case ($\psi = 10$ \$/MWh unless stated, 150 draws for $\sigma > 0$; σ in \$/MWh). Undeliverable energy: total absolute deviation between the day-ahead and the delivered ESS schedule. $\lvert\Delta\lambda\rvert$: mean price distortion. ESS profit: day-ahead settlement at λ^{CD} minus degradation, mean over draws. No draw sheds load.

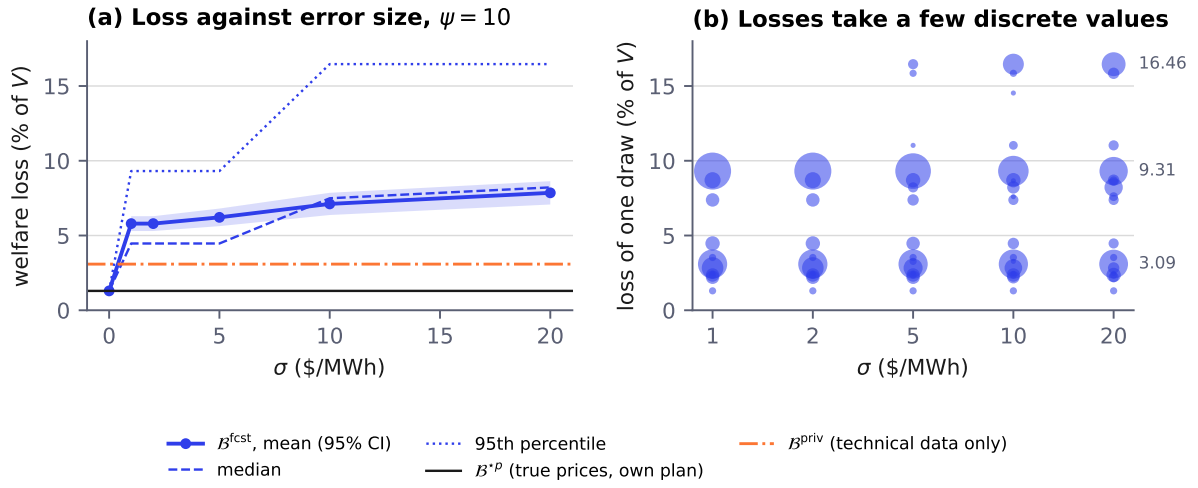


Figure 10.1: Stage 0. (a) Welfare loss of the forecast-based bids against the error size: mean with 95% confidence band, median and 95th percentile, with the losses of $\mathcal{B}^{\star p}$ and of the technical-data-only bids at $\psi = 10$ \$/MWh. (b) Loss of each draw; the area of a dot is proportional to the number of draws with that loss.

Paired comparison. Because all settings use the same 150 error shapes (Section 9.6), neighbouring error sizes can be compared draw by draw (Table 10.2). The paired confidence intervals are 2.5–3 times narrower than unpaired ones, and every step from $\sigma = 2$ upwards is resolved: a

larger error costs more. In 146 of the 150 draws the loss never decreases as σ grows; in 99 of them it is the same at every σ.

Step in σ (\$/MWh)	$1 \to 2$	$2 \to 5$	$5 \to 10$	$10 \to 20$
mean difference (points of V)	0.00	0.42	0.90	0.74
95% CI	[0.00, 0.00]	[0.13, 0.70]	[0.53, 1.27]	[0.39, 1.09]
draws in which the loss does not fall	100%	99%	100%	99%

Table 10.2: Stage 0: paired differences of the loss of $\mathcal{B}^{\text{fcst}}$ between neighbouring error sizes.

10.4 Stage 0: reading the results

The gate passes. With the benchmark multipliers and quantities, $\mathcal{B}^\star$ reproduces Baldick's dispatch through the whole pipeline (bid construction, day-ahead clearing, real-time repair and welfare) and loses nothing, to floating-point precision. This confirms H1 and the implementation. No setting gives a negative loss, and no ESS charges and discharges in the same interval.

Perfect prices with the owner's own plan: 1.30%. With the true prices but its own plan, the ESS loses 1.30% of V (4,411 \$). The plan selected by the tie-breaker charges 222.2 MW in the first interval and sells 200 MW in the second; the market accepts only the 100 MW of free surplus in the first interval, but in the second it takes 100 MW, the whole residual demand, while the 360 MWh stored allow only 90 MW: 40 MWh are undeliverable, and thermal generation covers them. In the evening, the day-ahead schedule sells 75 MW at 16–20h and 100 MW at 20–24h, where the benchmark sells 175 and 25 MW, so 25 MW more of thermal run in the scarce interval. Without the tie-breaker, the solver's own plan neither buys in the first interval nor sells in the second. Its spare-capacity block still buys the free surplus at 00–04h, but its offer at 04–08h sits at 50.01 \$/MWh and loses to the thermal unit at 50: the ESS carries 360 MWh it should have sold, and the loss is 6.32% of V. In both cases the loss is a coordination problem across intervals: buying at 00–04h and selling at 04–08h belong together, each is worth nothing to the ESS on its own, and an hourly market resolves the two ties independently. The SOC-based design sees both intervals and the SOC at once.

The selection rule does not matter at the true prices. At $\lambda^\star$ the plan leaves the ESS empty after 04–08h and full after 12–16h, so exactly two values of π are free, each anywhere in [1.11, 49] \$/MWh: the polyhedron of Proposition 6.5 projects to a square (Figure 10.2). Its corners include the benchmark multipliers, the pre-limit rule and the solver rule, and they produce visibly different bids. Yet all four give the same day-ahead schedule and the same loss of 1.30%: wherever their bids differ, the price stays on the same side of the bid. The whole $\mathcal{B}^{\star p}$ loss comes from the quantity plan, not from the bid prices, and no point of the polyhedron removes it.

Any error raises the loss to about 6%. At $\sigma = 1$ \$/MWh the mean loss jumps from 1.30% to 5.80%, and then grows slowly to 7.85% at $\sigma = 20$ \$/MWh. The jump is the cost of near-indifference: the tie-breaker resolves the exact ties of $\lambda^\star$, but an error of one dollar is already enough to make the ESS strictly prefer one of the tied plans, chosen by the sign of the error rather than by what the system needs. The tie-breaker changed the plan in none of the 750 forecast draws: a forecast error breaks the ties by itself.

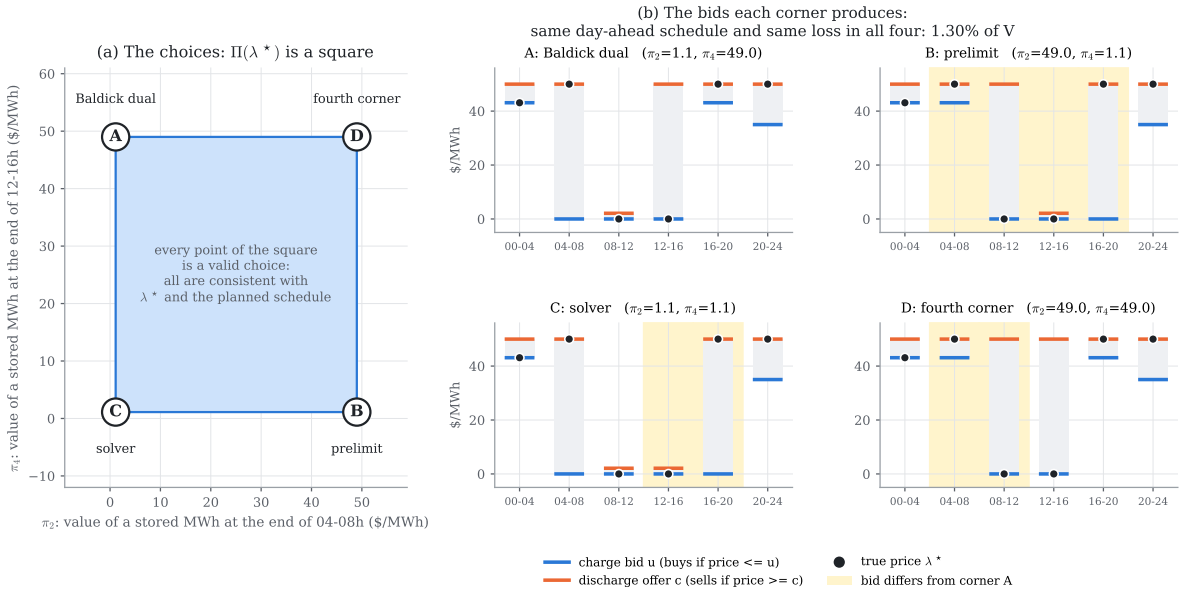


Figure 10.2: Multipliers at the true prices. (a) The two free values π_2, π_4 form a square of valid choices; A is the benchmark dual, B the pre-limit rule, C the solver rule. (b) The bids of each corner; shaded intervals differ from corner A. All four give the same day-ahead schedule and the same loss.

The shape of the error matters more than its size. In 99 of the 150 draws the loss is the same at every σ, and $\sigma = 1$ and $\sigma = 2\,\$/\text{MWh}$ give identical losses in every draw. Which plan the ESS chooses depends on which intervals the forecast ranks as cheaper or dearer, i.e. on the shape of the error; its size only matters once it is large enough to cross a threshold of the plan.

Losses take a few discrete values. The loss of a draw takes one of a small number of values (Figure 10.1b): 9.31% (the most frequent), 3.09%, 2.83%, 8.70%, 16.46% and a few others. This is a general feature of the experiments, not of this case only. Every problem of the chain is an LP with a vertex solution, so the ESS's day-ahead schedule changes only when the forecast crosses a threshold, and everything after the day-ahead schedule is deterministic. Grouping the draws by their day-ahead schedule gives a finite set of *regimes* k, each with a loss L_k and a probability p_k under the error model, and the mean loss decomposes exactly as $\mathbb{E}[\Delta SW] = \sum_k p_k L_k$. With common random numbers, the change of the mean between two settings is the net flow of draws between regimes. In an exploratory run with 400 error shapes, the loss was constant within every regime; there were 31 regimes in total, and between $\sigma = 1$ and $\sigma = 5\,\$/\text{MWh}$ 88% of the draws stayed in their regime. Most of the rise of the mean came from one flow: draws moving from a regime that keeps the morning charge and misses the morning sale (9.3% of V) to one that buys 200 MW from the thermal unit in the first interval (16.5%), which happens when the forecast of that interval is about $7\,\$/\text{MWh}$ too high. This reading is planned as part of the analysis of the full grid (Section 11.3).

Technical-data-only bids. With $\pi_t = w = 40\,\$/\text{MWh}$, the technical-data-only bids are $u_t = 35$ and $c_t = 41\,\$/\text{MWh}$ in every interval, at full power. They ignore the SOC limits and leave 1,051 MWh per day undeliverable. Their loss is exactly ψ times that energy: zero at $\psi = 0$, 3.09% at $\psi = 10$ and 15.43% at $\psi = 50\,\$/\text{MWh}$. The zero at $\psi = 0$ is specific to this system: the repaired schedule and the benchmark are forced through the same full SOC at 12–16h, and with no self-discharge the total generation and degradation then depend only on the totals, not on the timing. At $\psi = 10\,\$/\text{MWh}$ the technical-data-only bids lose less than the forecast-based bids at every error size, so there is no threshold σ^* in this case; forecast-based bids at $\psi = 0$ and 50

have not been run yet, so H6 remains open.

Decomposition. By (9.9), the 1.30% of $\mathcal{B}^{\star p}$ is the quantity-information part, and the price-forecast part grows from 4.5 to 6.6 points of V as σ goes from 1 to 20 \$/MWh.

Who bears the loss. The ESS's own day-ahead profit falls faster than welfare: from 38,308 \$ with $\mathcal{B}^{\star}$ to 30,244 \$ at $\sigma = 1$ and to $-4{,}871$ \$ at $\sigma = 20$ \$/MWh, while the mean price distortion grows from 1.2 to 11.2 \$/MWh. With larger errors the ESS moves prices against itself.

10.5 Stage 1: two storage owners

10.5.1 The case

Stage 1 keeps the system of Stage 0 and splits the 800 MWh ESS into two ESS of 400 MWh, A and B, owned separately and identical except for their terminal value: $w_A = 40$ and $w_B = 55$ \$/MWh. Total storage, and every other parameter, stay as in Stage 0, so any difference comes from having two owners.

The two values are placed on either side of a threshold of this system: 49 \$/MWh. Selling a stored MWh at the price of 50 earns $50 - \zeta = 49$, and the morning round trip buys at 43.1 and needs $(43.1 + \epsilon)/\alpha = 49$ to break even. An ESS with $w < 49$ sells its energy before the end of the day; with $w > 49$ it keeps it. A lies 9 \$/MWh below the threshold, B 6 \$/MWh above.

Benchmark. Table 10.3 and Figure 10.3 show the SOC-based design on this case. A is the trader: it runs the morning round trip (buy 100 MW, sell 90 MW, exactly break-even), fills up on free energy at 08–12h and empties at 16–20h, where it covers the 75 MW shortfall. B is the store: it fills on free energy at 12–16h and keeps its 400 MWh to the end of the day, because each MWh is worth 55 to it, more than the 49 that a sale would earn. The prices are those of Stage 0. Welfare is $SW^{\star} = -17{,}049$ \$ and $V = 342{,}951$ \$, both 2,400 \$ higher than in Stage 0, which is exactly 400 MWh $\times$ (55 − 49). Which ESS charges in which free interval is not unique; welfare and prices do not depend on that choice. With the benchmark multipliers and quantities of both ESS, $\mathcal{B}^{\star}$ again loses nothing.

Interval	00–04h	04–08h	08–12h	12–16h	16–20h	20–24h
Demand (MW)	100	200	300	300	200	100
Renewable used / available (MW)	200 / 200	100 / 100	411 / 500	411 / 500	25 / 25	0 / 0
Thermal (MW)	0	10	0	0	75	100
A: charge / discharge (MW)	100 / –	– / 90	111 / –	–	– / 100	–
A: SOC at the end (MWh)	360	0	400	400	0	0
B: charge (MW)	–	–	–	111	–	–
B: SOC at the end (MWh)	0	0	0	400	400	400
$\lambda^{\star}_t$ (\$/MWh)	43.1	50	0	0	50	50
$\pi^{\star}_{A,t}$ (\$/MWh)	49	49	1.11	1.11	49	40
$\pi^{\star}_{B,t}$ (\$/MWh)	49	1.11	1.11	55	55	55

Table 10.3: Stage 1: benchmark solution of the SOC-based design with two ESS.

Errors. Each owner builds its bids from its own price forecast, with a common and an owner-specific part (ω), and from its own estimate $\hat{w}_e$ of its terminal value (9.6). Because of the

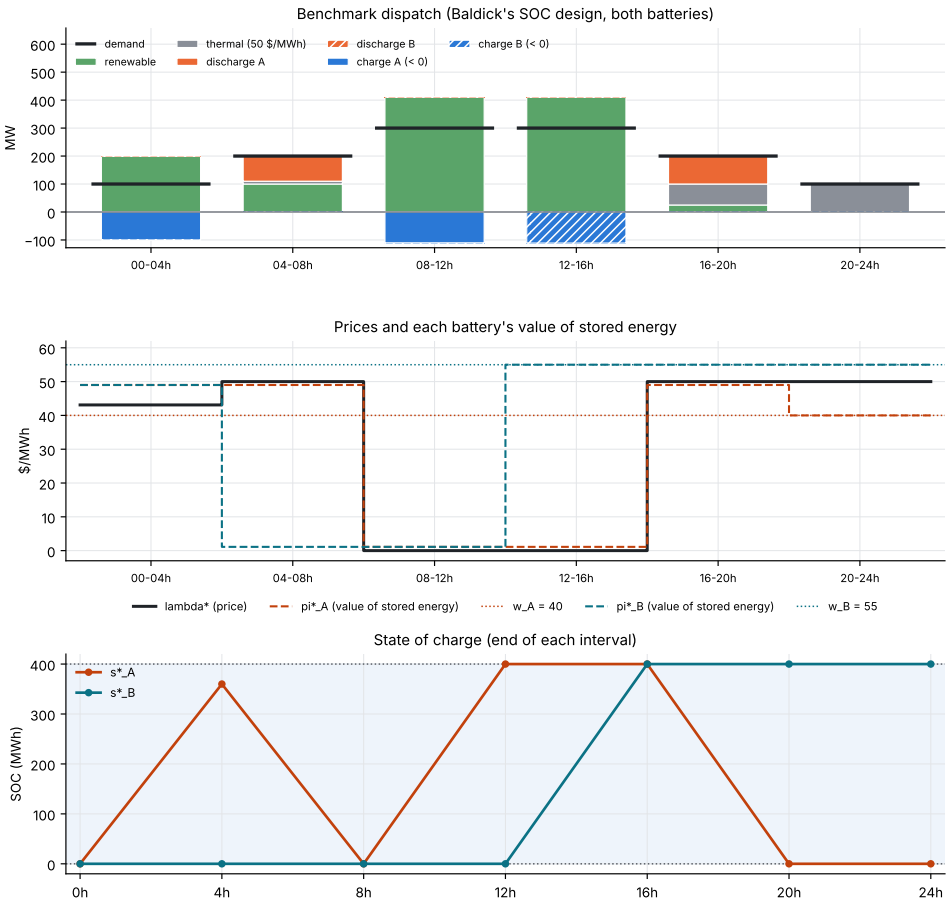


Figure 10.3: Stage 1: dispatch, prices and values of stored energy, and SOC of the two ESS in the SOC-based design.

threshold at 49 \$/MWh, an estimate on the wrong side of it flips the owner's end-of-day decision: an A that believes $\hat{w}_A > 49$ keeps its energy instead of covering the 16–20h shortfall, and a B that believes $\hat{w}_B < 49$ sells energy it should keep. At $\sigma_w = 5$ \$/MWh this happens with probability 3.6% for A and 11.5% for B.

10.5.2 First check at zero error: the two bids cross

Before any error, each owner builds $\mathcal{B}^{\star p}$, its bid from the true prices and its own plan. With one ESS this costs 1.30% of V. With two owners it costs **38.1%**, sheds 124 MWh of load and leaves 6,976 MWh of the day-ahead schedule undeliverable (Figure 10.4).

Mechanism. The quantity rule (9.3) offers, besides the planned quantity, all the remaining power. With Baldick's power limits of 1000 MW, this second block is up to 1000 MW regardless of how much energy the ESS holds. At 20–24h, A offers to sell at $c_A = 41$ \$/MWh ($\pi_A = w_A = 40$ plus ζ) and B bids to buy at $u_B = 43.1$ \$/MWh. The bids cross, and the market clears A selling 1000 MW and B buying 900 MW at 43.09 \$/MWh: the two ESS trade with each other. Neither can deliver, since A is almost empty and B almost full. In real time each ESS does what its SOC allows: A delivers 80 MW, while B still charges 111 MW, and 31 MW of load are shed to serve that charge, because the real-time stage treats an ESS's feasible charge as firm. With one ESS this cannot happen: an ESS cannot trade with itself.

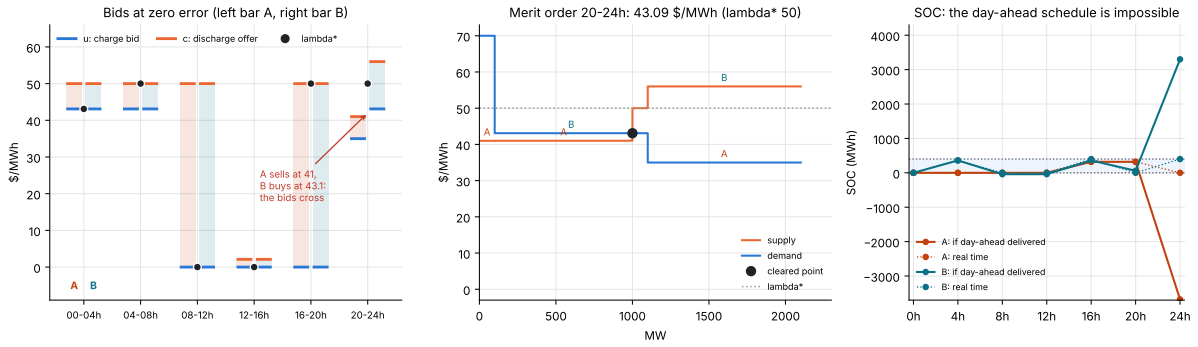


Figure 10.4: Stage 1, zero error ($\mathcal{B}^{\star p}$ for both owners). Left: bids and offers of A and B. Centre: merit order of 20–24h, where the two bids cross. Right: SOC if the day-ahead schedule were delivered, and in real time.

Alternatives. Table 10.4 repeats the zero-error check with three changes. Offering only the plan is worse, because the second blocks are also what lets the market move energy between the two ESS when their plans overlap. Capping the second block by what the ESS could still absorb or deliver from the SOC its own plan reaches,

$$\min\Big\{\bar{p}_e,\ \frac{\bar{s}_e - \gamma_e s^{\circ}_{e,t-1}}{\alpha_e \Delta T}\Big\} - p^{\circ}_{e,t} \quad \text{(charge)}, \qquad \min\Big\{\bar{g}_e,\ \frac{\gamma_e s^{\circ}_{e,t-1} - \underline{s}_e}{\beta_e \Delta T}\Big\} - g^{\circ}_{e,t} \quad \text{(discharge)}, \tag{10.1}$$

brings the loss down to 2.0% with no shedding. Realistic power limits of 100 MW have a similar effect.

Quantity rule / case (zero error, $\mathcal{B}^{\star p}$, $\psi = 10$)	loss (% of V)	shed (MWh)	undeliv. (MWh)
Current rule: second block = all remaining power	38.1	124	6,976
Plan only (no second block)	84.6	300	120
Second block capped by the SOC of the plan	2.0	0	120
Current rule, power limits of 100 MW	0.6	0	40

Table 10.4: Stage 1: zero-error check under alternative quantity rules.

Consequence. The crossing dominates every result with errors: in a first exploratory run with the current rule (40 draws per setting, $\omega = 0.5$), the mean loss ranged from about 20% to 100% of V, load was shed in most draws, and there was no clear pattern in σ or σ_w. The Stage 1 grid is therefore on hold until two decisions are taken: (i) whether to keep the current rule and report the crossing as a property of the current design with independent owners, to cap the second block by the SOC of the plan (an ESS should not offer energy it cannot have), or to use realistic power limits; and (ii) whether the real-time stage should curtail an ESS's charge before it sheds load.

10.6 Limitations and lessons

Limitations.

- One six-interval system with exact ties, in which the ESS is large compared with the load and the storage value is dominated by avoided load shedding.
- In Stage 0: $\rho = 0$, no bias and $\psi = 10$ \$/MWh only for the forecast-based bids; 150 draws per setting.

- The real-time model is simple: an L1 repair of each ESS schedule, a flat premium on every redispatched MWh, and an ESS's feasible charge treated as firm.

Lessons.

1. Report quantity and price losses separately: perfect prices are not enough, and at the true prices the choice of multipliers does not change the outcome.
2. Treat ψ as a parameter: it decides the ranking between forecast-based and technical-data-only bids.
3. Use common random numbers and paired comparisons; read the mean loss through the regimes that produce it.
4. With several owners, bids must not offer energy the ESS cannot have; the quantity rule must be revised before Stage 1 is run.
5. Move to a larger, tie-free system (Stage 3) before drawing conclusions on the size of the effects.

Chapter 11

Discussion and outlook

11.1 Answers so far

RQ1: equivalence. Yes. Interval bids built from the multipliers of the SOC-based design, $u_{e,t} = \alpha_e(\pi^\star_{e,t} - \bar{\nu}^\star_{e,t}) - \epsilon_e$ and $c_{e,t} = \zeta_e + \beta_e(\pi^\star_{e,t} + \underline{\nu}^\star_{e,t})$ with blocks at full capacity, reproduce its dispatch and nodal prices in the current design (Proposition 5.1). The result holds with a DC network, ramps, elastic demand and any number of ESS. The current design is a Lagrangian relaxation of the SOC-based design with respect to the SOC constraints, and the two optimal values differ by a known constant (Proposition 5.2). General stepped bids are equivalent if and only if the benchmark schedule lies on each ESS's bid and offer curves at the benchmark prices (Proposition 5.3).

RQ2: information. The equivalent bids depend on the instance only through the ESS's own technical data and the nodal prices at its own bus (Theorem 6.3). Each ESS can compute them by solving its own price-taking self-schedule and reading any optimal dual, independently of the other ESS (Corollary 6.15). The set of all equivalent bids is a polyhedron (Proposition 6.5). Prices pin the bids everywhere except at intervals where an SOC limit is reached and cannot be undone; there, a selection rule is needed, and the pre-limit rule is designed to protect against infeasible actions under price errors (Section 6.5). In closed form, the bids depend on the private parameters, on the prices of the marginal intervals of the SOC path, and on the binding intervals (Corollary 6.14). If no SOC limit binds, technical data alone suffice (Corollary 5.5), and in a current design that enforces the SOC, technical-data-only bids are always exact (Remark 5.8). Prices are not the whole story: where an ESS sets the price, the cleared quantity is also indeterminate, so exact reproduction requires quantity information too (Remark 5.6).

RQ3: cost of imperfect information (first results). The experiment pipeline is implemented and its gate passes: with the benchmark multipliers and quantities, the current design reproduces the SOC-based outcome with zero loss, with one and with two ESS (Chapter 10). On Baldick's base case with one ESS, perfect prices combined with the ESS's own plan lose 1.30% of the storage value, through quantities: an hourly market cannot coordinate a purchase in one interval with the sale it is meant for in the next. Any forecast error raises the loss to about 6%, and it grows to 7.9% at an error of 20 \$/MWh; the shape of the error matters more than its size, and the losses take a few discrete values that correspond to a few market regimes. Technical-data-only bids lose ψ times the energy they leave undeliverable, less than the forecast-based bids at $\psi = 10$ \$/MWh in this case. With two owners, bids built with the current quantity rule cross and the ESS trade with each other, which costs 38% of the storage value even with perfect prices; capping the bids by the SOC of the owner's plan reduces this to 2%. These numbers come from a six-interval system with exact ties and a storage value dominated by avoided load shedding; the size of the effects must be established on a larger system.

RQ4: use by the operator (proposed). The equivalent-bid map, evaluated by the operator with its own price forecast, gives an SOC-consistent competitive reference bid with a natural tolerance band (Section 6.7). Its evaluation against strategic bids and against CAISO-type default energy bids is future work.

11.2 Limitations of the analysis

- *Convexity.* The results rely on LP duality. Unit commitment, non-convex costs or SOC-dependent efficiencies would break the exact equivalence; the equivalence would then hold at best for the convex hull or at fixed commitments.
- *Price-taking.* All participants are price-takers. A large ESS that anticipates its effect on prices would not bid the equivalent bids; this is the subject of the optional strategic extension.
- *Terminal value.* The results assume a concave, piecewise-linear terminal benefit. For a nonlinear benefit, the final SOC enters the bids through $\pi_{e,T} = w_e(s_{e,T} - \bar{\bar{s}}_{e,T})$, which is one more endogenous quantity.
- *Exclusivity.* Charge–discharge exclusivity is relaxed; Section 7.3 shows that this is exact unless stored energy has a strongly negative value.
- *Price uniqueness.* Theorem 6.3 guarantees that $\lambda^\star$ is *a* vector of CD prices, not that the operator's pricing rule selects it. In all cases tested so far it does.
- *Quantities.* Interval bids cannot carry the quantity information that the SOC-based design uses where the ESS sets the price. The quantity rule of Section 9.5 uses the ESS's own plan, which need not match what the system needs, and with several owners its spare-capacity blocks can cross (Section 10.5.2).
- *Test system.* The experiments so far use a six-interval system with exact ties in the true prices; results there are driven by which tied plan the ESS chooses, and they may not carry over to a larger system with a smooth supply curve.

11.3 Next steps

1. *Decisions with the supervisors* (status page): quantity rule with several owners, priority of an ESS's charge against load shedding in real time, which Stage 1 grid to run, values of ψ, and the Stage 3 system.
2. *Stage 1*: two owners with price and terminal-value errors, or the full one-ESS grid, with the split of the loss by interval (H2), persistence (H3), the comparison of selection rules (H4) and the ψ grid (H6).
3. *Regime analysis*: group the draws of each setting by their day-ahead schedule, report each regime's loss and probability, and explain the change of the mean loss between settings as flows of draws between regimes (Section 10.4).
4. *Calibration*: backtest a naive and a state-of-the-art forecaster on Danish day-ahead prices to estimate $(b, \sigma, \rho, \omega)$, and prepare the day-block bootstrap of residuals (Section 9.7).
5. *Stage 2*: a fleet of ESS with common and independent errors (H5).
6. *Stage 3*: realistic system, many days, higher renewable shares, calibrated and empirical errors; multi-node version for congestion.
7. *Richer bids*: the loss at perfect prices is an inter-temporal coordination loss that no single interval bid can remove. Bids that spread the valid range of π, use a value function, or are built from scenarios of the owner's own forecast stay within the current design and measure how much of the gap is due to naive bidding; scenario-based bids are the most likely to help against forecast errors.

8. *Current design with SOC enforcement or SOC-dependent bids*: by Remark 5.8, technical-data-only bids are exact when the operator enforces the SOC; an intermediate design in which the market tracks the SOC and the ESS submits its value of stored energy as a function of the SOC would address the coordination loss directly. Comparing three designs (interval bids, SOC-dependent bids, SOC-based design) would measure the value of that intermediate step.
9. *Extension (RQ4)*: compare $\Phi_e(\theta_e, \hat{\lambda}^{\mathrm{ISO}})$ with a CAISO-type default energy bid as a screening reference, with strategic bid scenarios.

ID	Milestone or gate (project plan)	Target	Status
M1	Model formulation of both designs	Oct 2026	done (Chapter 3)
G1	Design variants, evaluation and price-selection rules	Oct 2026	implemented; decisions pending (Chapter 9)
M2	Equivalence and information analysis	Nov 2026	done (Chapters 5 and 6)
M3	Code base verified, Stage 0 run	Nov 2026	done (Chapters 8 and 10)
G2	Perfect forecast reproduces the exact equivalent bids	Dec 2026	passed in Stage 0 ($\mathcal{B}^\star$: zero loss)
M4	Stage 1 results	Dec 2026	case and first check done; grid on hold
M5	Stage 2 results	Jan 2027	–
G3	Decision on the optional strategic extension	Jan 2027	–
M6	Stage 3 results	Feb 2027	–
M7	Thesis submission	Mar 2027	–

Table 11.1: Milestones of the project plan and their status (November 2026).

Chapter A

Data of the test cases

A.1 Baldick's cases

One bus; $T = 6$ intervals of $\Delta T = 4\,\text{h}$; renewable unit at $0\,\$/\text{MWh}$ with availability $(200, 100, 500, 500, 25, 0)\,\text{MW}$; thermal unit of $100\,\text{MW}$ at $50\,\$/\text{MWh}$; inelastic demand $(100, 200, 300, 300, 200, 100)\,\text{MW}$; one ESS with $\alpha = 0.9$, $\beta = 1$, $\gamma = 1$, $\epsilon = \zeta = 1\,\$/\text{MWh}$, $\underline{s} = 0$, $s_0 = 0$, $\bar{p} = \bar{g} = 1000\,\text{MW}$ [8, §4.7].

Case	$\bar{s}$ (MWh)	w ($/MWh)
Base case	800	40
High willingness to pay	800	55
Increased maximum SOC	1200	40

A.2 Stage 1 case

Baldick's base case with the ESS split into two ESS, A and B, of $\bar{s} = 400\,\text{MWh}$ each, identical to the base-case ESS except for the terminal value: $w_A = 40$ and $w_B = 55\,\$/\text{MWh}$. Every other datum is as above.

Chapter B

Code and reproducibility

All results are produced by the code in the GitHub repository `https://github.com/ccanoyrac/ESS-equivalent-bids-from-SOC-DA-market` (folder `Code`). It requires Python 3.10 or later with NumPy, SciPy (HiGHS solver), pandas and Matplotlib; Jupyter for the notebooks.

File	Purpose
Verification of the analytical results (`src/`)	
`verify_equivalence.py`	Builds and solves both designs for a case (network in node pairs, ramps, elastic demand, ESS); extracts multipliers; KKT residuals; Baldick's cases.
`verify_price_based.py`	Self-schedule $\mathcal{P}_e^{\mathrm{SS}}(\lambda)$; polyhedron of Proposition 6.5; ranges of π; closed forms; pre-limit rule.
`numerical_check.py`	All checks of Table 8.1 and Figures 8.1 and 8.2.
`computational_burden.py`	Benchmark of Chapter 7.
Experiments (`src/experiments/`, Chapters 9 and 10)	
`config.py`	Settings (error model, bid rule, real time, seeds) and the stage grids.
`cases.py`	Test systems: Baldick's base case (Stage 0) and the two-owner case (Stage 1).
`benchmark.py`	SOC-based benchmark: $\lambda^\star$, schedules, multipliers, $SW^\star$, SW^{noESS}, V.
`forecasts.py`	Error model (9.4), (9.5) with price floor; terminal-value error (9.6).
`bids.py`	Bid families, self-schedule with tie-breaker (9.2), selection rules, quantity rule (9.3).
`market.py`	Day-ahead clearing of the current design with any number of blocks and ESS.
`realtime.py`	Real-time repair of each ESS and generator redispatch.
`welfare.py, metrics.py`	Realised welfare (9.7), ESS profit, loss, decomposition, paired differences, summaries.
`runner.py`	Experiment loop: one row per draw, resumable, with metadata.
Scripts, tests and notebooks	
`scripts/run_stage0_gate.py`	Runs Stage 0 and its gate (Table 10.1).
`scripts/make_stage1_figures.py`	Stage 1 case, benchmark, errors and zero-error check (Section 10.5).
`scripts/plot_multiplier_polyhedron.py`	Polyhedron figures (Figure 10.2).
`tests/`	21 tests: error model and seeds, bids and tie-breaker, clearing and real time, gates with one and two ESS.
`notebooks/`	`verify_equivalence.ipynb` (the checks of Chapter 8); `stage0_experiments.ipynb` (Stage 0 step by step, with an explorer of single draws).

From the `Code` folder,

```
python -m pytest tests    and    python scripts/run_stage0_gate.py
```

run the tests and reproduce Stage 0 (about a minute). The report folder contains `make_figures.py` (figures of Chapter 7) and `make_figures_experiments.py`, which regenerates the figures of Chap-

ters 9 and 10 from the repository. The five-ESS check of Section 8.3 was run with an earlier script (`multi_ess.py`) that is not yet in the repository.

Chapter C

Correspondence with the technical note

The technical note on the KKT equivalence contains the analytical results of Chapters 3, 5, 6 and 8 in compact form. Its numbering differs from that of this report.

Technical note	**This report**	**Technical note**	**This report**
Proposition 1	Proposition 5.1	Lemma 1	Lemma 6.10
Proposition 2	Proposition 5.2	Lemma 2	Lemma 6.11
Proposition 3	Proposition 5.3	Lemma 3	Lemma 6.12
Corollary 1	Corollary 5.4	Proposition 4	Proposition 6.13
Corollary 2	Corollary 5.5	Corollary 4	Corollary 6.14
Remark (uniqueness)	Remark 5.6	Lemma 4	Lemma 6.2
Remark (exclusivity)	Remark 5.7	Theorem 1	Theorem 6.3
Remark (SOC-aware CD)	Remark 5.8	Remark (claims)	Remark 6.4
Assumption 1	Assumption 6.7	Proposition 5	Proposition 6.5
Corollary 3	Corollary 6.8	Corollary 5	Corollary 6.6
Definition 1	Definition 6.9	Remark (who computes Φ_e)	Section 6.7
Assumption 2	Assumption 6.1	Section 5 (numerical check)	Chapter 8

New in this report: the SOC-based design as an MCP and competitive equilibrium (Chapter 4, which consolidates earlier separate notes on the MCP), the explicit result for several ESS (Corollary 6.15), the computational burden (Chapter 7), the experimental design (Chapter 9) and the first experiments (Chapter 10).

Chapter D

Work log

Period	Work and decisions
Sep 2026	MCP of the SOC-based design: participant problems, complete complementarity system, correspondence with the operator's KKT conditions (partition), equilibrium theorem and its assumptions, in particular the non-increasing end-of-horizon willingness to pay.
early Oct 2026	Technical project plan: research questions, hypotheses, stages, work packages and schedule. First version of the KKT equivalence (equivalent bids from multipliers, Lagrangian relaxation, stepped bids). Analysis of the non-uniqueness of π in Baldick's base case and of the robustness margins.
mid Oct 2026	Equivalent bids as functions of prices (segments, marginal intervals). Bids from technical data and prices (decomposition, Theorem, polyhedron). Supervisors' requests: network in node pairs, formal proof that the bids follow from technical data and prices, care with labelling results as lemmas and corollaries in presentations. Network rewritten in node pairs. Numerical checks restricted to Baldick's cases, with figures. Computational burden: formulas and benchmark. Project narrative: motivation for not adopting the SOC-based design directly, and the monitoring use. Code folder and notebook.
mid Oct 2026	Presentations to the supervisors restyled in the DTU Wind template.
20 Oct 2026	Several ESS: separability confirmed analytically and numerically (five ESS). Experimental design: bid families, error model, quantity rule, real-time evaluation, metrics; a first prototype of the pipeline. Design submitted for approval.
21 Oct 2026	Calibration of the error model from historical data proposed. This report assembled; MCP chapter added.
late Oct 2026	Experiment code built in the repository: bid families, clearing, real time, welfare, runner and tests. Stage 0 on Baldick's base case; price floor, throughput tie-breaker and common random numbers added after the first runs; Stage 0 gate passed. Analysis of the multiplier polyhedron at the true and forecast prices; diagnosis of the loss at perfect prices as a coordination loss; proposal of richer bid curves and of a regime analysis of the results.
28 Oct 2026	Stage 1 defined as two owners with an error in the terminal value. First check at zero error: the owners' bids cross; alternative quantity rules evaluated; decisions requested.
1 Nov 2026	Report updated to the experiment code: Chapter 9 rewritten, Chapter 10 replaces the earlier prototype results.

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